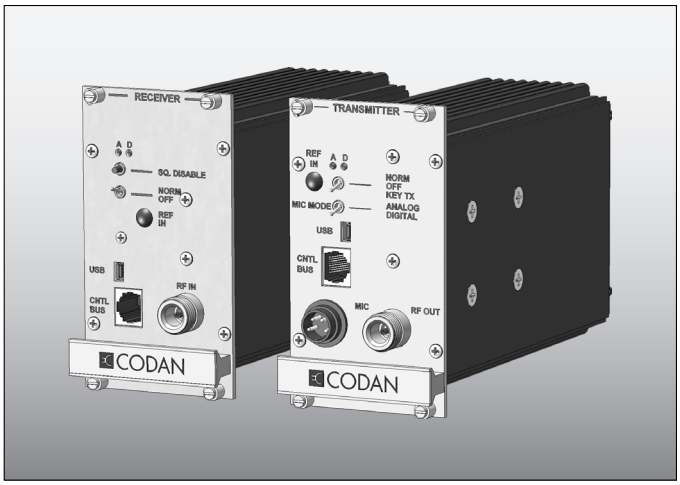
CODAN
COMMUNICATIONS



MT-4E VHF & UHF RECEIVER & TRANSMITTER INSTRUCTION MANUAL

136–174MHZ / 406–470MHZ / 470–520MHZ / 768–869MHZ / 896–960MHZ

Covers Models:	
VR-4E150-00-000	UR-4E800-00-000
VR-4E150-A0-000	UR-4E850-00-000
UR-4E420-00-000	UR-4E950-00-000
UR-4E420-A0-000	VT-4E150-00-800
UR-4E460-00-000	UT-4E450-00-800
UR-4E460-A0-000	UT-4E500-00-800
UR-4E500-00-000	UT-4E850-00-300
UR-4E768-00-000	UT-4E900-00-300

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The English version takes precedence over any translated versions.

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www.codancomms.com

NOTE

The user's authority to operate this equipment could be revoked through any changes or modifications not expressly approved by Codan Limited.

The design of this equipment is subject to change due to continuous development. This equipment may incorporate minor changes in detail from the information contained in this manual.

Operation is subject to the following two conditions: (1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

RF Exposure Warning

Exposure to radio frequency (RF) energy has been identified as a potential environmental factor that must be considered before a radio transmitter can be authorized or licensed. The FCC and IC have therefore developed maximum permissible exposure (MPE) limits for field strength and power density, listed in FCC 47 CFR § 1.1310 and IC RSS-102 Issue 5 Sect 4. The FCC has furthermore determined that determination of compliance with these exposure limits, and preparation of an Environmental Assessment (EA) if the limits are exceeded, is necessary only for facilities, operations and transmitters that fall into certain risk categories, listed in FCC 47 CFR § 1.1307 (b), Table 1. All other facilities, operations and transmitters are categorically excluded from making such studies or preparing an EA, except as indicated in FCC 47 CFR §§ 1.1307 (c) and (d).

KDB 447198 D01 General RF Exposure Guidance v06 and IC RSS-102 Issue 5 provide assistance in determining whether a proposed or existing transmitting facility, operation or device complies with RF exposure limits. In accordance with KDB 447198 , FCC 47 CFR § 1.1307 (b) and RSS-102 Issue 5 Sect 2.5, the Codan Communications transmitter manufactured in Canada is categorically excluded from routine evaluation or preparing an EA for RF emissions and this exclusion is sufficient basis for assuming compliance with FCC/IC MPE limits. This exclusion is subject to the limits specified in FCC 47 CFR §§ 1.1307 (b), 1.1310 and IC RSS-102 Issue 5 Sect 4. Codan Communications has no reason to believe that the excluded transmitter encompasses exceptional characteristics that could cause non-compliance.

Notes:

- The FCC and IC's exposure guidelines constitute exposure limits, not emission limits. They are relevant to locations that are accessible to workers or members of the public. Such access can be restricted or controlled by appropriate means (i.e., fences, warning signs and others).
- The FCC and IC's limits apply cumulatively to all sources of RF emissions affecting a given site. Sites exceeding these limits are subject to an EA and must provide test reports indicating compliance.

RF Safety Guidelines and Information

Base and Repeater radio transmitters are designed to generate and radiate RF energy by means of an external antenna, typically mounted at a significant height above ground to provide adequate signal coverage. To reduce potential radio interference to other users, the antenna type and its gain should be so chosen that the equivalent isotropically radiated power (EIRP) is not more than that permitted for successful communication. The following antenna installation guidelines must be adhered to in order to ensure RF exposure compliance:

Non-building-mounted Antennas:

- Height above ground level to lowest point of antenna ≥ 10m
- Power ≤ 1000W ERP (1640W EIRP)

Building-mounted Antennas:

Power ≤ 1000W ERP (1640W EIRP)

The following RF Safety Guidelines should be observed when working in or around transmitter sites:

- Do not work on or around any transmitting antenna while RF power is applied.
- Before working on an antenna, disable the appropriate transmitter and ensure a "DO NOT USE" or similar sign is placed on or near the PTT or key-up control.
- Assume all antennas are active unless specifically indicated otherwise.
- Never operate a transmitter with the cover removed.
- Ensure all personnel entering a transmitter site have electromagnetic energy awareness training.

For more information on RF energy exposure and compliance, please refer to the following:

1. FCC Code of Regulations; 47 CFR §§ 1.1307 and 1.1310
2. KDB 447198 D01 General RF Exposure Guidance v06
3. https://www.fcc.gov/general/radio-frequency-safety-0
4. IC RSS-102 Issue 5, "Radio Frequency Exposure Compliance of Radio Communication Apparatus"

RF Maximum Permissible Exposure (MPE)

Exhibit Requirements for Installations in the United States of America

FCC Part 1, Section 1.1307 table 1- Transmitters, Facilities and Operations Subject to Routine Environmental Evaluation states the following for Part 90 Devices:

- Part 90 devices Non-building-mounted antennas: height above ground level to lowest point of antenna <10 m and power >1000 W ERP (1640 W EIRP). Building-mounted antennas: power >1000 W ERP (1640 W EIRP).

Another way of wording this is that Part 90 devices are not Subject to Routine Environmental Evaluation when the antenna is installed at 10Meters or higher and operating total power level of all channels is less than 1640 Watts EIRP.

As an example, a 100W transmitter with a 10dB gain antenna with a low loss cable would translate into 1,000 Watts EIRP in the envelope lobe. If it is mounted 10 Meters or higher above where people could be walking, you have a safe installation and do not have to perform MPE calculations for safe distance.

If the antenna is lower than 10 Meters then you need to verify that your installation is at a safe distance for Exposure to the General Population.

For United States installations, you must ensure that your installation complies with the Maximum Permissible Exposure (MPE) requirements for general population that are specified under FCC Part 1 Section 1.1310 Table 1.

TABLE 1—LIMITS FOR MAXIMUM PERMISSIBLE EXPOSURE (MPE)

Frequency range (MHz)	Electric field strength (V/m)	Magnetic field strength (A/m)	Power density (mW/cm^2)	Averaging time (minutes)
(A) Limits for Occupational/Controlled Exposure				
0.3-3.0	614	1.63	*100	6
3.0-30	1842/f	4.89/f	*900/f^2	6
30-300	61.4	0.163	1.0	6
300-1,500			f/300	6
1,500-100,000			5	6
(B) Limits for General Population/Uncontrolled Exposure				
0.3-1.34	614	1.63	*100	30
1.34-30	824/f	2.19/f	*180/f^2	30
30-300	27.5	0.073	0.2	30
300-1,500			f/1500	30
1,500-100,000			1.0	30

f = frequency in MHz * = Plane-wave equivalent power density

For US Installations, the maximum power density resulting from the composite Effective Isotropic Radiated Power (EIRP) from the antenna connected to this equipment must be limited to the maximum permissible exposure as stated below:

- **Power density limit for the band 152 to 174MHz** = 0.2 mW/cm^2

MPE and Safe Distance Calculations for USA Installations

This Power Density value is determined by the combination of RF output, cable loss, antenna gain, and distance from the antenna when energized.
The MPE calculation for US installations is expressed as follows:

- **Power Density Pd (mW/cm^2) =** $\left(\frac{EIRP}{4*\pi*d^2}\right)$

Where

- **d** = distance from the antenna expressed in cm.
- **EIRP expressed in mW** = $10^{\left(\frac{[Tx\ pwr\ (dBm)+Ant\ Gain\ (dBi)-Cable\ Loss\ (dB)]}{10}\right)}$
- **Tx Power (dBm)** = 10*log[Tx Power (mW)]

As an example, with the transmitter running at 100 watts output into an antenna with a gain of 10 dBi using a short cable with 0dB loss, to verify if 650cm (6.5meters) is a safe distance from the antenna to ensure exposure compliance of 0.2mW/cm2:

1) 100 Watts Tx Power = 50dBm
2) EIRP (mW) = $10^{\left(\frac{[Tx\ pwr\ (50dBm)+Ant\ Gain\ (10dBi)-Cable\ Loss\ (0dB)]}{10}\right)} = 10^{\left(\frac{60}{10}\right)}$ = 1,000,000mW
3) Pd (mW/cm2) = $\left(\frac{\text{EIRP}}{4*\pi*d^2}\right) = \left(\frac{1{,}000{,}000}{4*\pi*650^2}\right) = \left(\frac{1{,}000{,}000}{5{,}309{,}291}\right)$=0.19 mW/cm^2

- 6.5 meters (21.125 Feet) is a safe distance for US installations when using a 10dBi Antenna

The minimum safe distance, from a radiating structure using different "Gain Antennas"

- For the Band 152 to 174MHz with 2dBi Gain Antenna: d (safe distance) = 2.6 m
- For the Band 152 to 174MHz with 6dBi Gain Antenna: d (safe distance) = 4.0 m
- For the Band 152 to 174MHz with 10dBi Gain Antenna: d (safe distance) = 6.5 m

RF Maximum Permissible Exposure (MPE)

Exhibit Requirements for Installations in Canada

Table 4: RF Field Strength Limits for Devices Used by the General Public (Uncontrolled Environment)

Frequency Range (MHz)	Electric Field (V/m rms)	Magnetic Field (A/m rms)	Power Density (W/m^2)	Reference Period (minutes)
$0.003\text{-}10^{21}$	83	90	-	Instantaneous*
0.1-10	-	$0.73/f$	-	6**
1.1-10	$87/f^{0.5}$	-	-	6**
10-20	27.46	0.0728	2	6
20-48	$58.07/f^{0.25}$	$0.1540/f^{0.25}$	$8.944/f^{0.5}$	6
48-300	22.06	0.05852	1.291	6
300-6000	$3.142f^{0.3417}$	$0.008335f^{0.3417}$	$0.02619f^{0.6834}$	6
6000-15000	61.4	0.163	10	6
15000-150000	61.4	0.163	10	$616000/f^{1.2}$
150000-300000	$0.158f^{0.5}$	$4.21 \times 10^{-4} f^{0.5}$	$6.67 \times 10^{-5} f$	$616000/f^{1.2}$

Note: f is frequency in MHz.
*Based on nerve stimulation (NS).
** Based on specific absorption rate (SAR).

MPE and Safe Distance Calculations for Canada Installations

For Canada installations, the maximum power density resulting from the composite Effective Isotropic Radiated Power (EIRP) from the antenna connected to this equipment must be limited to the maximum permissible exposure as stated below:

- **Power density limit for the band 152 to 174MHz** = 1.291 W/m²

The MPE calculation for US is expressed as follows:

- **Power Density Pd (W/m²)** = $\left(\frac{EIRP}{4*\pi*d^2}\right)$

Where

- **d** = distance from the antenna expressed in meters (m)
- **EIRP expressed in Watts (W)** = $10^{\left(\frac{[Tx\ pwr\ (dBW)+Ant\ Gain\ (dBi)-Cable\ Loss\ (dB)]}{10}\right)}$
- **Tx Power (dBW)** = 10*log[Tx Power (W)]

As an example, with the transmitter running at 100 watts output into an antenna with a gain of 10 dBi using a short cable with 0dB loss, to verify if 7.5meters is a safe distance from the antenna to ensure exposure compliance of 1.21W/m2:

4) 100 Watts Tx Power = 20dB

5) EIRP (W) = $10^{\left(\frac{[Tx\ pwr\ (20dBW)+Ant\ Gain\ (10dBi)-Cable\ Loss\ (0dB)]}{10}\right)} = 10^{\left(\frac{30}{10}\right)} = 1{,}000W$

6) Pd (W/m2) = Pd (W/m2) = $\left(\frac{EIRP}{4*\pi*d^2}\right) = \left(\frac{1{,}000}{4*\pi*8.5^2}\right) = \left(\frac{1{,}000}{907.9}\right) = 1.11\ W/m^2$

- 8.5 meters is a safe distance for Canada installations when using a 10dBi gain antenna.

When installing the antenna, the above relationship should be used to ensure the combination of power, antenna gain, and distance is such that the maximum permissible power density is not exceeded. Different combinations of output power and antenna gain will result in different minimum safe distances.

The minimum safe distance, from a radiating structure using different Gain Antennas"

- For the Band 152 to 174MHz with 2dBi Gain Antenna: d (safe distance) = 3.5 m
- For the Band 152 to 174MHz with 6dBi Gain Antenna: d (safe distance) = 5.5 m
- For the Band 152 to 174MHz with 10dBi Gain Antenna: d (safe distance) = 8.5 m

RoHS COMPLIANCE STATEMENT

The Restriction of Hazardous Substances (RoHS) Directive, also known as Directive 2002/95/EC, took effect on July 1, 2006 and must be complied with in order to sell electronic equipment in the European Union. RoHS restricts the use of six hazardous materials: lead, cadmium, hexavalent chromium, mercury, PBBs and PBDEs (brominated flame retardants) in new electrical and electronic equipment.

All Codan Communications products that are RoHS-compliant bear the RoHS logo.



WEEE COMPLIANCE STATEMENT

The Waste Electrical and Electronic Equipment (WEEE) Directive, also known as Directive 2002/96/EC, is closely linked with RoHS and, by European law, sets the targets for collection, treatment, recovery and environmentally sound disposal of electrical and electronic goods.

Codan Communications voluntarily complies with this directive and WEEE-compliant products bear the following international symbol.



IMPORTANT

RoHS-compliant lead-free solder alloys and processes are different from tin-lead solder alloys and processes. Prior to modification, identify which process is required.

- Printed Circuit Board (PCB) Series **42-xxxx**, **43-xxxx** and **5xxxx-xx** indicate a tin-lead manufacturing process
- Printed Circuit Board (PCB) Series **6xxxx-xx** indicate a RoHS-compliant manufacturing process using lead-free alloys

RoHS-compliant PCBs are further identified with the RoHS Logo where space permits.

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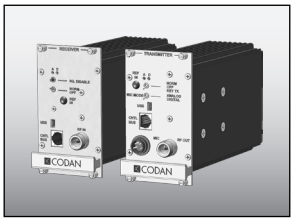
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GENERAL INFORMATION

The General Information and the Theory of Operation sections are common for all models of MT-4E Receivers and Transmitters. Information that is specific to individual models will be identified where relevant.

INTRODUCTION

The MT-4E Receiver and Transmitter are FM radio modules capable of analog operation in 12.5kHz (narrow band) or 25kHz (wide band) channels. A firmware upgrade may be purchased to allow P25 Digital operation. Modular design allows each of the receiver and transmitter's internal modules to be individually assembled and tested, which facilitates construction, tuning and general maintenance.

The MT-4E Receiver and Transmitter combine state of the art performance in a compact modular enclosure for applications ranging from remote mountain top repeaters to congested urban radio environments. Each module is characterized by dependable, low maintenance performance under the most severe environmental conditions.

The MT-4E Receiver and Transmitter are primarily software controlled radio modules, allowing tuning, programming and maintenance to be done via software service with few hardware adjustments required.

The MT-4E series is compatible with all subrack and base station enclosures. It supports a basic analog interface and may be used in a mixed system with MT-4 series receivers and transmitters.

P25 Digital Mode

When the P25 firmware upgrade is purchased, the MT-4E family of receivers and transmitters may be configured for P25 Digital operation, or mixed mode operation. In a repeater configuration, the receiver passes the complete P25 Digital voice packet directly to the transmitter so no P25 Digital information is lost.

All P25 specifications, operational description and information contained in this instruction manual require the P25 firmware upgrade to function. If the MT-4E Receiver and Transmitter are purchased without the P25 firmware upgrade, the radio will program and operate in analog mode only.

NOTE: This equipment is not to be operated within the frequency band *406–406.1MHz* unless specifically authorized by the Federal Communications Commission or Industry Canada in conjunction with COSPAS / SARSAT.

Secure Communications

The receiver and transmitter are capable of decoding and encoding secure communications if a DES-OFB / AES Encryption Module is installed. To successfully decode or encode a transmission, the encryption module must be programmed with a valid encryption key using the Motorola KVL 3000 Plus Key Loader in conjunction with a Key Loader Cable. The Key Loader Cable plugs into the front panel RJ45 jack on the front of the receiver or transmitter module.

Consult Key Loader instructions for details on loading a key. A loaded key may be zeroized by pulling the CLEAR KEYS 1, then the CLEAR KEYS 2 inputs to ground 500 ms apart. Alternately, the CI-BC-4E Base Controller or Universal Interface Card (UIC) may be used to clear the keys for all modules in a system with a single key press of the switch marked ZEROIZE KEY. The CI-BC-4E Base Controller can also be used to control whether the transmitter outputs a secure or a clear signal.

Firmware Upgrades

Receiver and transmitter firmware upgrading is performed with the PC-based firmware flashing software. A USB Type A to 5-pin Mini-B cable is used to connect the USB port of an IBM-compatible computer to the USB port on the front panel of the receiver or transmitter module.

Firmware upgrades can be found on the website through www.codancomms.com.

MT-4E Receiver Family Models

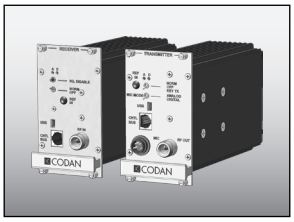
Following are the models in the MT-4E Receiver family:

VR-4E150-00-000	UR-4E500-00-000
VR-4E150-A0-000	UR-4E768-00-000
UR-4E420-00-000	UR-4E800-00-000
UR-4E420-A0-000	UR-4E850-00-000
UR-4E460-00-000	UR-4E950-00-000
UR-4E460-A0-000	

MT-4E Transmitter Family Models

Following are the models in the MT-4E Transmitter family:

VT-4E150-00-800
UT-4E450-00-800
UT-4E500-00-800
UT-4E850-00-300
UT-4E900-00-300



THEORY OF OPERATION

RECEIVER THEORY

The MT-4E Receiver is constructed with the Receiver Mainboard and the RF Preselector subassembly. The Synthesizer and the Universal Daughter Board (UDB), that contains the Digital Signal Processor (DSP), are mounted on the receiver mainboard. An optional Encryption Module may also be installed on the receiver mainboard.

The MT-4E Receiver can receive both an analog FM modulated carrier and a C4FM P25 digitally modulated carrier and outputs both a baseband audio signal to the motherboard and a Low Voltage Differential Signaling (LVDS) serial data signal to the front panel RJ45 jack.

Power Supply

A nominal 13.8VDC and a regulated +9.5VDC enters the receiver module from the motherboard.

Synthesizer

The synthesizer generates the LO frequency which is output through a small coax cable to the mixer in the receiver RF Preselector. Communication inputs and outputs exist between the DSP and synthesizer for control functions. The synthesizer is wide band and generates LO frequencies for mixing across the entire frequency band.

RF Preselector

The RF Preselector combines a low-noise amplifier with a high-selectivity filter structure. RF signals at the antenna input are filtered, amplified through a low-noise amplifier, filtered again, and then mixed with a local oscillator supplied by the synthesizer, to produce an IF frequency that is output to the receiver mainboard.

For receiver and transmitter block diagrams, see VHF & UHF Hardware Interfacing and 700 / 800 / 900MHz Hardware Interfacing sections.

Interconnections are made using quick connect SMB connectors. A high-selectivity narrow bandwidth, combined with high ultimate out-of-band signal rejection, greatly improves receiver spurious response and wide-band intermodulation rejection.

Power for the RF Preselector is supplied from the receiver's regulated +9.5VDC.

IF Demodulation

The receiver mainboard processes the intermediate frequency (IF) signal from the RF Preselector through selective crystal filtering and IF amplification. The signal is then passed through the IQ Demodulator and Digitizing Stage for demodulation to two quadrature-related baseband outputs. These outputs are represented in a digital stream which is passed to the UDB, where DSP techniques are used to further process the incoming sampled signal to extract P25 Digital voice signals and analog voice signals.

Analog / Digital Detection

The digital detector in the UDB looks for a specific P25-bit pattern called the Frame Sync. If the frame sync is detected, P25 Digital processing is applied. If no frame sync is detected, the signal is assumed to be analog, and analog processing is applied.

Analog FM Reception

When an analog FM signal is received, the sampled audio is sent from the demodulator to the UDB, where it is passed to a DSP for further processing. De-emphasis is applied using DSP techniques, and the sampled audio is output to both the LVDS Serial Data on the front panel RJ45 jack and to the motherboard through a D/A converter.

P25 Digital C4FM Reception

When a P25 Digital C4FM signal is received, the DSP performs bit recovery and the bit stream is sent to both the LVDS Serial Data on the front panel RJ45 jack and to the motherboard through a voice decoder and D/A converter.

Analog and P25 Digital Settings

The DSP controls the settings that are applied to the analog or P25 Digital signal, such as Continuous Tone-Controlled Squelch System (CTCSS), Digital Coded Squelch (DCS), de-emphasis, wide or narrow band selection, Network Access Code (NAC) and Talk-Group Identifier (TGID). These settings are configured for each individual channel using the Radio Service Software (RSS).

Audio Circuitry

Baseband analog audio is output through four separate outputs to the motherboard. An independent level adjustment is performed for the balanced audio output, the SM-3 speaker audio output, the discriminator audio output and discriminator low pass filtered output.

Carrier Operated Relay (COR) Circuitry

The MT-4E Receiver is able to distinguish between incoming analog signals and digital signals, and pulls the front panel ANALOG COR output low for an analog FM carrier, or it pulls the front panel DIGITAL COR output low for a P25 Digital C4FM carrier.

The front panel DIGITAL COR and ANALOG COR outputs are typically sent to the transmitter or universal interface card (UIC).

The receiver will also pull the COR output on the motherboard low when either an analog FM or P25 Digital C4FM carrier is detected.

When the receiver receives an analog or digital signal, the DSP will activate the front panel analog or digital LED.

Mute Circuitry

A Mute line from the motherboard will override all COR outputs, balanced audio output, SM-3 speaker audio output and the LVDS Serial Data output and keep them from activating.

The mute line is used to externally control squelching features of the receiver.

TRANSMITTER THEORY

The MT-4E Transmitter is constructed with the Transmitter Mainboard and the Amplifier module subassembly. The Synthesizer and the Universal Daughter Board (UDB) containing the Digital Signal Processor (DSP) are mounted on the transmitter mainboard. An optional Encryption Module may also be installed on the transmitter mainboard.

The MT-4E Transmitter can accept both a baseband audio signal from the motherboard or microphone, or a Low Voltage Differential Signaling (LVDS) Serial Data signal from the front panel RJ45 jack for processing and transmission as an FM modulated carrier or a C4FM P25 Digitally modulated carrier.

Power Supply

A nominal +13.8VDC and a regulated +9.5VDC enters the transmitter module from the motherboard.

Analog FM Transmission

When analog FM transmission is required, the MT-4E Transmitter accepts a baseband audio signal from the microphone or balanced audio input. A level adjustment is performed and an analog to digital (A/D) convertor samples the audio. The digitized signal is applied to the UDB and passed to a DSP for further processing.

The MT-4E Transmitter can also accept an LVDS serial data signal from the front panel RJ45 jack that is passed to the DSP for further processing.

Pre-emphasis and limiting functions are applied using DSP techniques, and the resulting signal is converted back to a baseband audio signal and applied to the voltage-controlled oscillator (VCO) and temperature compensated crystal oscillator (TCXO) on the synthesizer, producing an analog modulated carrier.

P25 Digital C4FM Transmission

When P25 Digital transmit mode is required, the MT-4E Transmitter sends the A/D sampled microphone or balanced audio through a Vocoder process in the DSP, which compresses the voice signal, and adds error protection information and other P25 Digital signaling information to produce a 9600 bps data stream.

Alternatively, the 9600 bps data stream can be obtained from the front panel RJ45 jack in the form of LVDS Serial Data.

Each pair of digital bits is used to generate one of four analog waveforms which is applied to the VCO and TCXO on the synthesizer, thus producing a C4FM P25 Digitally modulated carrier.

Push-To-Talk (PTT) Circuitry

For normal repeater operation, the transmitter may be keyed up in P25 Digital mode (if it is programmed for P25 Digital mode or mixed mode operation) by pulling the front panel DIGITAL PTT input low.

Similarly, it may be keyed up in analog mode (if it is programmed for analog mode or mixed mode operation) by pulling the front panel ANALOG PTT input low.

The front panel DIGITAL PTT and ANALOG PTT inputs are typically controlled by the receiver or Universal Interface Card (UIC).

Additionally, the transmitter may be keyed by:
- activating the microphone PTT switch
- moving the front panel power switch to the KEY TX position
- grounding one of the PTT inputs on the motherboard
- activating the relay

For each of these additional cases, if the transmitter is programmed for P25 Digital mode or for analog mode, it will be keyed in the appropriate mode. If it is programmed for mixed-mode operation, it will use the front panel analog / digital switch to select the mode of operation.

When the transmitter is keyed in analog or digital mode, the DSP will activate the front panel analog or digital LED and will also activate a PTT OUT signal line that can be used to signal externally when the transmitter is keyed (e.g., an Antenna Relay).

Analog and P25 Digital Settings

The DSP controls the settings that are applied to the analog or P25 Digital signal, such as CTCSS, DCS, pre-emphasis, wide or narrowband selection, NAC and TGID. These settings are configured in the RSS for each individual channel.

Synthesizer

The synthesizer modulates an RF frequency signal; with analog or P25 Digital signals. Modulation signals are input to the synthesizer from the DSP on the UDB via a D/A converter. The modulated RF signal is then output through a coaxial cable to the transmitter amplifier.

Communication inputs and outputs occur between the DSP and synthesizer for control functions. The synthesizer is operated in wide band, generating RF frequencies across the entire frequency band.

RF Power Amplifier

The Transmitter RF Power Amplifier provides the final stage of RF amplification and filtering for the transmitter. The amplifier has inputs for the RF signal from the synthesizer and for DC power. It has outputs for the amplified RF signal and for forward and reverse power alarms.

The amplifier output power is adjustable. See the Specification Data Sheets section for model information and power ranges.

This amplifier is mounted in a machined aluminum case that ensures mechanical integrity for the transmitter, provides a good ground and also acts as a heatsink.

Amplifier Alarms and Overload Protection

The amplifier is equipped with forward output power and voltage standing-wave ratio (VSWR) alarm circuitry. The forward output power alarm will activate when the forward power output drops below half of the output power setpoint. The VSWR alarm will activate at a 3:1 VSWR. The alarms are available on the motherboard.

The VSWR overload circuit protects the transmitter amplifier from excessive antenna VSWR by reducing the amplifier's gain (output power) when an overload condition occurs. The VSWR overload circuit is an extension of the VSWR alarm.

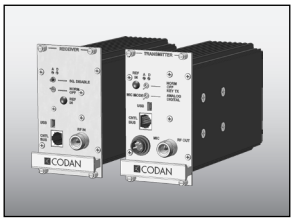
CHANNEL AND BANK SELECTION

Four channel select lines, CSEL0 – CSEL3, from the motherboard are brought into the receiver and transmitter modules by the 48-pin rear connector, allowing selection of 16 different channels. These signals are normally pulled low in the receiver and transmitter, but are typically set by jumpers on the motherboard to select Channel 1 by default. In addition, a bank select input is provided to switch between Bank A and B, each of which has 16 channels. The Bank select line normally floats high (+5V), selecting Bank A, but may be pulled low or high externally to select Bank B.

Channel and Bank Select Line Mapping

The table below shows the relationship between the states of the bank and channel select lines. Note that the channel select lines follow a binary pattern, but that the binary “0” represents Channel 1. The Bank A/B select line normally floats high (+5V), selecting Bank A, but may be pulled low or high externally via the Bank A/B select line.

BANK A/B	CSEL 3	CSEL 2	CSEL 1	CSEL 0	BANK SELECTED	CHANNEL SELECTED
HI	LO	LO	LO	LO	A	1
HI	LO	LO	LO	HI	A	2
HI	LO	LO	HI	LO	A	3
HI	LO	LO	HI	HI	A	4
HI	LO	HI	LO	LO	A	5
HI	LO	HI	LO	HI	A	6
HI	LO	HI	HI	LO	A	7
HI	LO	HI	HI	HI	A	8
HI	HI	LO	LO	LO	A	9
HI	HI	LO	LO	HI	A	10
HI	HI	LO	HI	LO	A	11
HI	HI	LO	HI	HI	A	12
HI	HI	HI	LO	LO	A	13
HI	HI	HI	LO	HI	A	14
HI	HI	HI	HI	LO	A	15
HI	HI	HI	HI	HI	A	16
LO	LO	LO	LO	LO	B	1
LO	LO	LO	LO	HI	B	2
LO	LO	LO	HI	LO	B	3
LO	LO	LO	HI	HI	B	4
LO	LO	HI	LO	LO	B	5
LO	LO	HI	LO	HI	B	6
LO	LO	HI	HI	LO	B	7
LO	LO	HI	HI	HI	B	8
LO	HI	LO	LO	LO	B	9
LO	HI	LO	LO	HI	B	10
LO	HI	LO	HI	LO	B	11
LO	HI	LO	HI	HI	B	12
LO	HI	HI	LO	LO	B	13
LO	HI	HI	LO	HI	B	14
LO	HI	HI	HI	LO	B	15
LO	HI	HI	HI	HI	B	16



RADIO SERVICE SOFTWARE PROGRAMMING

INTRODUCTION

This manual contains general information about Radio Service Software (RSS). Software revisions occur regularly. For current RSS versions and instructions, refer to up-to-date help files on the Codan website. Once RSS is downloaded with either the full or demo version, click **Help** to access the help files.

Receiver and transmitter programming is performed with the PC-based RSS. A USB Type A to 5-pin Mini-B cable (included with the software) is used to connect the USB port of an IBM-compatible computer to the USB port on the front panel of the receiver or transmitter module.

The RSS allows the receiver or transmitter to be programmed for:

- operating frequencies
- CTCSS and DCS signaling
- P25 Digital settings, such as NAC and TGID
- modulation type
- many other parameters

System Recommendations:

- Microsoft Windows 2000
- Microsoft Windows XP
- Windows 7

Recommended Minimum Specification:

- Pentium III Processor – 500MHz
- 256MB Memory (RAM)
- 1GB Free Disk Space

Installation

The RSS should install automatically once the CD is inserted. If not, run SETUP.EXE, located on the CD.

NOTE: The receiver and transmitter must be programmed separately.

Once the connections are made, the RSS may be run on the computer and the radio switched on. The first time a receiver or transmitter is connected, the USB drivers will need to be installed from the CD using the Hardware Update Wizard.

To test the connection:

1. Choose the appropriate module from the Select Device drop-down list (see Figure 1).

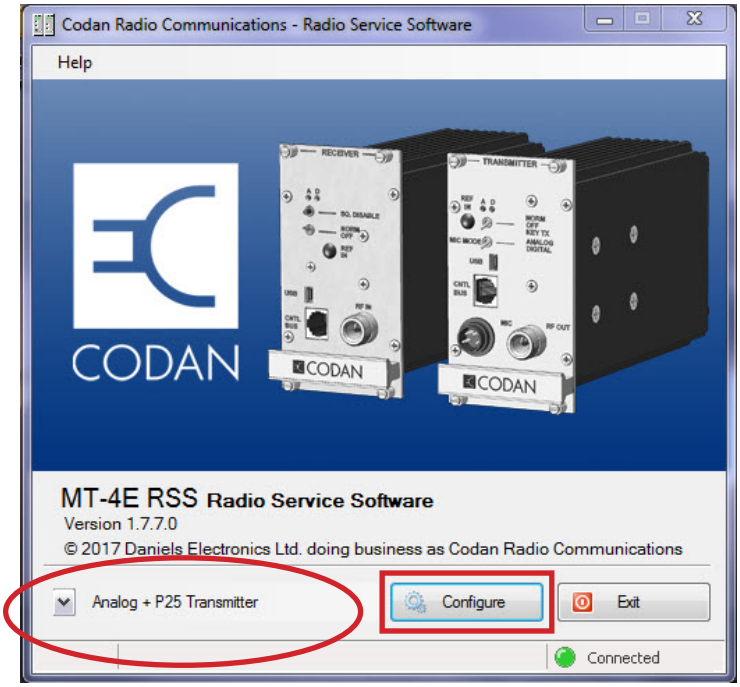
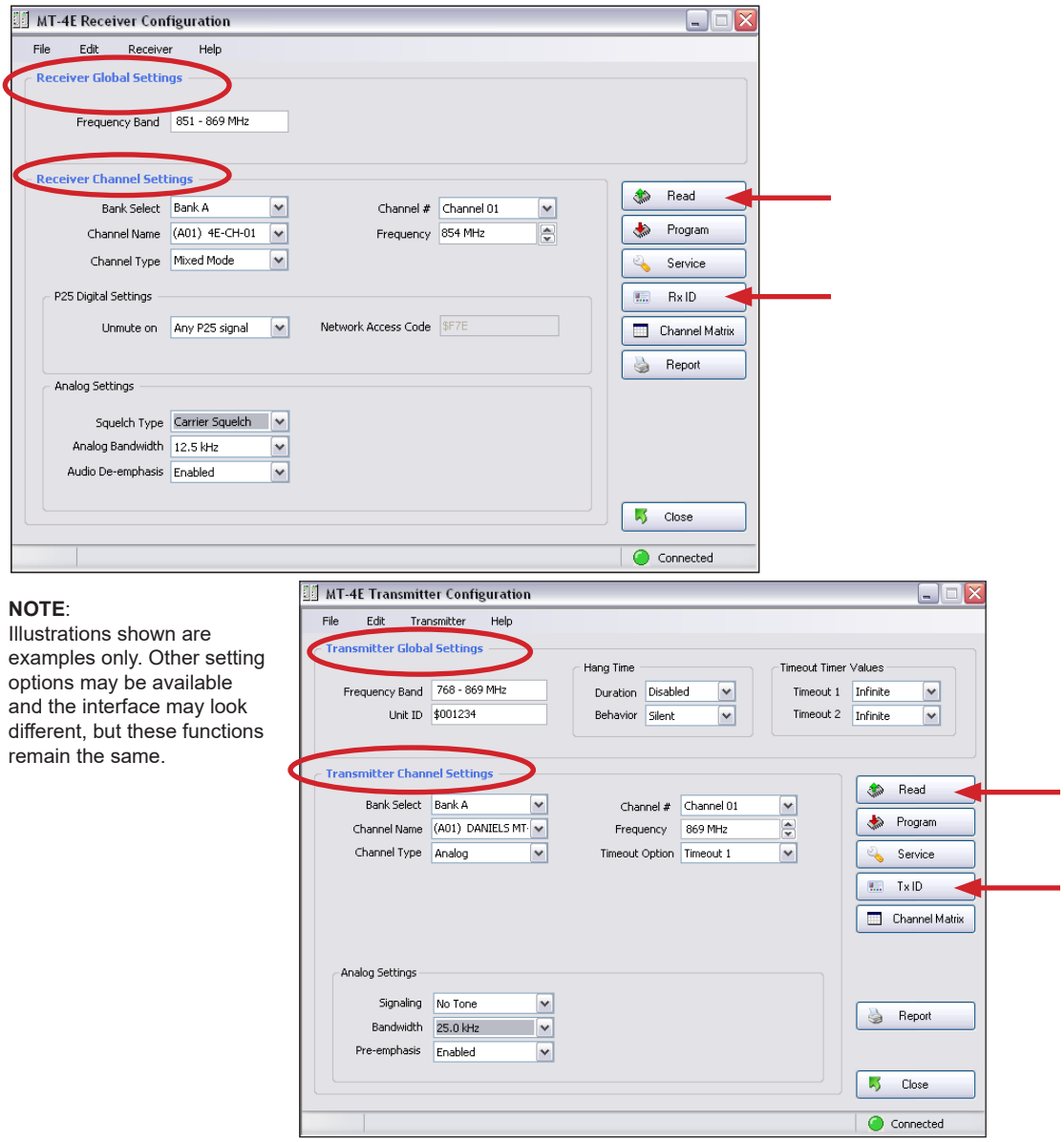


FIGURE 1: Radio Service Software Opening Screen

NOTE:
This illustration is an example only. The interface may look different, but the functionality remains the same.

2. Click **Configure** on the main screen to open the Receiver or Transmitter Configuration window (see Figure 2).
3. Click **Rx ID** (or **Tx ID)**.
4. Click **Read**.

The serial number, model number, firmware version and last programmed date should appear in the appropriate fields.



NOTE:
Illustrations shown are examples only. Other setting options may be available and the interface may look different, but these functions remain the same.

FIGURE 2: Receiver and Transmitter Configuration Windows

RSS Programming Mode

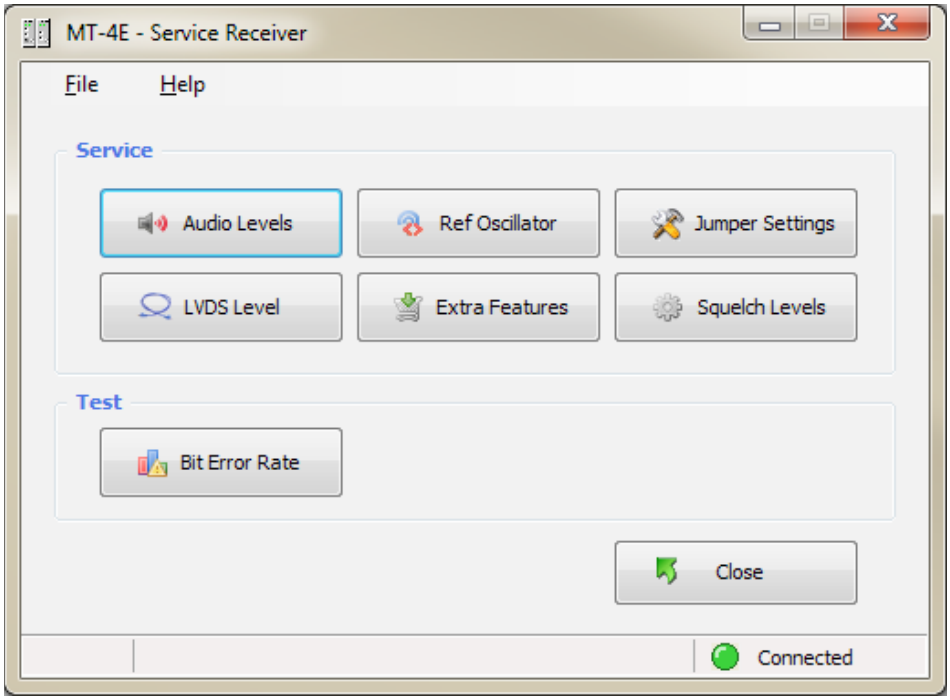
Programming options are divided into two main categories – Global Settings and Channel Settings (see Figure 2). When the receiver or transmitter menu is selected from the main screen, both the global and channel settings for the current channel are displayed.

The RSS may be used to save a receiver or transmitter configuration to disk. This function allows the user to save a standard configuration and use it as a starting configuration for all modules. An archive of configurations from each radio system in operation may also be kept so that replacement radios can be programmed easily.

NOTE: If the MT-4E Receiver or Transmitter is purchased without the P25 Digital firmware upgrade, the P25 Digital options will not be available in the RSS.

RSS Service Mode

RSS Service Mode allows tuning, testing and setup of the receiver and transmitter modules (see Figure 3). Analog and P25 Digital test modes, such as Bit Error Rate testing and Test Patterns generation can be selected by the RSS. Audio Levels, RF Power Output and Reference Oscillator adjustments can also be performed in RSS Service Mode.



NOTE: These Service images show units that are optioned for Analog **AND** Digital. Analog only units would show less options.

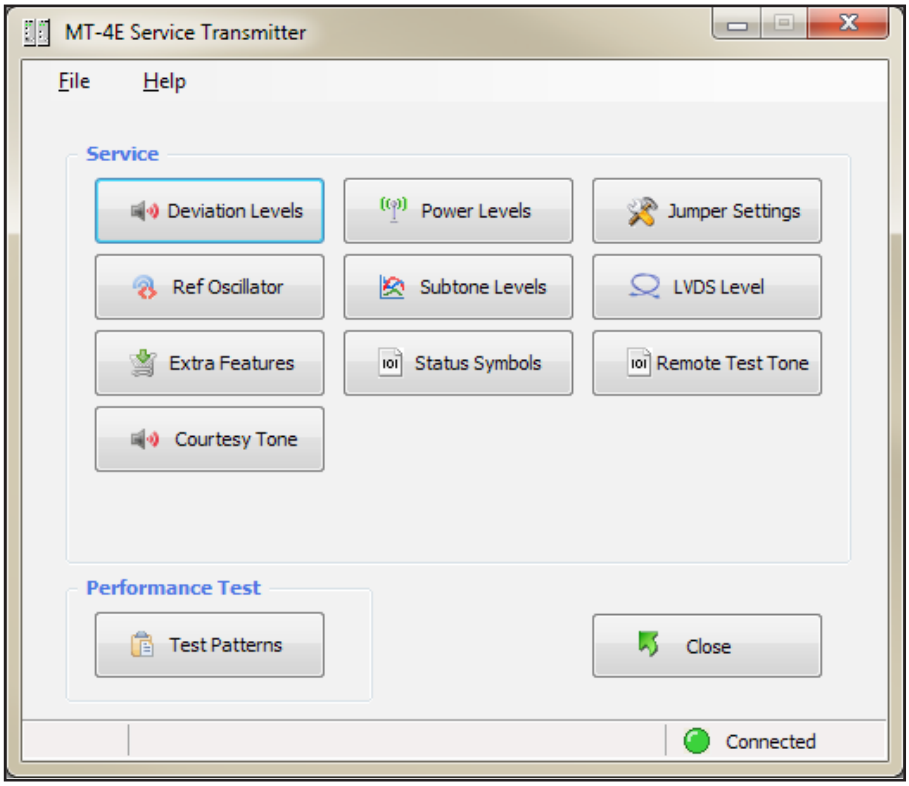
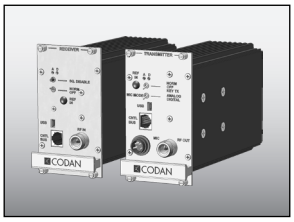


FIGURE 3: RSS Receiver and Transmitter Service Modes



RADIO SYSTEM CONFIGURATIONS

REPEATER

The MT-4E series receivers and transmitters may be configured in a repeater configuration. The standard way of setting up a repeater system is to connect the receiver to the transmitter through a Radio Interconnect Cable plugged into the RJ45 CNTL BUS connectors on the front panels of each radio module. This cable routes analog and digital COR / PTT signals and LVDS Serial Data from the receiver to the transmitter in both analog and P25 Digital mode. The Radio Interconnect Cable may be connected directly from the receiver to the transmitter, or may be routed through a Repeater Control Card.

A typical configuration is to set up the repeater to re-transmit an incoming signal in the same mode as it is received (i.e., normally, you would want a received analog signal to be repeated as analog and a received digital signal to be repeated as digital). This requires the receiver and transmitter modules to be programmed in Mixed Mode of operation through the RSS. The repeater may also be programmed to operate in analog or P25 Digital mode only.

Mixed Mode Repeat operation occurs as follows:

1. The receiver determines whether an incoming signal is analog or digital and asserts either the ANALOG COR or DIGITAL COR signal line on the front panel RJ45 CNTL BUS connector. LVDS Serial Data is sent to the RJ45 CNTL BUS connector as well.

2. The COR signal and LVDS Serial Data are transferred over the Radio Interconnect Cable directly to the transmitter or are routed through a Repeater Control Card.

3. The transmitter senses the incoming ANALOG PTT or DIGITAL PTT signal from the front panel RJ45 CNTL BUS connector and keys up in analog mode or P25 Digital mode. LVDS Serial Data is routed through the transmitter for transmission.

An alternate way of setting up a repeater system is to connect the receiver to the transmitter through an Audio Control Card or Base Controller. These control cards route analog audio and a single COR / PTT signal from the receiver to the transmitter. This configuration is recommended for analog repeater systems only.

The receiver and transmitter have no connection between them apart from the audio (Audio Control Card or Base Controller only), COR / PTT signals and LVDS Serial Data lines. The transmitter does not get any information from the receiver about which frequency or channel number to key up on. The receiver and transmitter channel is determined by the state of the channel select lines and bank select line.

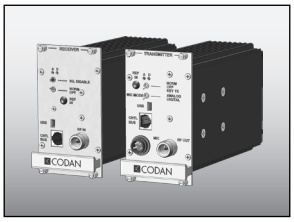
ANALOG FIXED (BASE) STATION INTERFACE

The MT-4E series receivers and transmitters may be configured in a fixed (or base) station configuration with an analog interface. An Audio Control Card or Base Controller is added to the system for E&M control. For tone remote control of the system, a Tone Remote Adapter is added to the system along with the base controller. The audio control card or the base controller will also allow for repeat capability of the system using analog audio as the repeat path.

The base controller is used for system control of encryption and decryption of the P25 Digital voice messages. Optional decryption or encryption modules are required in the receiver and transmitter modules.

DIGITAL FIXED (BASE) STATION INTERFACE

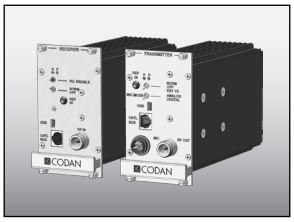
The MT-4E series receivers and transmitters may be configured in a fixed (or base) station configuration with a digital interface. A Universal Interface Card (UIC) is added to the system for Ethernet control.



VHF & UHF RECEIVERS AND TRANSMITTERS

The following section provides information that is primarily for VHF / UHF Receiver and Transmitter modules.

For specific information on 700 / 800 / 900MHz Receiver and Transmitter modules, see the 700 / 800 / 900MHz section in this manual.



VHF & UHF HARDWARE TUNING AND TROUBLESHOOTING

REPAIR NOTE

MT-4E Receivers and Transmitters employ a high percentage of surface mount components which should not be removed or replaced using an ordinary soldering iron. Removal and replacement of surface mount components should be performed only with specifically designed surface mount rework and repair stations complete with Electrostatic Discharge (ESD) protection.

When removing surface mount solder jumpers, it is recommended to use solder wick braid in place of vacuum-type de-soldering tools. This will help prevent damage to the circuit boards.

RECOMMENDED TEST EQUIPMENT

Alignment of the receiver and transmitter requires the following test equipment or its equivalent.

Power Supply – Regulated +9.5VDC at 2A	Phillips PM 2811
Power Supply – Regulated +13.8VDC at 2A	Topward TPS-4000
Oscilloscope / Multimeter	Fluke 97 Scopemeter
Current Meter	Fluke 75 Multimeter
Communications Service Monitor (Analog)	Marconi Instruments 2965A or equivalent
Communications Service Monitor (P25 Digital and Analog)	IFR 2975
Alignment Tools	Codan A-TK-04

It is recommended that the radio communications test set be frequency-locked to an external reference (WWVH, GPS, Loran C) so that the high-stability local oscillator may be accurately set.

Complete Receiver and Transmitter Alignment

A complete receiver and transmitter alignment is performed at the factory and should not be required under normal circumstances. A large change in receiver or transmitter operating frequency, as discussed in the next section, or a replacement of major receiver or transmitter subassembly modules, may require a complete realignment operation.

RECEIVER ASSEMBLY

The UDB, synthesizer and optional encryption module are mounted on the receiver mainboard. The RF Preselector is attached with two front panel screws and one screw through the rear Type-F connector. An extruded aluminum shell that slides over the receiver assembly forms an enclosure. The enclosure is completed by the installation of side and front panel screws (see Figure 5: VHF / UHF Receiver Exploded View Diagram).

Receiver hardware alignment is performed solely on the RF Preselector. Alignment is simplified by using a SR-3 Subrack, SM–3 System Regulator, and extender card or cable to provide receiver power and signal interconnection.

RECEIVER FREQUENCY CHANGE

The receiver is initially aligned at the factory for the frequency stamped on the “Factory Set Operating Frequency” label (see Figure 4). This label should list the frequency at which the last complete receiver alignment was performed.

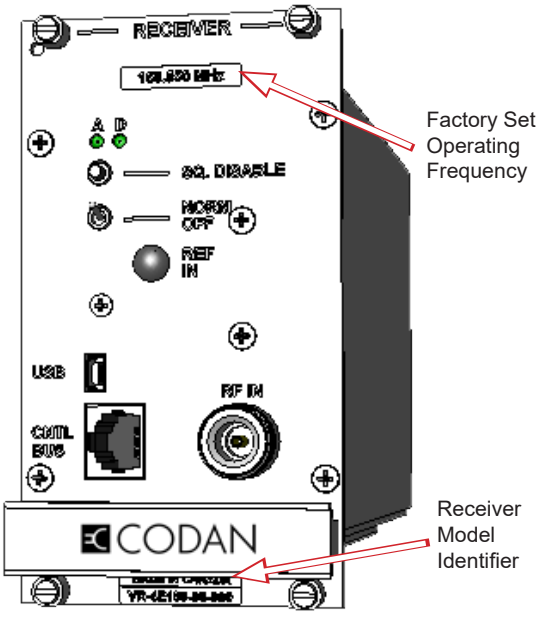


FIGURE 4: MT-4E Receiver Labels

A small frequency change of less than ± 2.0MHz from the frequency stamped on the frequency label will generally not require any adjustment.

Changes greater than ± 2.0MHz from a previously tuned working receiver frequency will require RF Preselector alignment using the Receiver RF Preselector Tuning Procedure outlined after Figure 6. Failure to perform a realignment after a large frequency change could result in unreliable receiver operation or operation that does not conform to the published specifications.

Change the frequency using the RSS. Verify that the receiver sensitivity is approximately the rated sensitivity for 12dB SINAD (see VHF and UHF Receiver Performance Specifications). A slight adjustment of the RF Preselector tuning capacitors may be performed to maximize sensitivity at the new frequency (see Figure 6: VHF / UHF Receiver Top Component View [1]).

NOTE: The preferred RF Preselector tuning procedure requires swept frequency response measurement of the RF Preselector filter in order to provide a maximally flat response over a range of input frequencies, with the primary channel frequency centered in the filter passband. This alignment approach is not mandatory for single-channel operation.

Alignment for the RF Preselector consists of tuning the five-section preselector filters only. No tuning is required for frequency changes within the 3dB bandwidth of its originally tuned frequency (5MHz for VHF and 7MHz for UHF) although, without tuning at the 3dB band edges, the receiver’s sensitivity will be degraded.

For frequency changes outside of the RF Preselector’s 3dB bandwidth, use the Receiver RF Preselector Tuning Procedure outlined after Figure 6.

VHF / UHF RECEIVER EXPLODED VIEW DIAGRAM

To Remove the Receiver Case:

1. Remove the four Front Panel screws (A - 5812-3M0VP08S).
2. Remove the two Side Case screws (B - 5812-3M0VP08S).

To Remove the RF Preselector Assembly:

1. Remove the two Front Panel screws (C - 5812-3M0VP06S).
2. Remove the RF Preselector screw (D - 5812-2M5PP14S).
3. Remove the SMB connector from the Receiver Mainboard IF IN.
4. Remove the SMB connector from the Synthesizer LO OUT.
5. De-solder the +9.5 VDC Power Supply wire from the Receiver Mainboard.

To Remove the Synthesizer:

Remove the three Synthesizer screws (E - 5812-2M5PP06S).

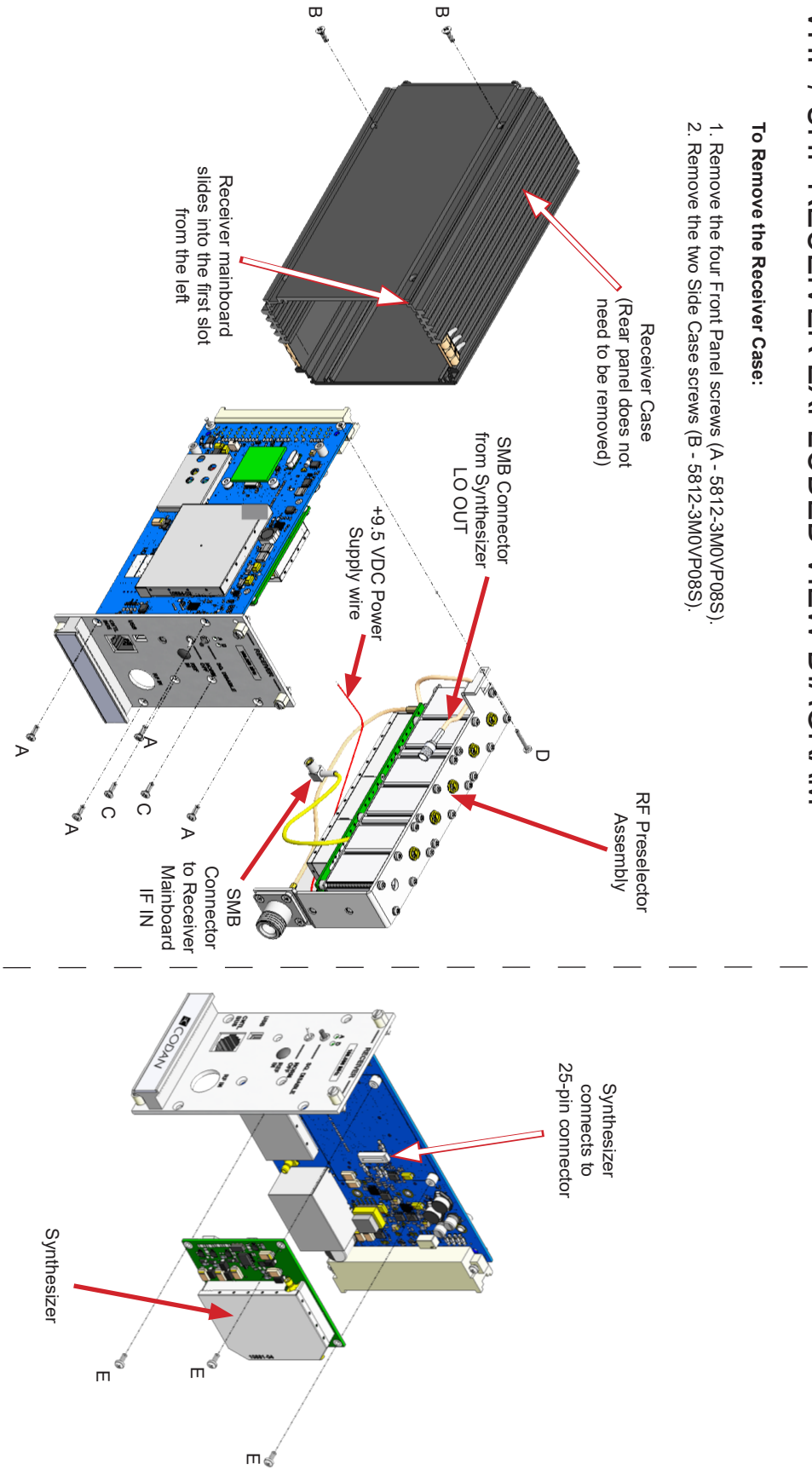


FIGURE 5: VHF / UHF Receiver Exploded View Diagram

VHF / UHF RECEIVER TOP COMPONENT VIEW (1)

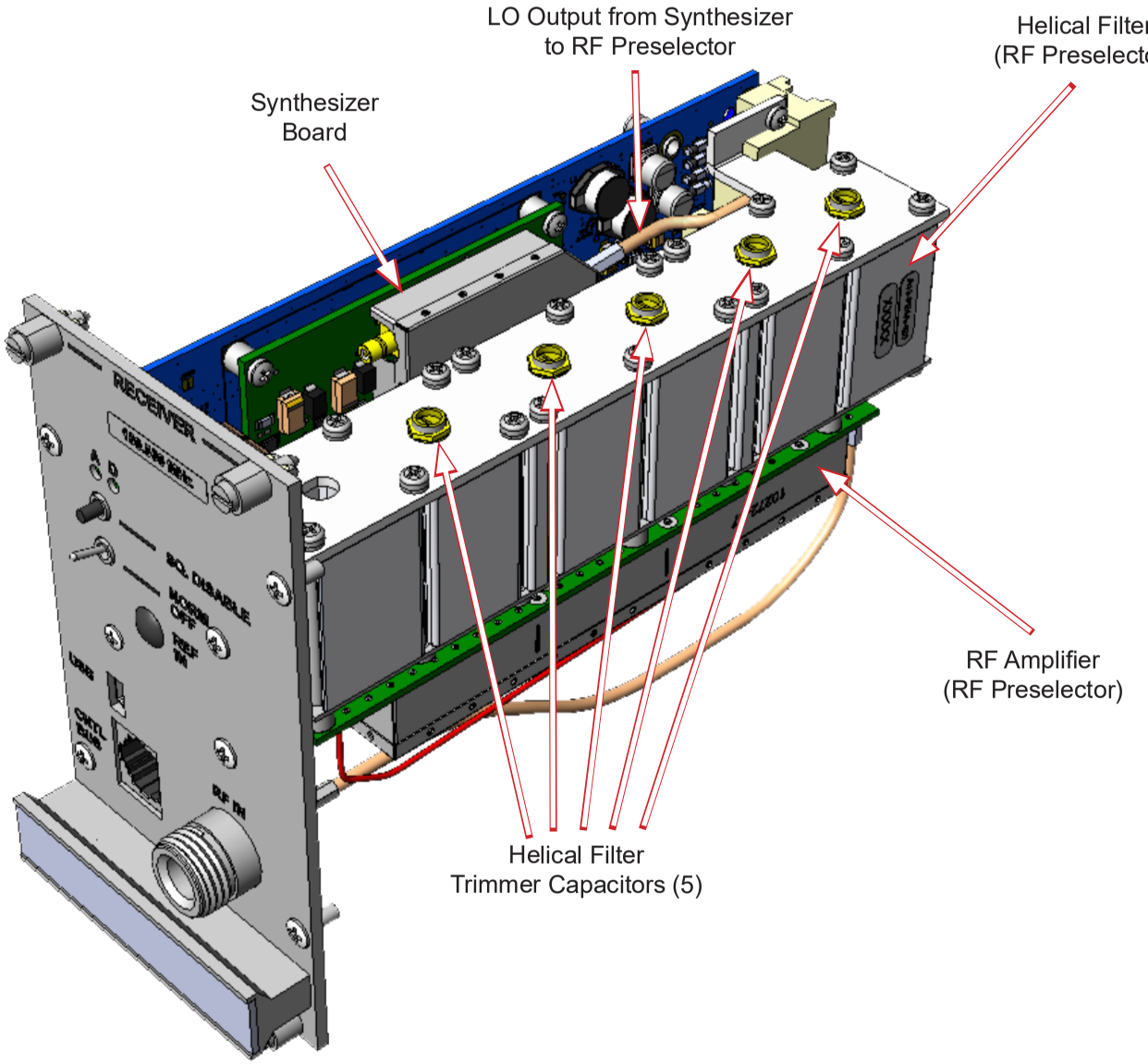


FIGURE 6: VHF / UHF Receiver Top Component View (1)

RECEIVER RF PRESELECTOR TUNING PROCEDURE

The best way to tune the RF Preselector is to use a Spectrum Analyzer with a tracking generator. The frequency response of the preselector filter can be seen at the IF output.

Use the following procedure for swept frequency tuning:

1. Check that the +9.5VDC red wire is connected to the RF Preselector from the receiver mainboard.

2. Connect the Tracking Generator output to the RF Preselector's RF input (the RF IN, Type-N connector on the receiver front panel).

   The tracking generator should have an output level of -20dBm and a frequency span of 50MHz. If the preselector filter is grossly misaligned, a larger frequency span may be required.

3. Disconnect the RF Preselector's IF output from the receiver mainboard and connect it to the Tracking Generator input.

   The IF output is an SMB connector and will likely require an adapter to connect to the Tracking Generator. The Tracking Generator frequency will be the receiver's RF frequency.

4. Adjust the helical filter trimmer capacitors for a flat response centered at the desired RF frequency.

   The measured input level should be greater than -60dBm for Class A and -65dBm for Class B. The 3dB bandwidth should be ranged approximately 5MHz for VHF and 7MHz for UHF respectively.

5. Once the RF Preselector is tuned, reconnect the RF Preselector's IF output back to the receiver mainboard.

RECEIVER ALTERNATE TUNING PROCEDURE

The alternate method of tuning the RF Preselector is to monitor receiver SINAD. This tuning procedure is typically used when the receiver only has a single RF frequency.

Use the following procedure for SINAD frequency tuning:

1. Check that the +9.5VDC red wire is connected to the RF Preselector from the receiver mainboard.

2. Inject the desired RF frequency into the RF IN connector on the receiver front panel at a level of the rated sensitivity (see VHF and UHF Receiver Performance Specifications).

3. Ensure the synthesizer is connected to the LO input of the RF Preselector.

4. Adjust the helical filter trimmer capacitors for best receiver SINAD (better than the rated sensitivity – see VHF and UHF Receiver Performance Specifications).

TRANSMITTER ASSEMBLY

The UDB, synthesizer and optional encryption module are mounted to the transmitter mainboard. The RF Amplifier is mounted on an aluminum heatsink with the mainboard mounted on top (see Figure 8: VHF / UHF Transmitter Exploded View Diagram, Figure 9: VHF / UHF Transmitter Top Component View and Figure 10: VHF / UHF Transmitter Bottom Component View). An extruded aluminum shell slides over this assembly to complete the case.

This shell also serves as a heatsink to remove heat from the amplifier module. It is important that the four screws that bond the shell to the amplifier module be installed before prolonged operation of the transmitter. Moreover, the surface of the amplifier module that contacts the shell should be clean and free of foreign material.

Transmitter hardware alignment is performed solely on the amplifier. Alignment is simplified by using a SR-3 Subrack, SM-3 System Regulator, and extender card or cable to provide transmitter power and signal interconnection.

TRANSMITTER FREQUENCY CHANGE

The transmitter is initially aligned at the factory for the frequency stamped on the “Factory Set Operating Frequency” label (see Figure 7). This label should list the frequency at which the last complete transmitter alignment was performed.

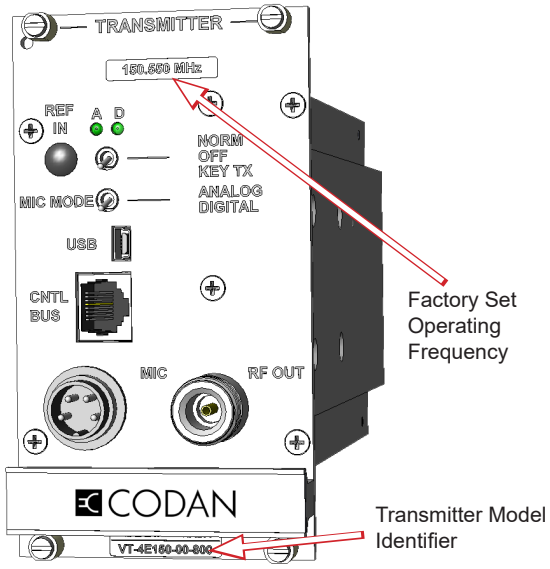


FIGURE 7: MT-4E Transmitter Labels

A small frequency change of less than ± 0.5MHz from the frequency stamped on the frequency label will generally not require any adjustment.

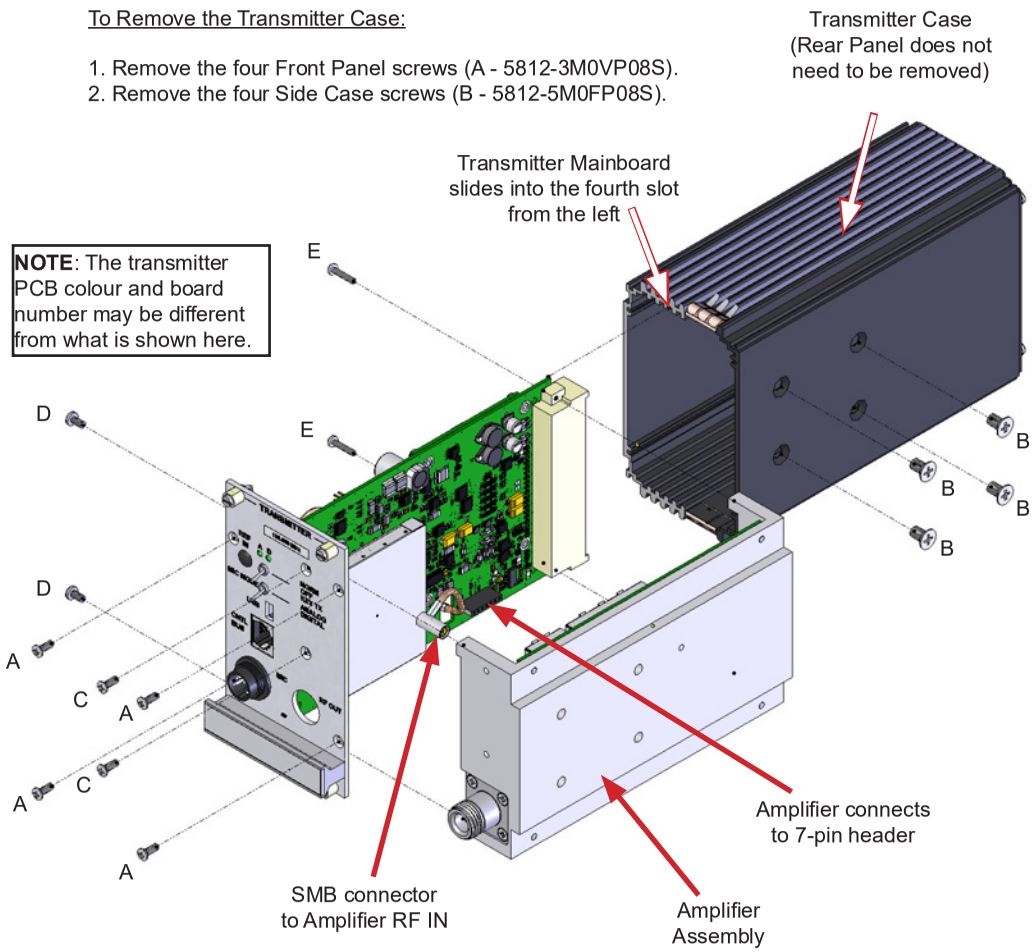
Changes greater than ± 0.5MHz from a previously tuned working transmit frequency will require VSWR alarm / overload alignment using the Transmitter Amplifier Tuning Procedure outlined after Figure 10. Failure to perform a realignment after a large frequency change could result in unreliable transmitter VSWR alarm / overload operation.

When the VSWR alarm / overload circuit is properly set, the amplifier is protected from excessive antenna VSWR by reducing the amplifier’s gain when an overload condition occurs. If the VSWR alarm is not set, the amplifier is still protected by a current limiter circuit. The circuit limits amplifier current draw to 2.0A and protects the amplifier from damage.

VHF / UHF TRANSMITTER EXPLODED VIEW DIAGRAM

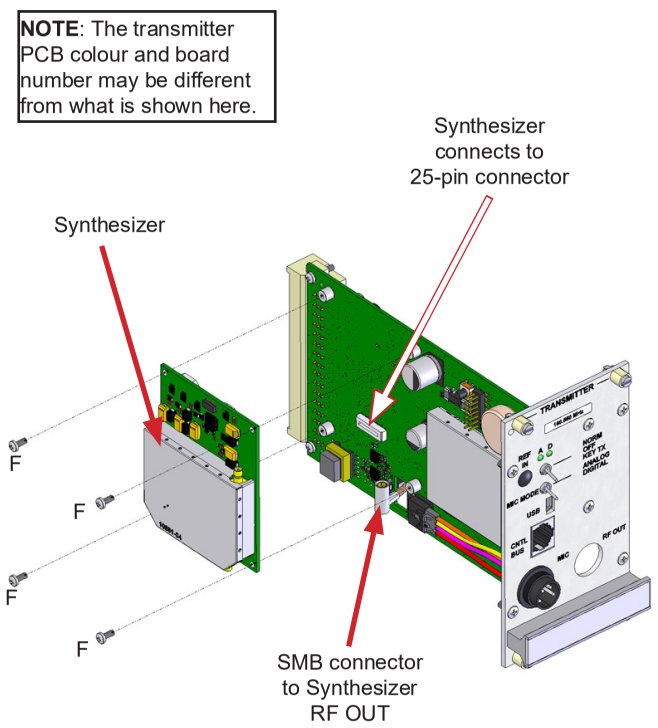
<u>To Remove the Transmitter Case:</u>

1. Remove the four Front Panel screws (A - 5812-3M0VP08S).
2. Remove the four Side Case screws (B - 5812-5M0FP08S).



<u>To Remove the Amplifier Assembly:</u>

1. Remove the two Front Panel screws (C - 5812-3M0PP06S).
2. Remove the two front Amplifier Assembly screws (D - 5812-3M0PP06S).
3. Remove the two rear Amplifier Assembly screws (E - 5812-2M5PP12S).
4. Remove the SMB connector from the Amplifier RF IN.



<u>To Remove the Synthesizer:</u>

1. Remove the four Synthesizer screws (F - 5812-2M5PP06S).
2. Remove the SMB connector from the Synthesizer RF OUT.

FIGURE 8: VHF / UHF Transmitter Exploded View Diagram

VHF / UHF TRANSMITTER TOP COMPONENT VIEW

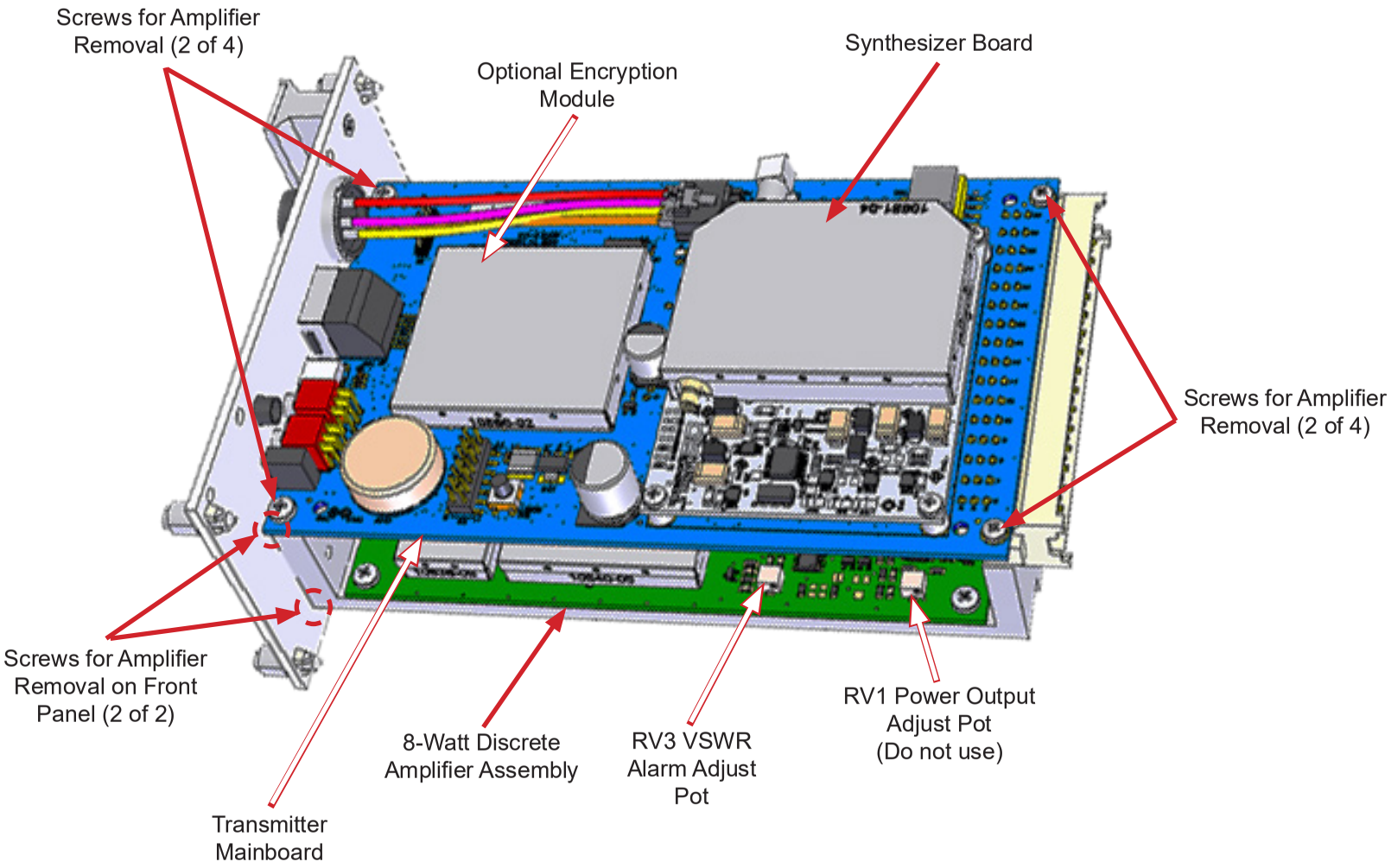


FIGURE 9: VHF / UHF Transmitter Top Component View

VHF / UHF TRANSMITTER BOTTOM COMPONENT VIEW

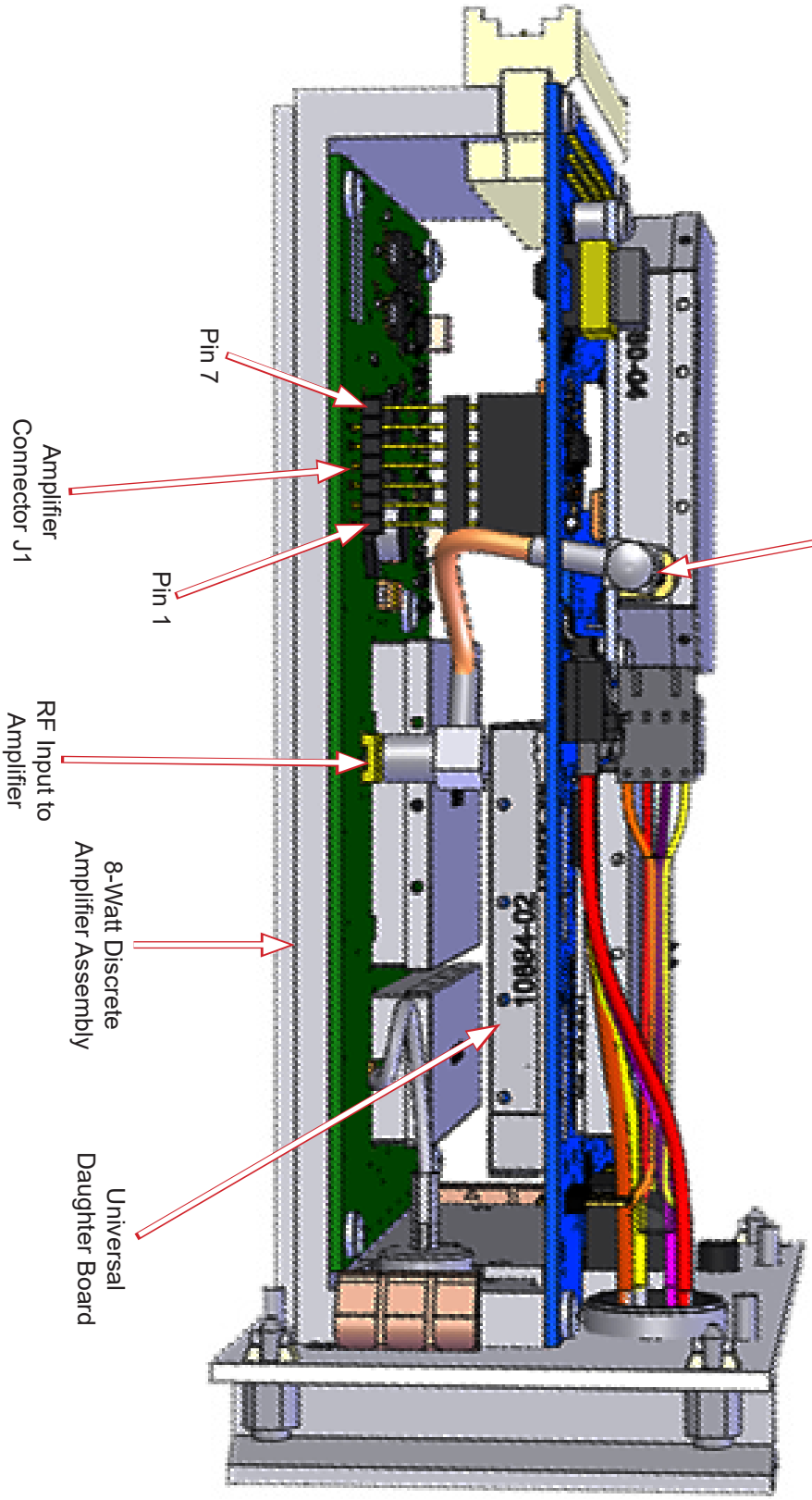


FIGURE 10: VHF / UHF Transmitter Bottom Component View

TRANSMITTER AMPLIFIER TUNING PROCEDURE

The transmitter amplifier requires 0dBm of input power and is continuously adjustable over its power range (see the VHF and UHF Transmitter Performance Specifications data sheet). The amplifier provides Output Power and Antenna VSWR Alarm outputs (open drain; active low) through the transmitter to the motherboard. The amplifier's output power level and alarm level can be set without detaching the amplifier from the transmitter board.

Transmitter Amplifier Adjustment

The RF output and VSWR alarm threshold level adjustments are easily accessible so that fine adjustments can be made in the field. The forward power alarm does not require alignment. Depending on user requirements, the VSWR alarm threshold level should be checked whenever a significant change in operating frequency (± 0.5MHz) is made. As the antenna VSWR alarm is dependent on the output power, the output power should always be set first. The order of adjustment should be:

1. Set the output power.
2. Set the desired transmitter VSWR alarm level.
3. Test the transmitter overload.
4. Verify transmitter current draw.

Initial Set-Up

Step	Reference	Action	Desired Results	Notes
1.	Transmitter RF output connector	Connect to radio communications test set.		Use short section of low-loss 50Ω coaxial cable.
2.	Potentiometers RV1 and RV3	Turn fully counter clockwise (CCW).		Initial starting point.
3.	MT-4E RSS	Click **Key Tx**. Adjust slider.	Desired Transmitter RF Output Power	Transmitter RSS Service > Power Levels

Transmitter VSWR Alarm (Reverse Power)

Step	Reference	Action	Desired Results	Notes
1.	Transmitter	Disconnect the test set and terminate the transmitter with a 3:1 mismatch load.		
2.	Connector J1	Connect a voltmeter to Pin 5 of Connector J1.		
3.	RV3	Adjust clockwise (CW).	Until Pin 5 reads +2.5VDC, or a noticeable drop (10% of total current or more) in current on the +13.8VDC line occurs.	
4.	Transmitter RF Output Connector	Connect to radio communications test set.	Pin 5 should read approximately 0VDC.	Use short section of low-loss 50Ω coaxial cable.

Transmitter Overload Test

Step	Reference	Action	Desired Results	Notes
1.	Amplifier	Disconnect radio communications test set.	Amplifier terminated with an open circuit.	
2.	+13.8VDC Supply	Monitor current.	Noticeable drop (10% of total current or more) in +13.8VDC current.	
3.	Amplifier	Reconnect radio communications test set.	+13.8VDC current should return to previous level.	

Transmitter Current Draw Verification

Step	Reference	Action	Desired Results	Notes
1.	+13.8VDC supply		Current < 2.0A	When transmitting.
2.	+9.5VDC supply		Current < 0.2A	When transmitting.

RECEIVER JUMPERS

VHF / UHF Receiver Mainboard

PCB 60021

Jumper	Default Position	Function / Description
JU1	OUT	Enable Clear Keys 1
JU2	OUT	Enable Clear Keys 2
JU3	IN	Enable Power to Discriminator Circuit
JU4	IN	Enable Pullup Resistor on RX_Mode
JU5	OUT	Bypass 11uF Audio Cap
JU6	IN	Enable Pullup Resistor on RX_Sec/Clr
JU7	OUT	Enable Power to Discriminator LPF Circuit
JU8	OUT	Enable Pullup Resistor on COR
JU9	X	Audio Transformer (X=Active, Y=Bypass)
JU10	X	Audio Transformer (X=Active, Y=Bypass)
JU11	IN	Enable Power to Balanced Audio Amplifier
JU12	OUT	Increase Gain Bal Aud Gain by 2.7dB
JU13	OUT	Bypass 600 Ohm on Aud Amp Output
JU14	X	Audio Transformer (X=Active, Y=Bypass)
JU15	OUT	Jumper for D&D Testing (enable DC pwr on RF Line)
JU16	IN	Jumper for D&D Testing (isolate xtal filter stage from cct)
JU17	IN	Jumper for D&D Testing (disconnect SSI Data Line)
JU18	IN	Jumper for D&D Testing (disconnect SSI Data Line)
JU19	IN	Jumper for D&D Testing (disconnect SSI Data Line)
JU20	IN	Jumper for D&D Testing (disconnect SPI Data Line)
JU21	IN	Jumper for D&D Testing (disconnect SPI Data Line)
JU22	IN	Jumper for D&D Testing (disconnect SPI Data Line)
JU23	IN	Jumper for D&D Testing (disconnect SPI Data Line)
JU33	IN	Jumper for D&D Testing (Remove pwr from TCXO)
JU34	X	Jumper for D&D Testing (X=TCXO, Y=12MHz Ref)
JU35	OUT	Jumper for D&D Testing (for install of reset push button)

Jumpers are shown on the VHF / UHF Receiver Jumper and Test Point Locator Illustrations (see Figure 11 and Figure 12).

VHF / UHF RX TOP JUMPER AND TEST POINT LOCATOR

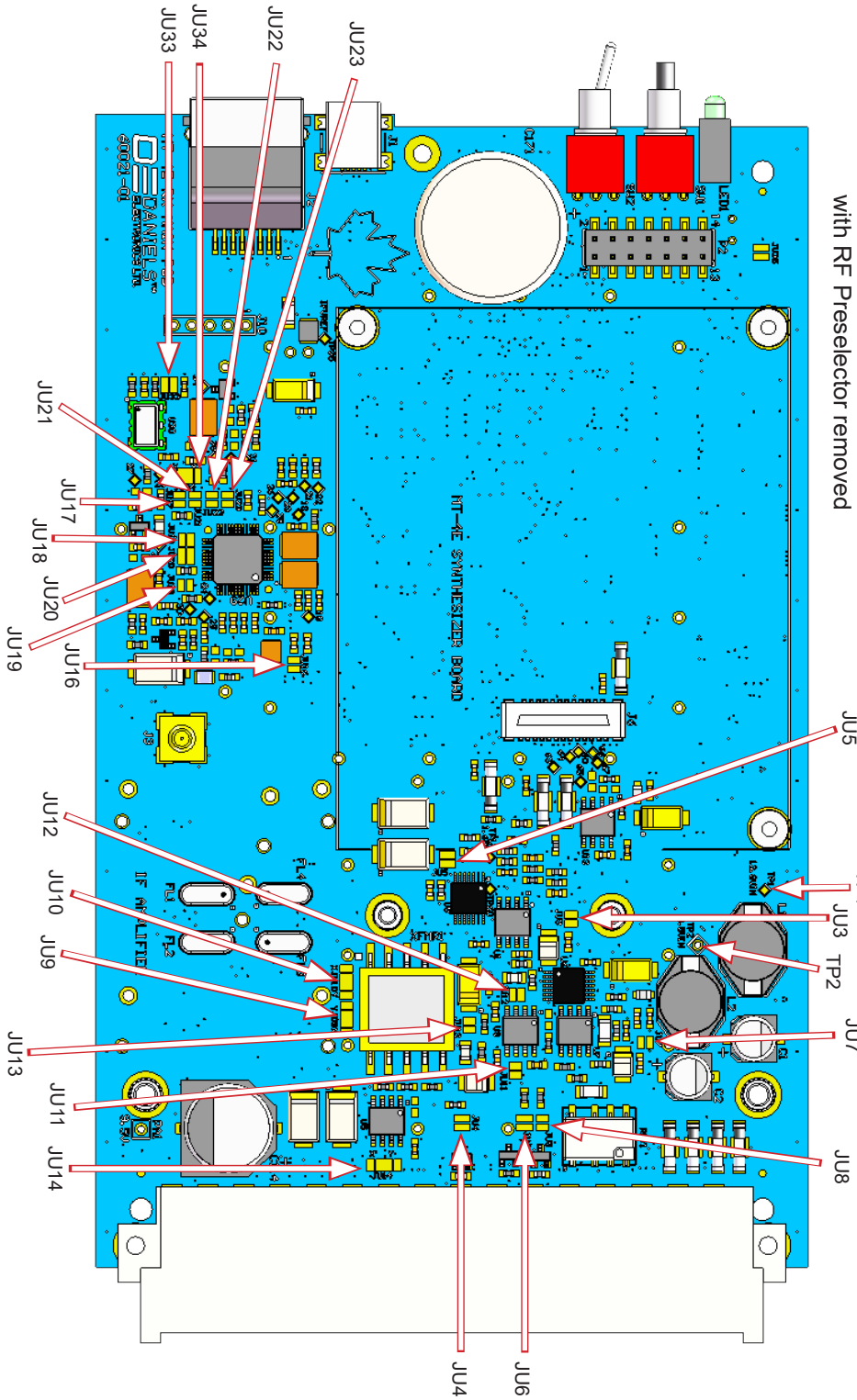


FIGURE 11: VHF / UHF RX Top Jumper and Test Point Locator

VHF / UHF RX BOTTOM JUMPER AND TEST POINT LOCATOR

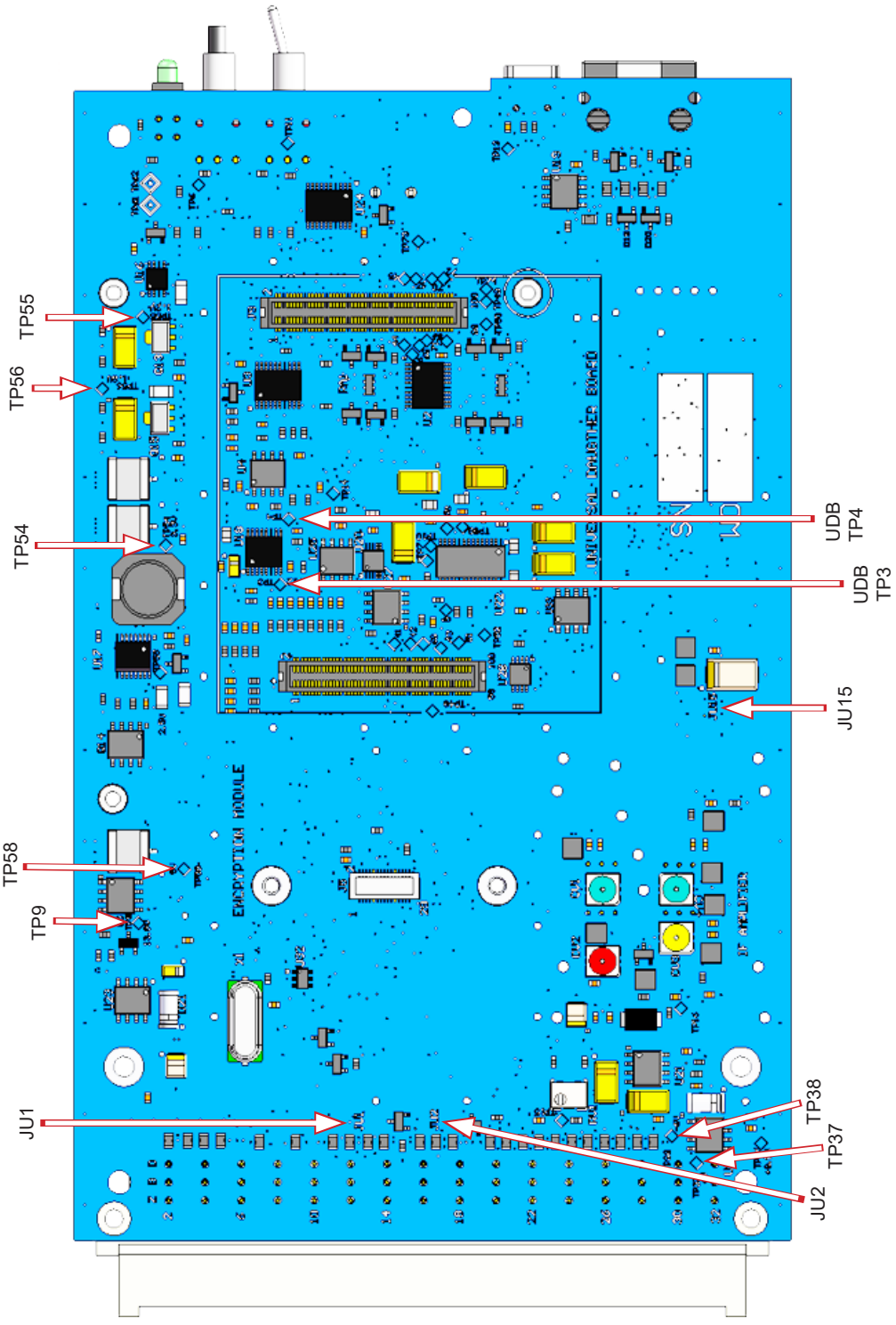


FIGURE 12: VHF / UHF RX Bottom Jumper and Test Point Locator

TRANSMITTER JUMPERS

VHF / UHF Transmitter Mainboard

PCB 50156 and PCB 60012

Jumper	Default Position	Function / Description
JU1	X	X = A/D Mode front Panel Switch Select / Y = A/D Mode External Input Select
JU2	Y	X = Mic Output Audio Select / Y = Mic Input Audio Select
JU3	X	X = 600 Ohm Audio transformer Select / Y = Transformer Bypass
JU4	IN	600 Ohm resistor Input enable
JU5*	OUT	When installed enables Clear Keys 1 Input
JU6*	OUT	When Installed enables Clear Keys 2 Input
JU7	X	X = 600 Ohm Audio transformer Select / Y = Transformer Bypass
JU8	X	X = 600 Ohm Audio transformer Select / Y = Transformer Bypass
JU9	OUT	When Installed Bypass C coupling on Subtone / Direct Mod Input
JU10	X	X = ENCRYPT-MT4E-XX / Y = ENCRYPT-MT4E-XX1 (Select which UCM Installed)
JU11	OUT	Gain control for TCXO / MOD input to synthesizer; Installed for P25CC simulcast
JU12	X	X = Direct MOD input / Y = Spare; Neither X nor Y installed for P25CC simulcast
JU13	X	X = Direct MOD input / Y = DSP Bypass
JU14	OUT	When installed disables AC coupling for direct MOD input
JU15**	X	X=Enable SM3_SPKR_MIC / Y=Enable +13.8V for DTMF_MIC option
JU16**	OUT	Installed for P25CC simulcast
JU17**	OUT	Installed for P25CC simulcast

*Installed only if encryption is enabled.
**Added to TX Mainboard for Version 60012.

Jumpers are shown on the VHF / UHF Transmitter Jumper and Test Point Locator illustrations (see Figure 13 and Figure 14).

FRONT PANEL RJ45 JACK PINOUTS

A single, 8-position RJ45 jack is mounted on the front panel of the receiver and transmitter. The following are the connections on the RJ45 jack.

Pin	Signal
1	CRYPTO MODULE KEYLOAD
2	UNUSED SPARE
3	ANALOG COR / PTT
4	LVDS DATA A
5	LVDS DATA B
6	DIGITAL COR / PTT
7	NO CONNECTION
8	NO CONNECTION
Shield	GROUND

VHF / UHF TX TOP JUMPER AND TEST POINT LOCATOR

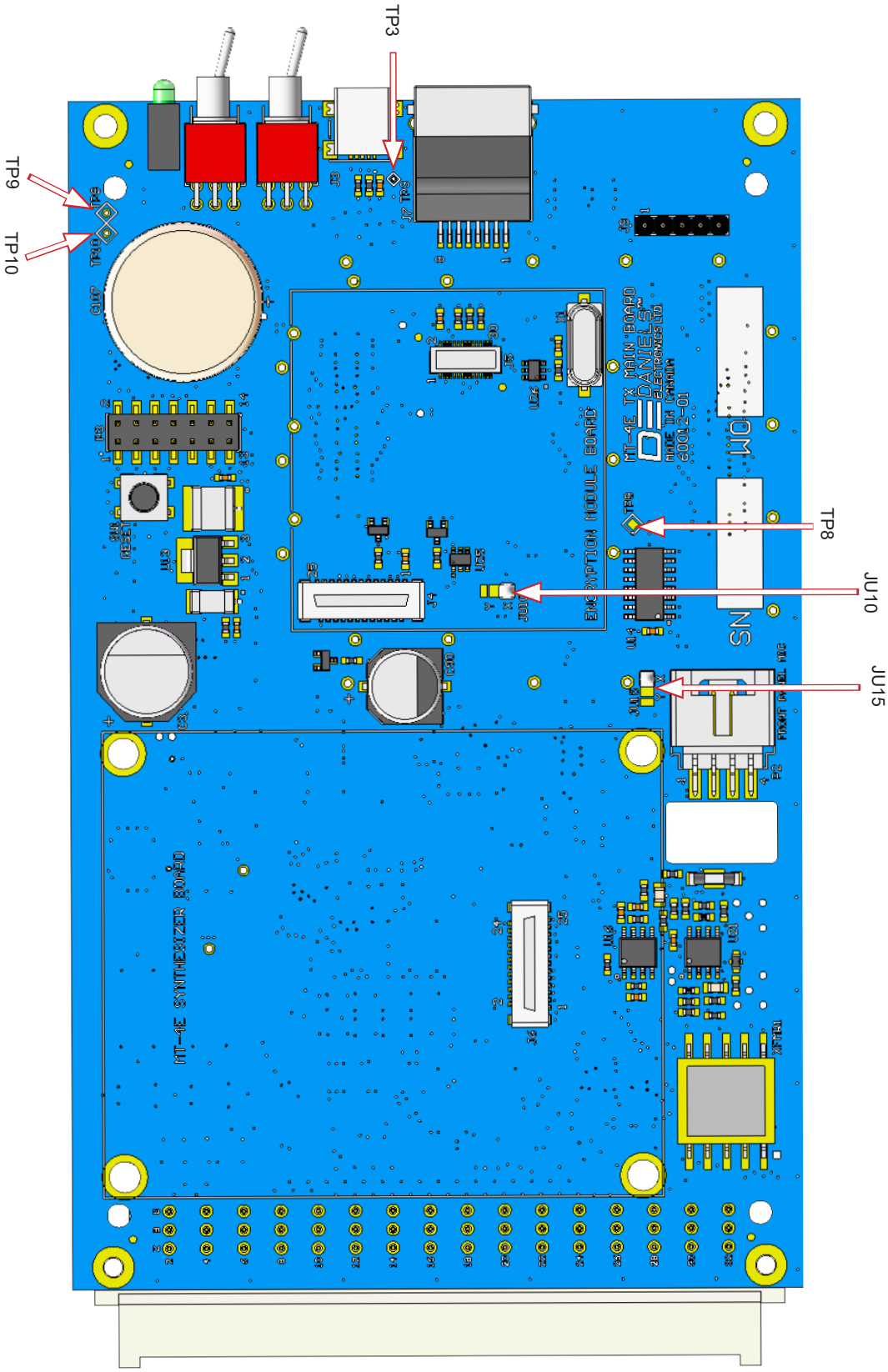


FIGURE 13: VHF / UHF TX Top Jumper and Test Point Locator

VHF / UHF TX BOTTOM JUMPER AND TEST POINT LOCATOR

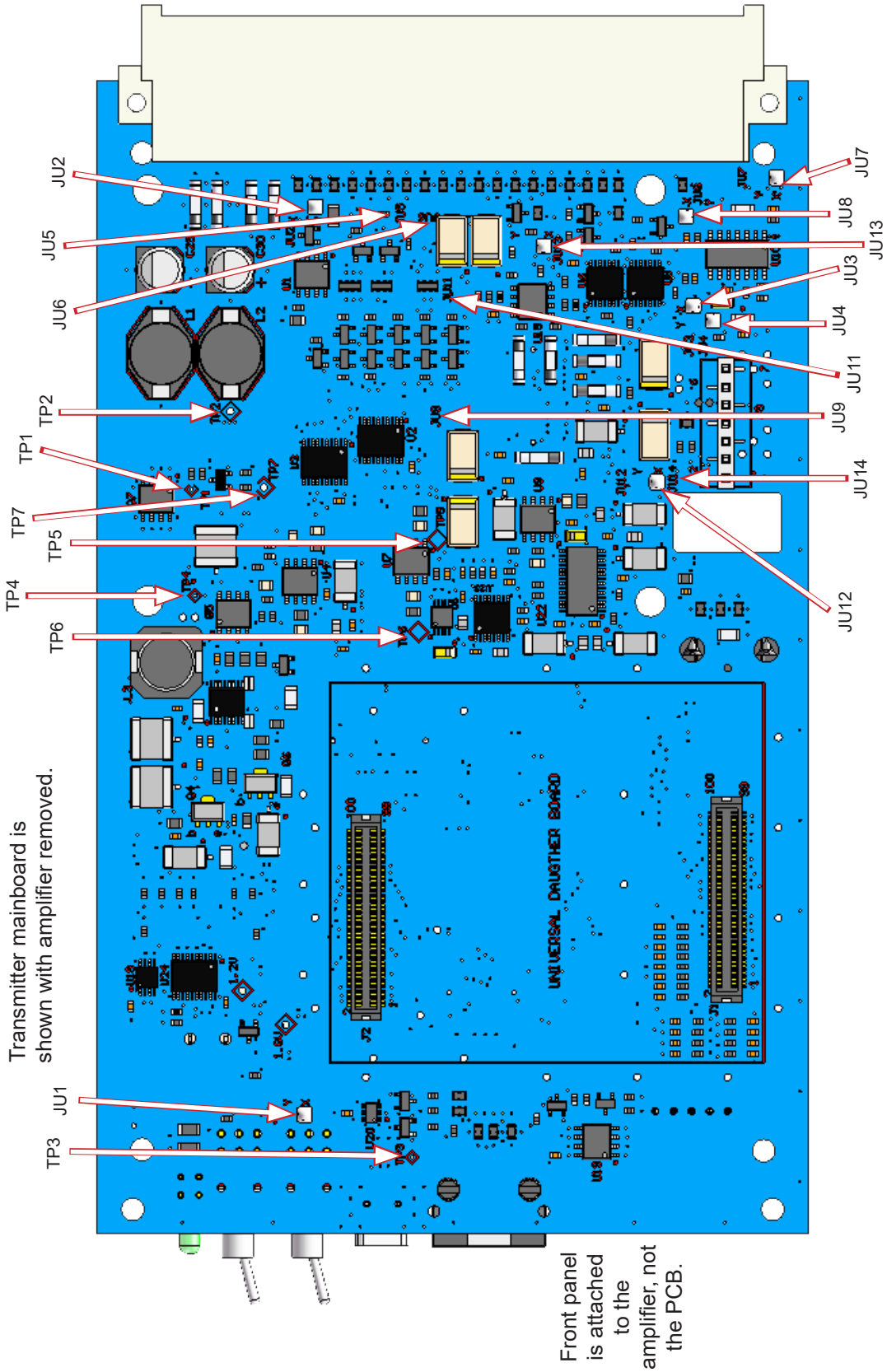


FIGURE 14: VHF / UHF TX Bottom Jumper and Test Point Locator

RECEIVER TROUBLESHOOTING

The receiver is composed of four main components; the receiver mainboard, the synthesizer module, the RF Preselector and the universal daughter board (UDB). An optional decryption module can be installed in the receiver. See Figure 15: VHF / UHF Receiver Bottom Component View.

The following steps will help you determine the most likely subassembly that contains a fault and may help you determine why the fault is occurring.

NOTE: The subassembly or the entire receiver can be sent back to Codan Service department for repair. Contact the Codan Service department before returning any product.

Before testing, the receiver should be in the subrack or connected to the subrack by means of an Extender Card or kit. The main power to the system (+13.8VDC) and the +9.5VDC regulated supply should be checked via the front panel of the system regulator.

Testing the Receiver

1. Flip the receiver front panel ON/OFF switch from the OFF position to the NORM position.
2. Check the front panel LED lights.
   Typically the front panel LED lights will turn ON for less than 10 seconds and then turn OFF. On occasion these lights may stay on for up to one minute.

   If this is not the case, the most likely fault is on the UDB, on the synthesizer, or on the receiver mainboard. Make sure that the UDB and the synthesizer module are properly seated on the receiver mainboard.
3. Connect the receiver to the PC with the USB cable and run the Radio Service Software (RSS).
4. Select [P25 or Analog] Receiver.
5. Click **Configure**.
6. Click **Read** to read the receiver configuration.

   If the RSS returns "Device Not Responding," physically inspect the USB cable and connection, check the USB driver and check that the USB is recognized by the Device Manager. If the RSS still does not recognize the connection, the most likely fault is on the UDB.
7. Click **Rx ID**.
8. Click **Read** to read the Receiver ID.

   Confirm that the Frequency Band in the Synthesizer ID Model Number matches the Frequency Band in the Receiver Global Settings. If the settings do not match, the synthesizer will require replacement or the programming will need to be changed to match the synthesizer hardware.

If the receiver is still not functioning properly, the module level check should be conducted in the following order.

VHF / UHF RECEIVER BOTTOM COMPONENT VIEW

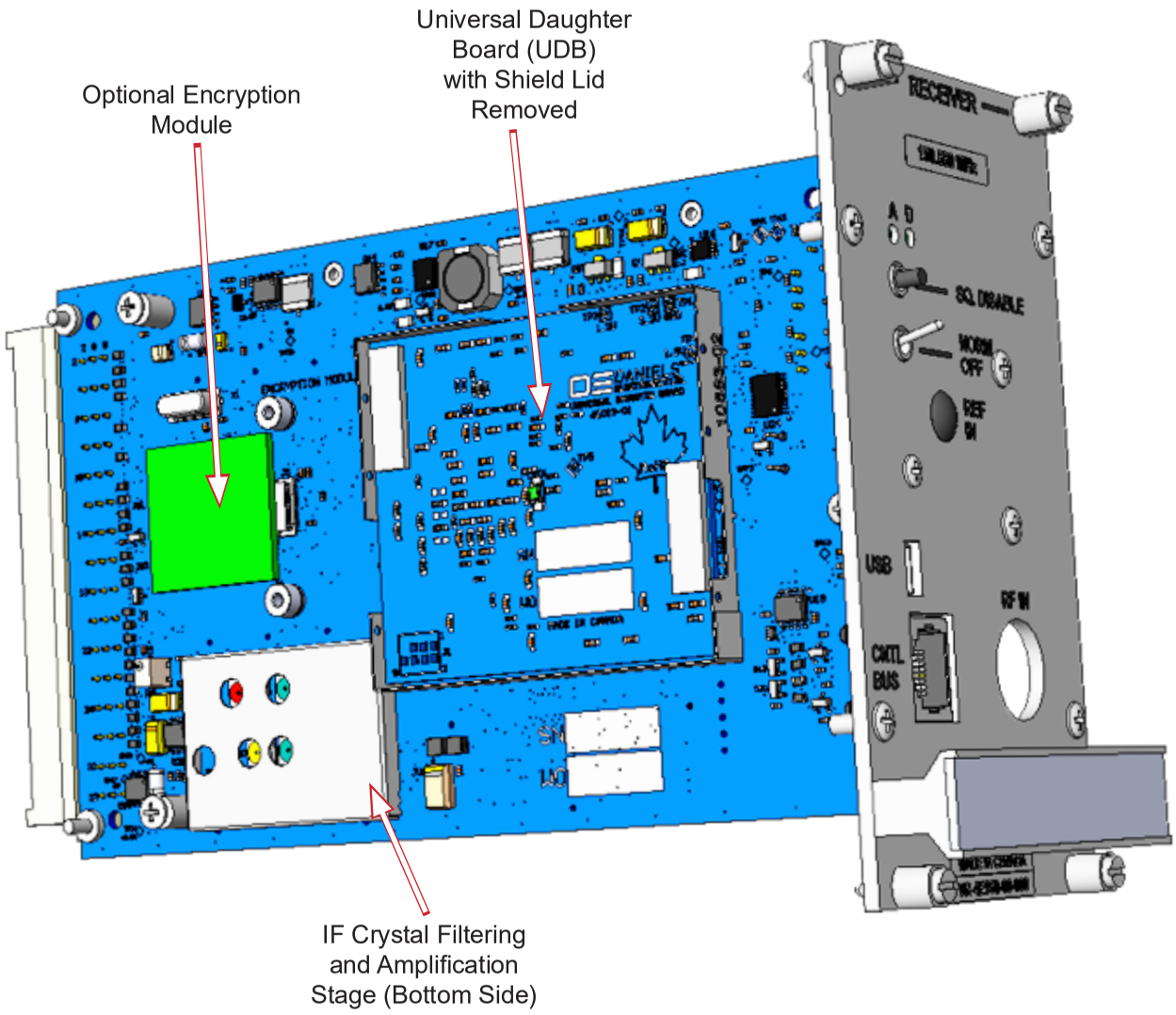


FIGURE 15: VHF / UHF Receiver Bottom Component View

Receiver Mainboard

Check the following test points on the receiver mainboard:

Test Point	Signal Monitored	Typical Voltage
TP1	Voltage from motherboard +13.8VDC line	+13.8VDC**
TP2	Regulated +9.5VDC from motherboard	+9.5VDC ± 5%
TP9	Switched +13.8VDC line from front panel switch	+13.8VDC**
TP37	+6.0VDC supply	+6.0VDC ± 5%
TP38	-6.0VDC supply	-6.0VDC ± 5%
TP54	+3.3VDC supply	+3.3VDC ± 5%
TP55	+1.2VDC supply	+1.2VDC ± 5%
TP56	+1.8VDC supply	+1.8VDC ± 5%
TP58	+5.0VDC supply	+5.0VDC ± 5%

Test Points are shown on the VHF / UHF Receiver Jumper and Test Point Locator illustrations (see Figure 11 and Figure 12).

If the test point voltages are within the specified range, the following tests further verify that the receiver mainboard may not be the source of problem.

Apply a -30dBm IF signal of the frequency specified (see the VHF and UHF Receiver Performance Specifications data sheet) to the receiver mainboard IF Input SMB connector. (See Figure 16: VHF / UHF Receiver Top Component View [2].) If the signal is properly demodulated it proves that the receiver mainboard and UDB are in working condition. Proceed to troubleshooting the synthesizer and UDB module.

Synthesizer and UDB Module

The Synthesizer Module and UDB are bound together through their communication link. At this level of fault diagnosis it is not feasible to isolate the problem between the two units.

If the receiver front panel ON/OFF switch is set from the OFF position to the NORM position and the two front panel LEDs stay on, check that the synthesizer and the UDB are properly seated on the receiver mainboard and check the following test points on the UDB. To access the UDB, the shield lid covering the UDB will need to be removed. (See Figure 15: VHF / UHF Receiver Bottom Component View.)

Test Point	Signal Monitored	Typical Voltage
TP1	Ground	0VDC
TP2	Logic power	+3.3VDC ± 5%
TP3	DSP core supply	+1.2VDC ± 5%
TP4	CPLD core supply	+1.8VDC ± 5%

Test Points are shown on the VHF / UHF Receiver Jumper and Test Point Locator illustrations. See Figure 11 and Figure 12.

If the UDB voltages are not within the specified tolerances, it is most likely that the receiver mainboard (which supplies these voltages to the UDB) is faulty or there is a short within the UDB.

If the UDB voltages are within the specified tolerances, inject a -70dBm carrier on the receiver frequency into the RF input and check the RSSI meter reading in RSS Service mode. If the RSSI meter reading is low (< 30), perform the following test to verify that the synthesizer is locked on frequency:

Disconnect the LO output SMB connector from the synthesizer to the RF Preselector (see Figure 6: VHF / UHF Receiver Top Component View [1]) and connect the synthesizer LO output to a radio communication test set. Confirm that the synthesizer LO frequency matches the Target Synthesizer RF OUT found in the “Reference Oscillator” field of RSS Service mode. The LO output should be approximately +7dBm.

The frequency of the synthesizer LO output will be above or below the receiver frequency by the IF frequency (see the VHF and UHF Receiver Performance Specifications data sheet). Check that this output is within the tolerance for the local oscillator frequency stability (see the VHF and UHF Receiver Performance Specifications data sheet). Also ensure that the RF frequency will change with programmed channel changes in the RSS.

Receiver RF Preselector

Check the +9.5VDC power supply wire to the RF Preselector. (See Figure 16: VHF / UHF Receiver Top Component View [2].) The supply voltage should be +9.5VDC ± 5%. Inject a -30dBm carrier on the receiver frequency into the RF Input and check that the IF Output of the RF Preselector is equal to the specified IF frequency (see the VHF and UHF Receiver Performance Specifications data sheet) and the output can be demodulated through the communication test set. If this is not the case, the synthesizer or the RF Preselector, or both, are faulty.

To further isolate the fault, the LO output (see Figure 6: VHF / UHF Receiver Top Component View [1]) can be disconnected from the synthesizer and an LO input can be applied from an external source to the RF Preselector. The frequency of the external LO input should be the Target Synthesizer RF OUT found in the “Reference Oscillator” field of RSS Service mode. Check the 21.4MHz output of the module using a communication test set.

VHF / UHF RECEIVER TOP COMPONENT VIEW (2)

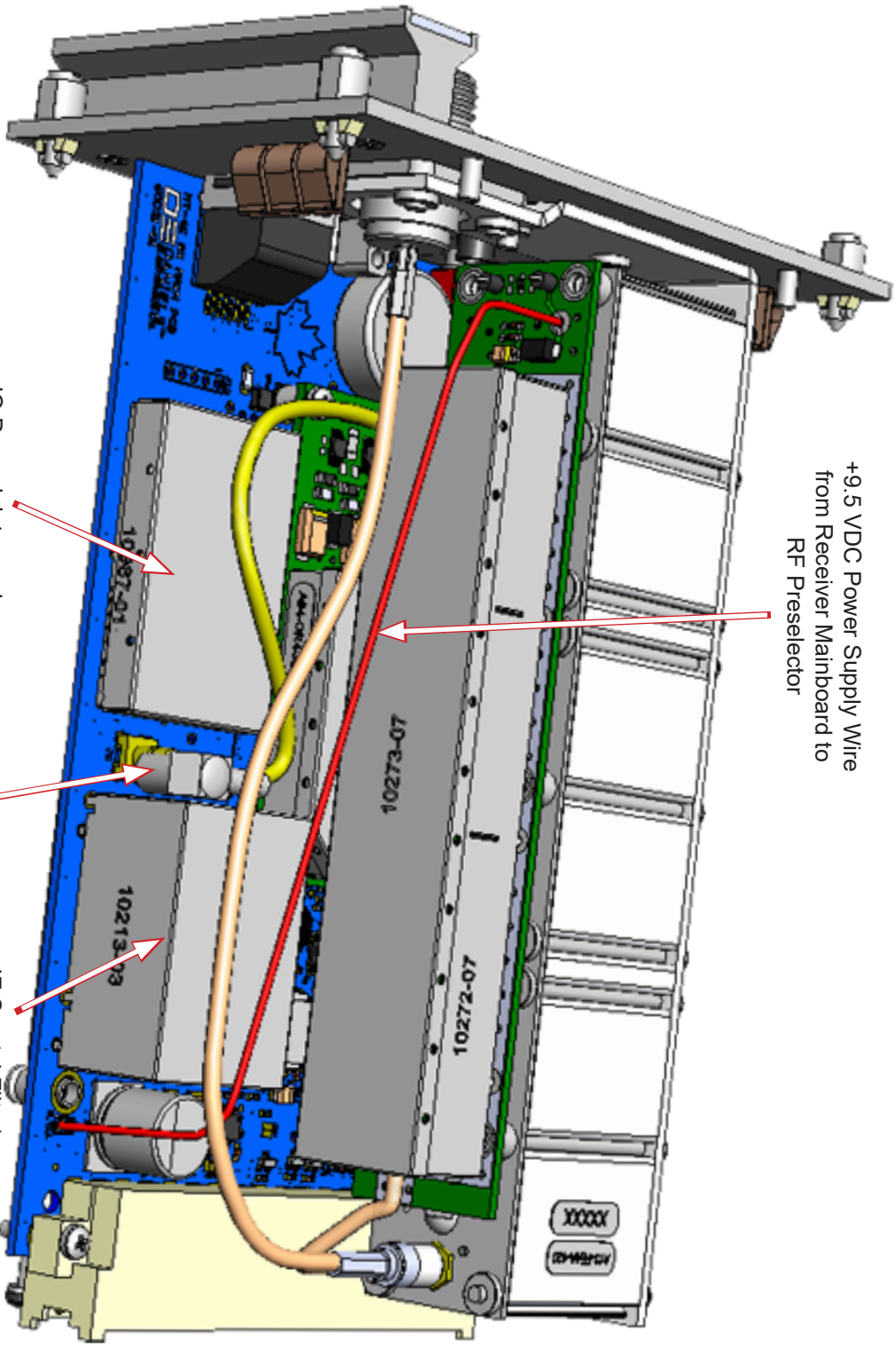


FIGURE 16: VHF / UHF Receiver Top Component View (2)

TRANSMITTER TROUBLESHOOTING

The transmitter is composed of four main components: the transmitter mainboard, the synthesizer module, RF power amplifier and universal daughter board (UDB). An optional encryption module can also be installed in the transmitter.

The following steps will help you determine the most likely subassembly that contains a fault and may help you determine why the fault is occurring.

NOTE: The subassembly or the entire transmitter can be sent back to Codan Service department for repair. Contact the Codan Service department before returning any product.

Before testing, the transmitter should be in the subrack or connected to the subrack by means of an extender card or kit, and the main power to the system (+13.8VDC) and the +9.5VDC regulated supply should be checked via the front panel of the system regulator. The transmitter RF output should be terminated with a 50Ω load of appropriate wattage or communications test set.

Testing the Transmitter

1. Flip the transmitter front panel NORM/ OFF/ KEY TX switch from the OFF position to the NORM position
2. Check the two front panel LED lights.
   Typically the front panel LED lights will turn ON for less than 10 seconds and then turn OFF. On occasion these lights may stay on for up to one minute.

   If this is not the case or the transmitter is not functioning properly, the following steps can be taken.
3. Connect the transmitter to the PC with the USB cable.
4. Run the Radio Service Software (RSS).
5. Select [P25 or Analog] Transmitter.
6. Click **Configure**.
7. Click **Read** to read the transmitter configuration.

   If the RSS returns "Device Not Responding," physically inspect the USB cable and connection, check the USB driver and check the connection through the Device Manager. If the RSS still does not recognize the connection, the most likely fault is on the UDB.
8. Click **Tx ID**.
9. Click Read to read the transmitter ID.

   Confirm that the frequency band in the Synthesizer ID Model Number matches the frequency band in the Transmitter Global Settings. If the settings do not match, the synthesizer will require replacement or the programming will need to be changed to match the synthesizer hardware.

If the transmitter is still not functioning properly, the module level check should be conducted in the following order.

Transmitter Mainboard

Check the following test points on the transmitter mainboard:

Test Point	Signal Monitored	Typical Voltage
TP1	Voltage from Motherboard +13.8VDC Line	+13.8VDC**
TP2	Regulated +9.5VDC from Motherboard	+9.5VDC ± 5%
TP3	USB Connector Power	+3.0VDC or +5.0VDC
TP4	Switched +13.8VDC Line from Front Panel Switch	+13.8VDC**

Test Points are shown on the VHF / UHF Transmitter Jumper and Test Point Locator illustrations (see Figure 13 and Figure 14).

If the test point voltages are within the specified range, it is most likely that the transmitter mainboard is not the source of problem. Proceed to the following step.

Synthesizer and UDB Modules

The synthesizer module and UDB are bound together through their communication link. At this level of fault diagnosis it is not feasible to isolate the problem between two units.

If the transmitter front panel NORM/OFF/ KEY TX switch is set from the OFF position to the NORM position and the two front panel LEDs stay on for typically longer than 10 seconds, but occasionally longer than one minute, check that the synthesizer and the UDB are properly seated on the transmitter mainboard and check the following test points on the UDB. To access the UDB, the amplifier subassembly and the shield lid covering the UDB will need to be removed (see Figure 17).

Test Point	Signal Monitored	Typical Voltage
TP1	Ground	0VDC
TP2	Logic power	+3.3VDC ± 5%
TP3	DSP core supply	+1.2VDC ± 5%
TP4	CPLD core supply	+1.8VDC ± 5%

Test Points are shown on the VHF / UHF Transmitter Jumper and Test Point Locator illustrations (see Figure 13 and Figure 14).

If the UDB voltages are not within the specified tolerances, the transmitter mainboard (that supplies these voltages to the UDB) is most likely faulty. If the UDB voltages are within the specified tolerances, it is likely that the synthesizer or the UDB, or both, are faulty.

If the transmitter front panel NORM/OFF/ KEY TX switch is set from the OFF position to the NORM position; the two front panel LEDs both turn ON for typically less than 10 seconds but occasionally up to one minute and then turn OFF; the RF output power control is set at its maximum in the “Power Levels” field of RSS Service mode, but the RF output power is considerably less than 8.0W (or 6.0W for UHF 470-520MHz T-Band models):

- Ensure that Synthesizer Reference is set to Internal in the “Jumper Settings” field of RSS Service mode.
- Ensure that the output of the synthesizer module on the RF cable as shown on Figure 10: VHF / UHF Transmitter Bottom Component View is 0dBm (± 2dBm) and that the RF frequency will change with programmed channel changes in the RSS. If this is not the case, it is likely that the synthesizer or the UDB, or both, are faulty.

**Refer to the Specification Data Sheets (located in this manual) for the full Supply Voltage range.

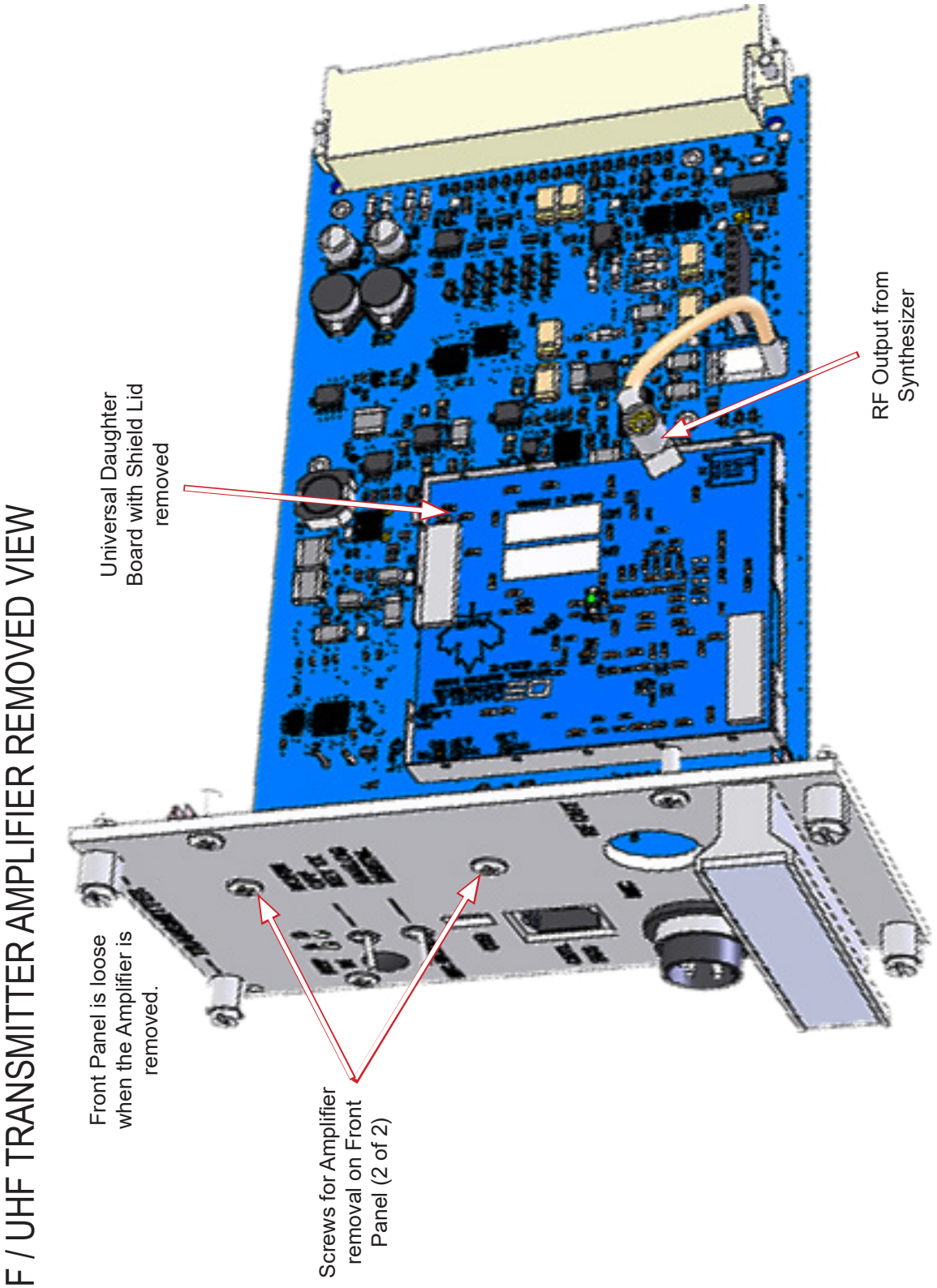


FIGURE 17: VHF / UHF Transmitter Amplifier Removed View

Transmitter Amplifier

Set the front panel switch on the transmitter to KEY TX and check that the transmitter output power is approximately 0.5 to 8.0W for VHF and UHF 406–470MHz models and 0.5 to 6.0W for UHF 470–520MHz T-Band models.

Check the following pins on Connector J1 (while the transmitter is keyed):

Pin	Signal Monitored	Typical Voltage
1	RF Enable / Power Control (+0.5 to +3.0VDC corresponds to 0.5 to 8.0W or 6.0W for UHF 470-520MHz T-Band RF Power Output)	+0.5 to +3.0VDC
2	Regulated +9.5VDC from motherboard	+9.5VDC ± 5%
3	Voltage from motherboard +13.8VDC line	+13.8VDC**

Pins are shown on the VHF / UHF Transmitter Bottom Component View (see Figure 10).

If Pin 1 is not within tolerance, the most likely fault is in the UDB or synthesizer module. If Pins 1, 2 and 3 are within the specified tolerances and there is no RF output power, the most likely fault is in the amplifier.

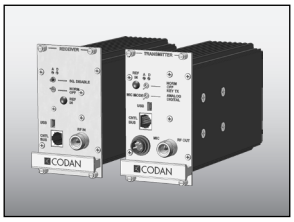
To test the RF power amplifier subassembly, set the output power to maximum using “Power Level Adjustments” in RSS Service mode, and drive the input of the amplifier with a 0dBm RF signal. Check that the RF power output is 8.0W (or 6.0W for UHF 470–520MHz T-Band models) when the front panel switch is set to KEY TX.

SYSTEM TROUBLESHOOTING

If the radio system is not receiving or transmitting on the right frequency, check the Channel and Bank Select lines of the receiver and transmitter modules. The motherboard has jumpers that allow one of 16 channels to be selected on a receiver or transmitter. The factory default is to have them all set to ground the channel select lines CSEL0 through CSEL3, forcing the radio to operate on Channel 1. On the receiver and transmitter there is also a Bank A/B line which usually floats high – selecting Bank A. The receiver and transmitter have separate control of their channel select lines, so changing receiver channels does not affect the transmitter channel.

If the Radio Service Software is failing partway through a read or write of the receiver, check that the receiver is not squelching and unsquelching repeatedly during the read or write operation. Disconnect the RF signal source from the receiver during the read or write operation.

** Refer to the Specification Data Sheets (located in this manual) for the full Supply Voltage range.



VHF & UHF HARDWARE INTERFACING

RECEIVER AND TRANSMITTER INTERFACING

Interface connection to the MT-4E Receiver and Transmitter is available through the 48-pin motherboard interface connectors. These connectors allow interfacing to the audio, COR-PTT, Channel / Bank Select signals and many other signal lines in the receiver and transmitter.

Easy access to most of these signal lines can be achieved through the use of the optional A-PNL-AUX96-3 Auxiliary Connector, which uses plug-in screw connectors to secure 22AWG to 16AWG wires for external interface. Some of the interface lines are accessible through both jumpers and solder points on the motherboard. More information is available in the Subrack / Motherboard Instruction Manual.

User interfacing to the RJ45 jack on the front of the receiver and transmitter is not recommended. Signal interference may impair the operation of the module(s).

For the VHF / UHF Receiver Interface Components Schematic Diagram, see Figure 18. For the VHF / UHF Transmitter Interface Components Schematic Diagram, see Figure 20.

VHF / UHF Receiver (see Figure 19) and VHF / UHF Transmitter (see Figure 21) block diagrams are also included in this manual.

VHF / UHF RX INTERFACE COMPONENTS SCHEMATIC DIAGRAM

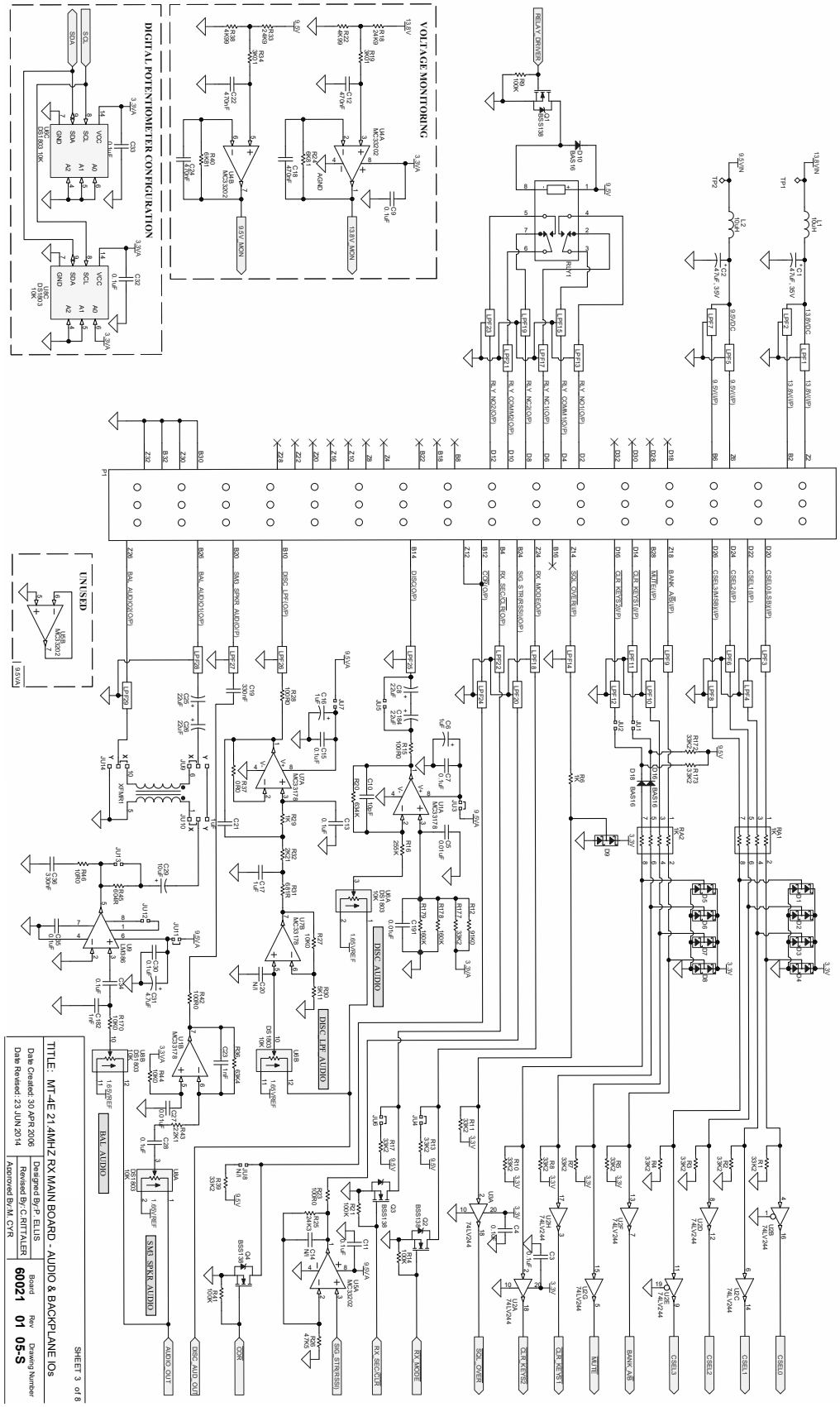


FIGURE 18: UHF / VHF Receiver Interface Components Schematic Diagram

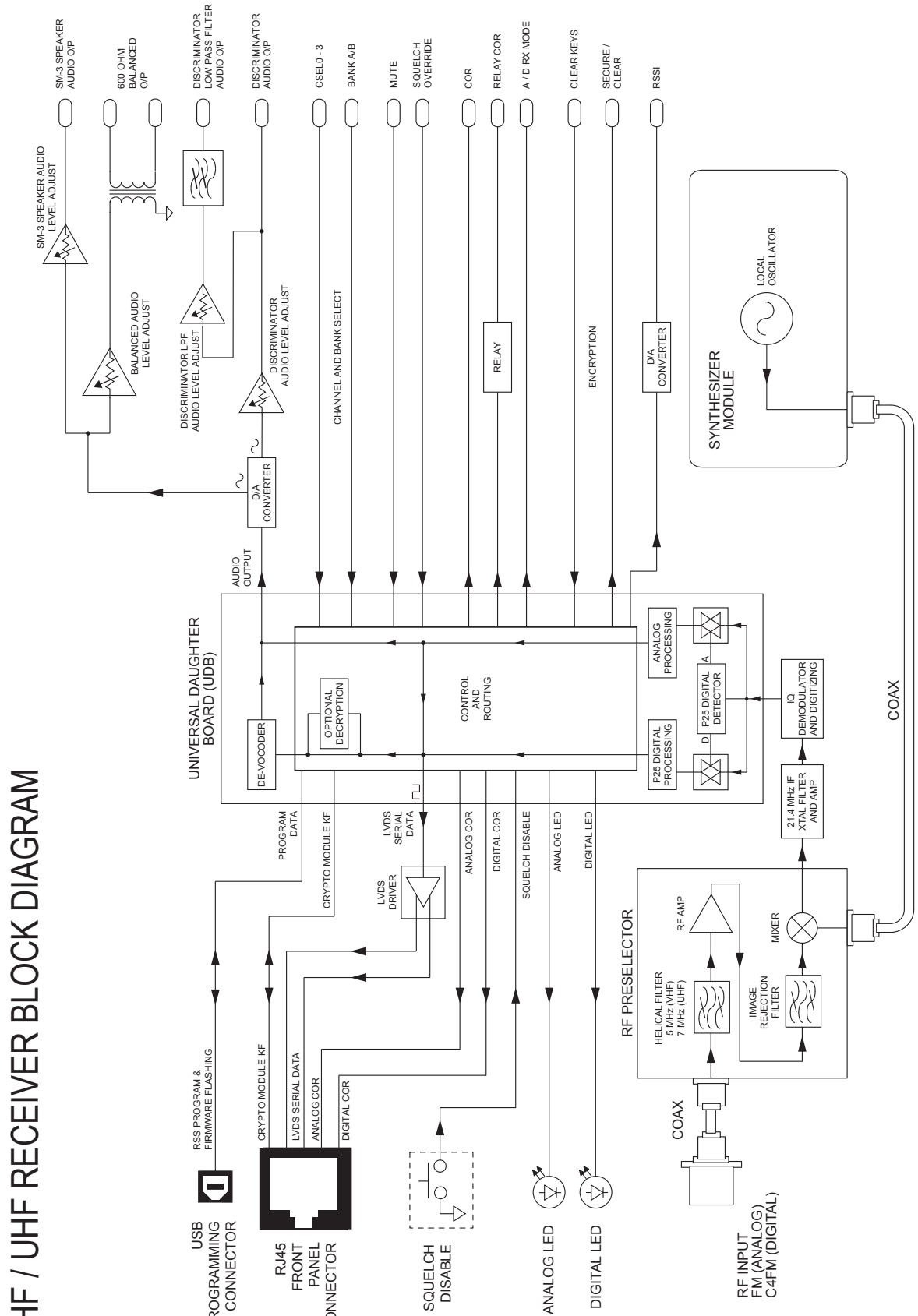


FIGURE 19: VHF / UHF Receiver Block Diagram

VHF / UHF TX INTERFACE COMPONENTS SCHEMATIC DIAGRAM

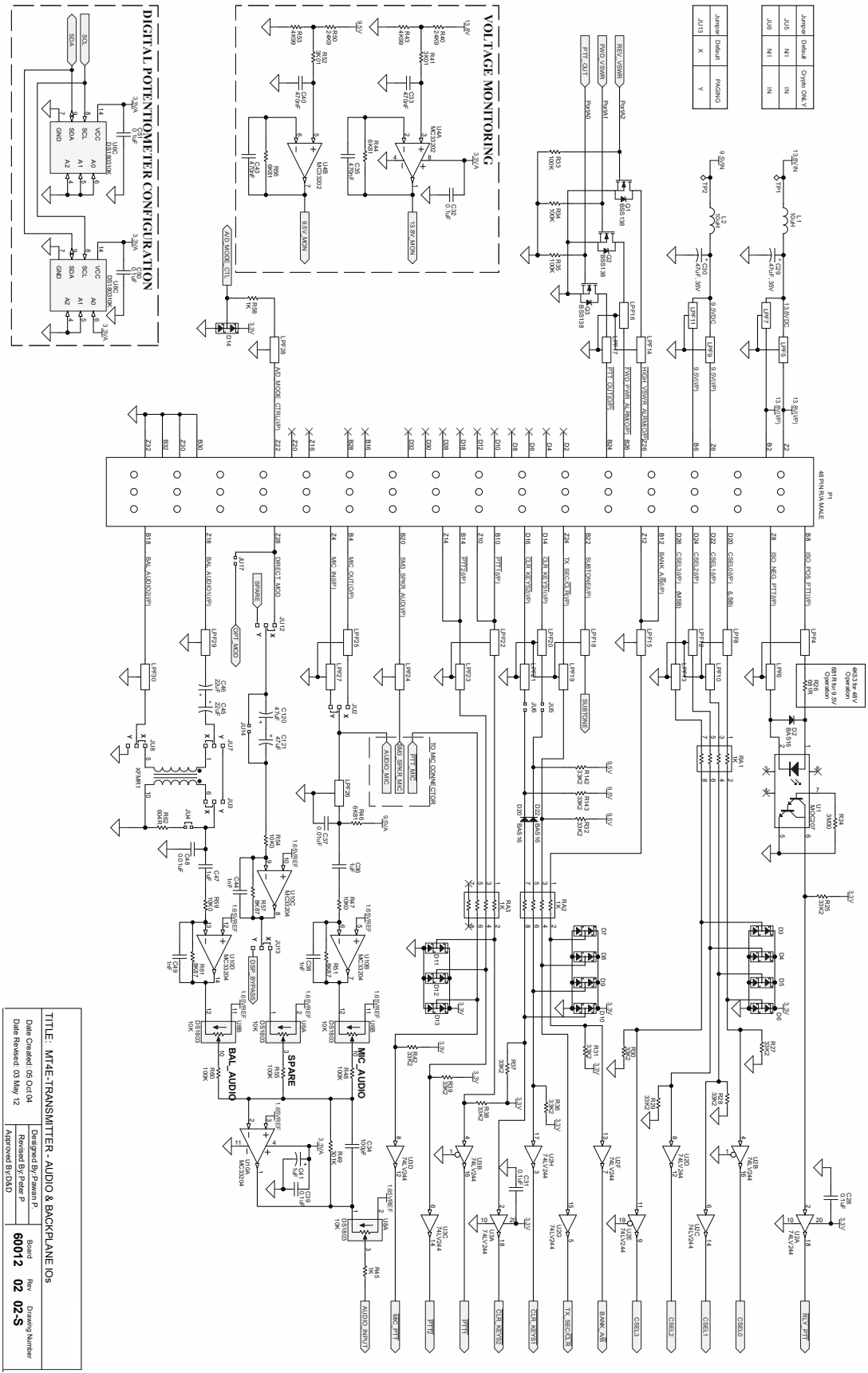


FIGURE 20: VHF / UHF Transmitter Interface Components Schematic Diagram

VHF / UHF TRANSMITTER BLOCK DIAGRAM

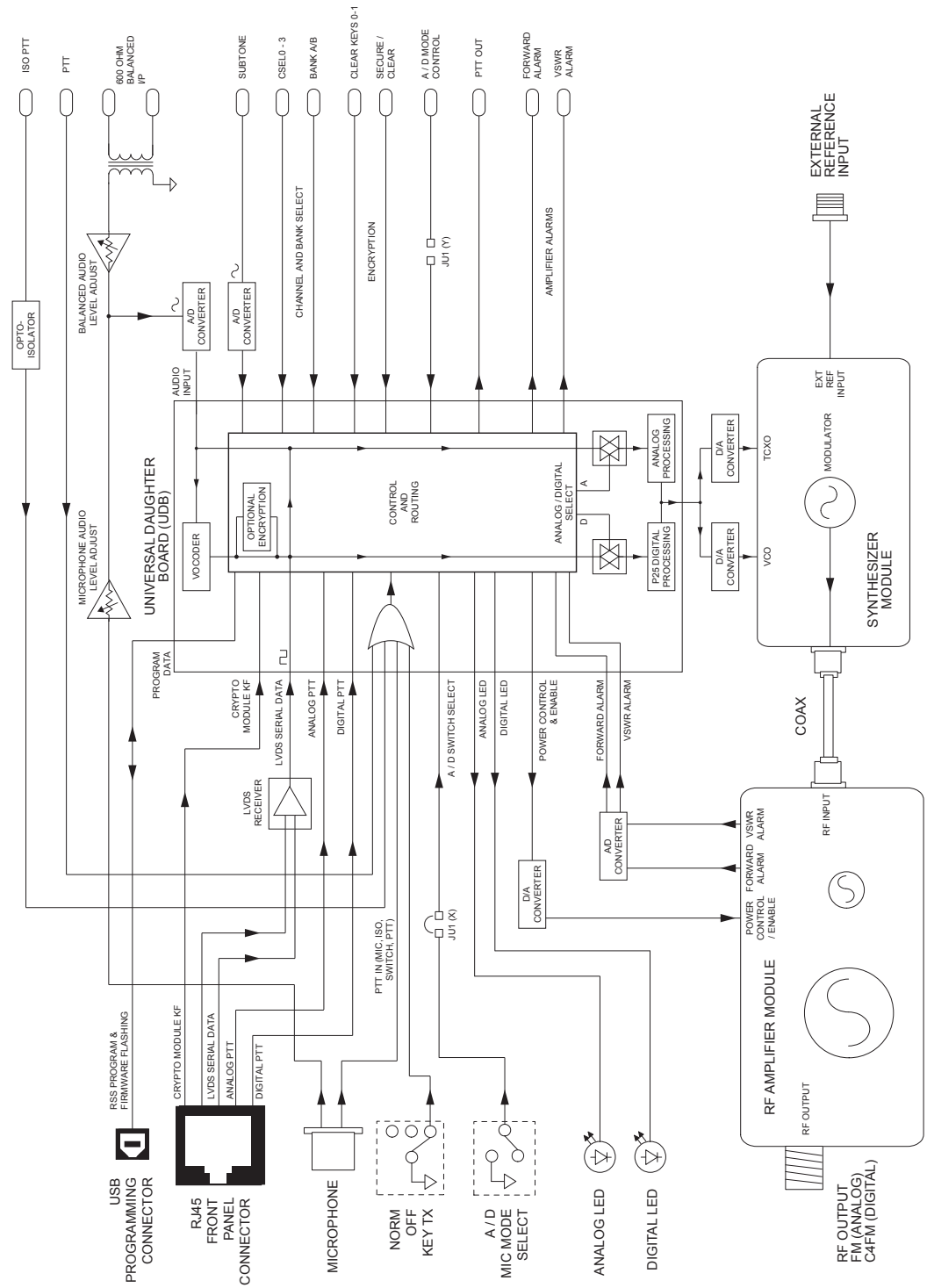


FIGURE 21: VHF / UHF Transmitter Block Diagram

48-PIN MOTHERBOARD INTERFACE CONNECTOR

A 48-pin connector is used for interfacing the MT-4E Motherboard to the receiver and transmitter.

Receiver

Pin	Name	Pin	Name	Pin	Name
D2	Relay Normally Open 1 [O/P]	B2	+13.8VDC [I/P]	Z2	+13.8VDC [I/P]
D4	Relay Common 1 [O/P]	B4	Receive Secure / Clear* [O/P]	Z4	No Connect
D6	Relay Normally Closed 1 [O/P]	B6	+9.5VDC [I/P]	Z6	+9.5VDC [I/P]
D8	Relay Normally Closed 2 [O/P]	B8	No Connect	Z8	No Connect
D10	Relay Common 2 [O/P]	B10	Discriminator LPF [O/P]	Z10	No Connect
D12	Relay Normally Open 2 [O/P]	B12	COR* [O/P]	Z12	COR* [O/P]
D14	Clear Keys 1* [I/P]	B14	Discriminator [O/P]	Z14	Squelch Override* [I/P]
D16	Clear Keys 2* [I/P]	B16	No Connect	Z16	No Connect
D18	No Connect	B18	No Connect	Z18	Bank A/B* [I/P]
D20	Channel Select 0 (LSB) [I/P]	B20	SM-3 Speaker Audio [O/P]	Z20	No Connect
D22	Channel Select 1 [I/P]	B22	No Connect	Z22	No Connect
D24	Channel Select 2 [I/P]	B24	Signal Strength (RSSI) [O/P]	Z24	RX Mode [O/P]
D26	Channel Select 3 (MSB) [I/P]	B26	Balanced Audio 1 [O/P]	Z26	Balanced Audio 2 [O/P]
D28	No Connect	B28	Mute* [I/P]	Z28	No Connect
D30	No Connect	B30	Ground	Z30	Ground
D32	No Connect	B32	Ground	Z32	Ground

Transmitter

Pin	Name	Pin	Name	Pin	Name
D2	No Connect	B2	+13.8VDC [I/P]	Z2	+13.8VDC [I/P]
D4	No Connect	B4	Microphone Audio Output [O/P]	Z4	Microphone Audio Input [I/P]
D6	No Connect	B6	+9.5VDC [I/P]	Z6	+9.5VDC [I/P]
D8	No Connect	B8	Isolated Positive PTT [I/P]	Z8	Isolated Negative PTT [I/P]
D10	No Connect	B10	PTT* [I/P]	Z10	PTT* [I/P]
D12	No Connect	B12	Bank A/B* [I/P]	Z12	Bank A/B* [I/P]
D14	Clear Keys 1* [I/P]	B14	PTT* [I/P]	Z14	PTT* [I/P]
D16	Clear Keys 2* [I/P]	B16	No Connect	Z16	No Connect
D18	No Connect	B18	Balanced Audio 2 [I/P]	Z18	Balanced Audio 1 [I/P]
D20	Channel Select 0 (LSB) [I/P]	B20	SM-3 Speaker Audio [I/P]	Z20	No Connect
D22	Channel Select 1 [I/P]	B22	Subtone [I/P]	Z22	A*/D Mode Control [I/P]
D24	Channel Select 2 [I/P]	B24	PTT Output* [O/P]	Z24	Transmit Secure / Clear* [I/P]
D26	Channel Select 3 (MSB) [I/P]	B26	Forward Power Alarm* [O/P]	Z26	High VSWR Alarm* [O/P]
D28	No Connect	B28	No Connect	Z28	Direct Modulation
D30	No Connect	B30	Ground	Z30	Ground
D32	No Connect	B32	Ground	Z32	Ground

* Indicates an active low signal.

RECEIVER PIN INTERFACE INFORMATION

COR Relay Outputs

D2	Relay Normally Open 1 [O/P]
D4	Relay Common 1 [O/P]
D6	Relay Normally Closed 1 [O/P]
D8	Relay Normally Closed 2 [O/P]
D10	Relay Common 2 [O/P]
D12	Relay Normally Open 2 [O/P]

The COR dual output is a general purpose output that can be used to drive various connection requirements, such as E&M. This output can be enabled and disabled in RSS.

The COR outputs are accessible through solder points on the motherboard. Solder connections for motherboard Version 50016-02 are labelled J41 through J46 for Receiver A and J35 through J40 for Receiver B. Solder connections for motherboard Version 50016-03 are labelled J17, J20, J24, J29, J36 and J40 for Receiver A and J16, J22, J26, J28, J37 and J42 for Receiver B.

Motherboard Version 50016-07 and newer contain three jumpers (JU80, JU81 and JU82) that can be enabled in the A position to allow Receiver A Relay 1 outputs to be routed to the A-PNL-AUX96-3 Auxiliary Connector for easier access on Pins J6-9 (N/O), J6-2 (COM) and J6-6 (N/C).

COR Output

B12	COR* [O/P]	Z12	COR* [O/P]

The COR output is an active low indication of an incoming decoded RF signal (analog or digital). This output pulls to ground through an open collector output Q4. The COR output also has jumper JU8 that can be installed to provide a pull up through R39 to 9.5VDC.

Mute Input

B28	Mute* [I/P]

The Mute input is an active low input that will keep the receiver from unsquelching when grounded. This input is internally pulled high through R7 to 3.3VDC to allow the receiver to unsquelch when this line is not connected to an external input.

When the receiver RSS is configured for internal CTCSS or DCS decode, this line should not be used or a conflict between internal and external mute functions will occur.

Squelch Override Input

Z14	Squelch Override* [I/P]

The Squelch Override input is an active low input that will unsquelch the receiver to monitor any RF signal present on the receiver frequency. This input is internally pulled high through R11 to 3.3VDC when this line is not connected to an external input.

* Indicates an active low signal.

Channel Select Inputs

D20	Channel Select 0 (LSB) [I/P]
D22	Channel Select 1 [I/P]
D24	Channel Select 2 [I/P]
D26	Channel Select 3 (MSB) [I/P]

The Channel Select inputs are used as binary logic inputs that allow the receiver to change up to 16 channels. A pull to ground represents a Logic 0 and a pull to high (5.0VDC – 9.5VDC) is a Logic 1. When there are no connections to these inputs, the channel select lines are internally pulled to ground.

Important Note: Channel Select lines must be set before the receiver unsquelches or the receiver will not change channel.

Bank A/B Input

Z18	Bank A/B* [I/P]

The Bank A/B input is used as a binary logic input that allows the receiver to change between two banks of 16 channels. A pull to ground represents a Logic 0 (Bank B) and a pull to high (5.0VDC–9.5VDC) is a Logic 1 (Bank A). When there are no connections to these inputs, the channel select lines are internally pulled to high (Bank A).

Important Note: The Bank A/B line must be set before the receiver unsquelches or the receiver will not change bank.

Clear Keys Inputs

D14	Clear Keys 1* [I/P]
D16	Clear Keys 2* [I/P]

The Clear Keys inputs are used to zeroize all encryption keys in a receiver with an optional decryption module installed. Jumpers JU1 and JU2 need to be installed in order to enable the Clear Keys inputs. The activation sequence to zeroize encryption keys is generated by the CI-BC-4E Base Controller when the zeroize keys push button is pushed.

NOTE: The duration of the key press must be greater than one second to ensure that the keys have been zeroized.

Receive Secure / Clear Output

B4	Receive Secure / Clear* [O/P]

The receive Secure / Clear output is a logic output that indicates whether the received signal is encrypted or clear. A logic low level (Ground) is an indication of a non-encrypted signal received.

* Indicates an active low signal.

When an encrypted signal has been received this output will be floating or pulled to +9.5VDC through R17. The jumper JU6 will determine if the output is pulled up to +9.5VDC (JU6 IN) or floating (JU6 OUT).

When the receiver is not receiving, the signal line sits low (Ground). This signal line should be used in conjunction with the COR output.

Analog / Digital Receive Mode Output

Z24 RX Mode [O/P]

The receiver mode output is a logic output that indicates whether the received signal is analog or P25 Digital. A logic low level (Ground) is an indication of an analog signal received. When a P25 Digital signal has been received this output will be floating or pulled to +9.5VDC through R13. The jumper JU4 will determine if the output is pulled up to +9.5VDC (JU4 IN) or floating (JU4 OUT).

When the receiver is not receiving, the signal line sits low (Ground). This signal line should be used in conjunction with the COR output.

Receive Signal Strength Indicator (RSSI) Output

B24 Signal Strength (RSSI) [O/P]

The Receive Signal Strength Indicator (RSSI) output is a 0VDC to +5VDC voltage indication of the received RF signal strength. The RSS also has an RSSI meter built into the Service mode that gives a numerical representation from (0 to 5000) of the received RF signal strength. These values are not calibrated to any standard and may vary from receiver to receiver.

The graph below (see Figure 22) shows a typical representation of RF Signal Strength versus RSSI voltage. (Data results are from receiver PCB mainboard 50164-05 and Firmware Version 2.1.1.)

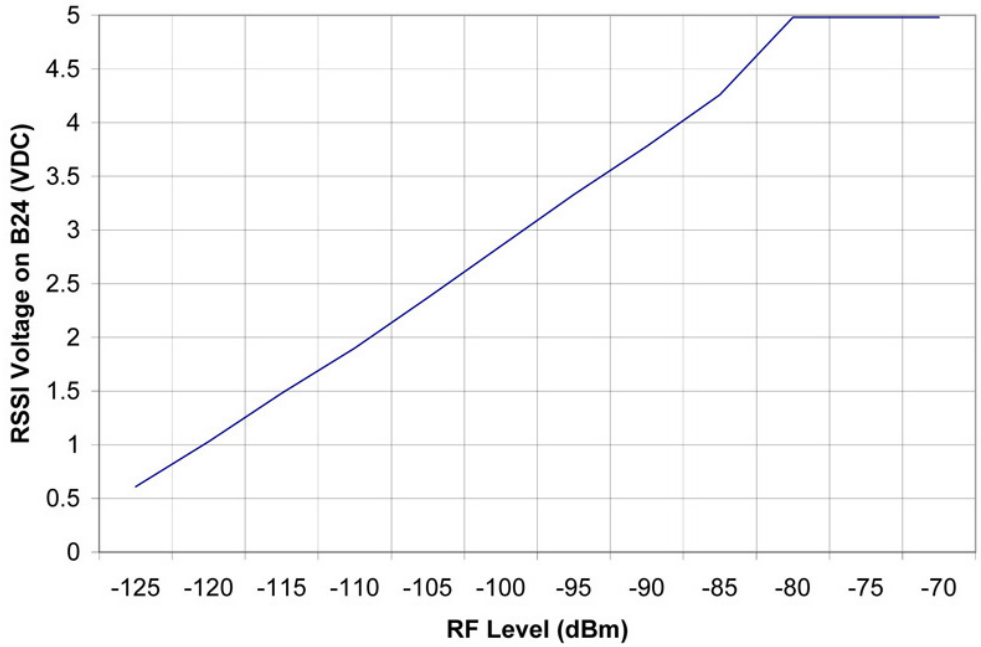


FIGURE 22: RF Signal Strength Versus RSSI Voltage

Discriminator Audio Output

B14	Discriminator [O/P]

The Discriminator Audio output is an audio output that is not gated by the squelch circuitry. Audio, CTCSS and DCS present on the received signal will be output. The Discriminator Audio output is typically used to connect to external CTCSS, DCS or DTMF decoder modules. The Discriminator Audio output level is adjustable via a softpot in the RSS.

Discriminator LPF Output

B10	Discriminator LPF [O/P]

The Discriminator Low Pass Filter (LPF) output is a 250Hz low pass filtered audio output that is not gated by the squelch circuitry. CTCSS and DCS present on the received signal will be output. The Discriminator LPF output is typically used to pass CTCSS tones directly through to the transmitter or to connect to external CTCSS or DCS decoder modules. The Discriminator LPF output level is adjustable via a softpot in the RSS.

SM-3 Speaker Audio Output

B20	SM-3 Speaker Audio [O/P]

The SM-3 Speaker Audio output is an unbalanced audio output that is used to drive the speaker audio on the SM-3 System Regulator. The SM-3 Speaker Audio output level is adjustable via a softpot in the RSS.

Balanced Audio Output

B26 Balanced Audio 1 [O/P]	Z26 Balanced Audio 2 [O/P]

The Balanced Audio output is a 600-ohm balanced (via transformer) audio output that is capable of various configurations. The Balanced Audio output can be configured for de-emphasis or flat audio response via RSS programming options. De-emphasis (6dB / octave) audio is the standard setting and is typically used for all voice communications. Flat audio (de-emphasis disabled) is typically used for analog data formats such as frequency shift keying (FSK).

The Balanced Audio output can be configured by setting jumpers JU9, JU10, JU13 and JU14 (JU9, JU10 and JU14 can be installed in position X or Y; JU13 is installed or not installed):

	JU9	JU10	JU13	JU14
Balanced 600 ohm with transformer (standard configuration)	X	X	OUT	X
Balanced low impedance with transformer	X	X	IN	X
Unbalanced 600 ohm without transformer	Y	Y	OUT	Y
Unbalanced low impedance without transformer	Y	Y	IN	Y

NOTE: Z26 Balanced Audio 2 [O/P] is grounded when the receiver is set for unbalanced.

The Balanced Audio output level is adjustable via a softpot in the RSS. Maximum audio output level for the 600-ohm balanced output configuration is +4.0dBm with a 1.0kHz tone @ 1.5kHz deviation at the receiver RF input.

Status Tone Encoder Option

The Balanced Audio output can be configured with a status tone encoder option for “Voting.” The Status Tone Encoder is not a standard feature; the receiver is ordered or upgraded to enable this custom feature.

In the “Extra Features” field of RSS Service mode, the frequency and amplitude of the status tone can be adjusted. The typical frequency of a status tone is 1950Hz or 2175Hz, but the frequency can be selected for any non-standard tone (300Hz–3000Hz). The status tone is active when the receiver is squelched and will revert to audio only when the receiver is unsquelched.

TRANSMITTER PIN INTERFACE INFORMATION

Isolated PTT Inputs

B8	Isolated Positive PTT [I/P]	Z8	Isolated Negative PTT [I/P]

The Isolated PTT (Push-To-Talk) input is a general purpose input for keying the transmitter that is capable of handling many different voltage inputs. To activate the Isolated PTT input, connect the positive voltage to pin B8 and the open collector output of the external device being used to activate the Isolated PTT will connect to pin Z8. When the open collector input is pulled to ground the transmitter PTT will be activated through U1.

When using lower voltages such as +9.5VDC on the positive pin, R26 will remain as a standard 681 ohm resistor. When the positive input is connected to higher voltage such as an E&M +48VDC or - 48VDC, R26 will need to be changed to a 4K53 ohm resistor to prevent damage by limiting the current draw through U1.

PTT Inputs

B10	PTT* [I/P]	Z10	PTT* [I/P]
B14	PTT* [I/P]	Z14	PTT* [I/P]

The PTT inputs are the standard inputs for keying the transmitter. These inputs are typically controlled using an open collector input from a receiver COR output. A pull to ground will activate the transmitter PTT.

PTT Output

B24	PTT Output* [O/P]

The PTT output is a logic output that indicates when the transmitter is active (keyed). This is an open collector output that will pull to ground through Q3 when the transmitter is active.

Alarm Outputs

B26	Forward Power Alarm* [O/P]	Z26	High VSWR Alarm* [O/P]

The Alarm outputs are open collector outputs that will pull to ground with a failure condition. The Forward Power Alarm will pull to ground through Q2 with a condition of half power or less out of the transmitter amplifier. The high VSWR Alarm will pull to ground through Q1 with a high VSWR condition of 3:1 or more reflected power.

* Indicates an active low signal.

Channel Select Inputs

D20	Channel Select 0 (LSB) [I/P]
D22	Channel Select 1 [I/P]
D24	Channel Select 2 [I/P]
D26	Channel Select 3 (MSB) [I/P]

The Channel Select inputs are used as binary logic inputs that will allow the transmitter to change up to 16 channels. A pull to ground represents a Logic 0 and a pull to high (5.0VDC–9.5VDC) is a Logic 1. When there are no connections to these inputs, the channel select lines are internally pulled to ground.

Important Note: Set the Channel Select lines before the transmitter is keyed (PTT) or the transmitter will not change channel.

Bank A/B Inputs

B12	Bank A/B* [I/P]	Z12	Bank A/B* [I/P]

The Bank A/B inputs are used as a binary logic input that allows the transmitter to change between two banks of 16 channels. A pull to ground represents a Logic 0 (Bank B) and a pull to high (5.0VDC–9.5VDC) is a Logic 1 (Bank A). When there are no connections to these inputs, the channel select lines are internally pulled to high (Bank A).

Important Note: The Bank A/B line must be set before the transmitter is keyed (PTT) or the transmitter will not change bank.

Clear Keys Inputs

D14	Clear Keys 1* [I/P]
D16	Clear Keys 2* [I/P]

The Clear Keys inputs are used to zeroize all encryption keys in a transmitter with an optional encryption module installed. Jumpers JU5 and JU6 need to be installed in order to enable the Clear Keys inputs. The activation sequence to zeroize encryption keys is generated by the CI-BC-4E Base Controller when the zeroize keys push button is pushed.

NOTE: The duration of the key press must be greater than one second to ensure that the keys have been zeroized.

* Indicates an active low signal.

Transmit Secure / Clear Input

Z24	Transmit Secure / Clear* [I/P]

The Transmit Secure / Clear input is a logic control input for a transmitter in base station configuration with an encryption module installed. To transmit in secure mode, the Transmit Secure / Clear input needs to be pulled high using a 5.0VDC–9.5VDC input. To transmit in clear mode, the Transmit Secure / Clear input needs to pulled low (Ground). If the input is not connected, the transmitter will transmit in clear mode.

Analog / Digital Mode Control Input

Z22	A*/D Mode Control [I/P]

The Analog / Digital Mode Control input provides added control over the transmitter when used in a base station configuration. To use this input, the transmitter must be programmed by the RSS for mixed mode.

If the transmitter is keyed up through the connections on the front panel RJ45 jack, the incoming PTT signal is separated into Analog PTT and Digital PTT. This is typically being driven by the mode the receiver received the RF signal in, and is used for most repeat functions. When the transmitter is keyed up by the 48-pin motherboard interface PTT or Isolated PTT, or by the Microphone PTT or test switch “Key TX” on the front panel, the transmitter mode needs to be selected. When JU1 is in the X position, the front panel “MIC MODE” switch position will determine the transmitter mode. When JU1 is in the Y position, a ground on the Analog / Digital Mode Control input will key the transmitter in analog mode. To key the transmitter in P25 Digital mode, leave the input open and the internal pull up Resistor R72 will provide the logic high required.

When the transmitter is configured for paging applications, JU1 will be in the Y position and “Paging**”** will be selected in RSS for the Transmitter Channel Type. With this paging configuration, the following changes occur:

- The front panel A/D LEDs no longer indicate the transmitter’s true A/D status.
- The front panel Analog / Digital switch SW2 is disabled.
- The A/D_MODE_CTRL (I/P) at Z22 is now externally controlled by the paging modulator. When this line goes ‘LOW’, the transmitter balanced input is turned off.

Important Note: The Analog / Digital Mode Control line must be set before the transmitter is keyed (PTT) or the transmitter will not change mode.

* Indicates an active low signal.

Balanced Audio Input

B18	Balanced Audio 2 [I/P]	Z18	Balanced Audio 1 [I/P]

The Balanced Audio input is a 600-ohm balanced (via transformer) audio input that is capable of various configurations. The Balanced Audio input can be configured for pre-emphasis or flat audio response via RSS programming options. Pre-emphasis (6dB / octave) audio is the standard setting and is typically used for all voice communications. Flat audio (pre-emphasis disabled) is typically used for analog data formats such as frequency shift keying (FSK).

The Balanced Audio input can be configured the following ways by setting jumpers JU3, JU4, JU7 and JU8 (JU3, JU7 and JU8 can be installed in position X or Y; JU4 is installed or not installed).

	JU3	JU4	JU7	JU8
Balanced 600 ohm with transformer (standard configuration)	X	IN	X	X
Balanced low impedance with transformer	X	OUT	X	X
Unbalanced 600 ohm without transformer	Y	IN	Y	Y
Unbalanced low impedance without transformer	Y	OUT	Y	Y

NOTE: B18 Balanced Audio 2 [I/P] is ground when the receiver is set for unbalanced.

The Balanced Audio input level is adjustable via a softpot in the RSS. Audio input level for the 600-ohm balanced input configuration is adjustable from -18dBm to +4.0dBm with a 1.0kHz tone @ 1.5kHz deviation at the transmitter RF output.

Microphone Audio Input / Output and SM-3 Speaker Audio Input

B4	Microphone Audio Output [O/P]	Z4	Microphone Audio Input [I/P]
B20	SM-3 Speaker Audio [I/P]		

The Microphone Audio input / output is an unbalanced high impedance audio input. Using jumper JU2 in the X position, the microphone audio output can be directly routed out to the subrack. Using jumper JU2 in the Y position, the microphone audio input can be used as a spare high-impedance unbalanced audio input. The SM-3 Speaker Audio input is used as an input for custom microphones.

The Microphone Audio input level is adjustable via a softpot in the RSS. Audio input level for the unbalanced high-impedance input is adjustable from -20dBm to +4.0dBm with a 1.0kHz tone @ 1.5kHz deviation at the transmitter RF output.

Subtone Input

B22	Subtone [I/P]

The Subtone input supports external modulation sources for CTCSS, DCS and Logic Trunked Radio (LTR®).

The RSS enables the Subtone input in the “Analog Settings” field of the MT-4E Transmitter configuration screen. Set the “Signaling” field to “External Input” on a per-channel basis.

The Subtone input level is adjustable via a softpot in the RSS. Level adjustments can also be made on the external encoder device. Subtone input level for the CTCSS configuration is -18dBm input level for a 100Hz tone @ 0.35kHz deviation (narrow band) or 0.50kHz deviation (wide band) at the transmitter RF Output.

CTCSS

Continuous Tone Coded Squelch System (CTCSS) can be injected from an external device rather than using the internal CTCSS generator in the RSS.

DCS / LTR®

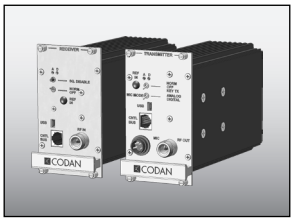
Digitally Coded Squelch (DCS) and LTR® can be connected to the transmitter Subtone input.

Direct Modulation Audio

Z28	Direct Modulation

The Direct Modulation Audio is configured for a flat audio response. This input can be used for audio or analog in-band data formats (such as 2400 baud FSK) or Flex™ paging formats. The Direct Modulation input should be AC coupled with JU14 (OUT – default) and JU13 in the Y position for paging applications. For formats such as FSK, JU13 can be set in the X position.

LTR® is a registered trademark of E.F. Johnson Co.
FLEX™ is a trademark of Motorola, Inc.



VHF AND UHF PARTS LIST

VHF / UHF RECEIVER ELECTRICAL PARTS

Part Number	Description	Product	Qty
A11-RX4E-MAIN-01	RX MB|MT4E|136-520MHz|INCL UDB	UHF and VHF	1
A11-UDB	Included in A11-RX4E-MAIN-01 subassembly	ALL	1
A11-FE4-150	PRESELECTOR/4E|CLASS B|136-174	VHF CLASS B	1
A11-FE4A-150	PRESELECTOR/4E|CLASS A|136-174	VHF CLASS A	1
A13-FE4-420	PRESELECTOR/4E|CLASS B|406-430	UHF (406-430) CLASS B	1
A13-FE4A-420	PRESELECTOR/4E|CLASS A|406-430	UHF (406-430) CLASS A	1
A13-FE4-460	PRESELECTOR/4E|CLASS B|450-470	UHF (450-470) CLASS B	1
A13-FE4A-460	PRESELECTOR/4E|CLASS A|450-470	UHF (450-470) CLASS A	1
A13-FE4-500	PRESELECTOR/4E|CLASS B|470-512	UHF (470-520) CLASS B	1
A64-OR4-150	SYNTHESIZER| RX/4E|VHF|136-174	VHF	1
A64-OR4-440	SYNTHESIZER| RX/4E|UHF|406-470	UHF (406-430,450-470)	1
A64-OR4-500	SYNTHESIZER| RX/4E|UHF|470-512	UHF (470-520)	1

VHF / UHF RECEIVER MECHANICAL PARTS

Part Number	Description	Product	Qty
3702-62501010	CASE,14HP RF PLUG-IN,MT-4 RX	ALL	1
3702-10000120	FASTENER, QUICK RELEASE, GRAY	ALL	4
5630-12023250	GASKET, BeCu,3FINGER,.71",CLIP	ALL	2
3702-10000614	HANDLE, FRONT PANEL,14HP, GRAY	ALL	1
3702-33160011	HOLDER, PCB/EXTRACTOR,DIE CAST	ALL	1
5671-250N062B	HOLE PLUG,.250" HOLE,NYLON,BLK	ALL	1
5783-15110977	INSULATOR,2.56" x 2.27",MYLAR	ALL	1
5814-3M0LK00S	LOCKWASHER,SPLIT, M3, A2 STEEL	ALL	4
5813-2M5SQ50S	NUT, M2.5, SQUARE-5mm, A2 S/S	ALL	2
3702-63001101	PANEL,REAR,POS1,14HP EXTRUSION	ALL	1
3802-61001082	PANEL/FRONT,W/IDENT: MT-4E RX	ALL	1
3802-12141030	PLATE,BLACK IDENT,14HP MODULES	ALL	1
5812-2M5PP12S	SCREW,M2.5 x12 PAN/PHILLIPS,A2	ALL	1
5812-2M5FP14S	SCREW,M2.5x14 FLAT/PHILLIPS,A2	ALL	2
5812-2M5PP06S	SCREW,M2.5 X 6,PAN/PHILLIPS,A2	ALL	4
5812-2M5VP08S	SCREW,M2.5x8,OVAL CS/PHILPS,A2	ALL	1
5812-3M0PP08T	SCREW, M3x8,PAN/PHILLIPS,BLACK	ALL	4
5812-3M0VP08S	SCREW,M3 x8,OVAL C/S,PHILPS,A2	ALL	6

VHF / UHF RECEIVER P25 DIGITAL FIRMWARE

Part Number	Description	Product	Qty
APP-FIRM01-W-XX	FIRMWARE/MT4E RX/DIGITAL UPGRD	ALL	1

VHF / UHF RECEIVER AND TRANSMITTER ENCRYPTION

Part Number	Description	Product	Qty
ENCRYPT-MT4E-XX1	ENCRYPTION/MODULE, P25	ALL	1

VHF / UHF RECEIVER AND TRANSMITTER RSS

Part Number	Description	Product	Qty
APP-RSS03-WC-XX	RADIO SERVICE SOFTWARE|USB|4E	ALL	1

VHF / UHF TRANSMITTER ELECTRICAL PARTS

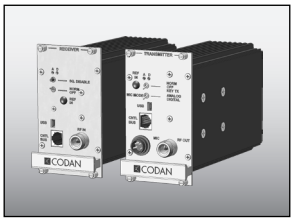
Part Number	Description	Product	Qty
A21-TX4E-MAIN-01	TX MAIN| MT4E|136-869MHz|W/UDB	ALL	1
A11-UDB	UNIVERSAL DAUGHTER BOARD	ALL	1
A21-VPA155-08	POWER AMP,TX2-8W,VHF130-180MHz	VHF	1
A23-UPA450-08	POWER AMP,TX 8W,UHF 380-520MHz	UHF	1
A64-OT4-150	SYNTHESIZER| TX/4E|VHF|136-174	VHF	1
A64-OT4-440	SYNTHESIZER| TX/4E|UHF|406-470	UHF (406-470)	1
A64-OT4-500	SYNTHESIZER| TX/4E|UHF|470-512	UHF (470-512)	1

VHF / UHF TRANSMITTER MECHANICAL PARTS

Part Number	Description	Product	Qty
A89-MIC4-08	CABLE/CONN ASSY,MICROPHONE CON	ALL	1
7910-WP0WP011	CABLE,SMB PLUG-PLUG,RG316,11cm	ALL	1
3702-62502010	CASE,14HP RF PLUG-IN,MT-4 TX	ALL	1
3702-10000120	FASTENER, QUICK RELEASE, GRAY	ALL	4
5630-12023250	GASKET, BeCu,3FINGER,.71",CLIP	ALL	2
3702-10000614	HANDLE, FRONT PANEL,14HP, GRAY	ALL	1
5671-250N062B	HOLE PLUG,.250" HOLE,NYLON,BLK	ALL	1
5783-15110977	INSULATOR,2.56" x 2.27",MYLAR	ALL	1
5814-3M0LK00S	LOCKWASHER,SPLIT, M3, A2 STEEL	ALL	4
5813-2M5SQ50S	NUT, M2.5, SQUARE-5mm, A2 S/S	ALL	2
3802-12141030	PLATE,BLACK IDENT,14HP MODULES	ALL	1
3702-63002101	PANEL,REAR,POS4,14HP EXTRUSION	ALL	1
3802-61002121	PANEL,FRONT, W/IDENT: MT-4E TX	ALL	1
5812-2M5PP12S	SCREW,M2.5 x12 PAN/PHILLIPS,A2	ALL	2
5812-2M5FP14S	SCREW,M2.5x14 FLAT/PHILLIPS,A2	ALL	2
5812-2M5PP06S	SCREW,M2.5 X 6,PAN/PHILLIPS,A2	ALL	4
5812-3M0PP06S	SCREW, M3 X 6, PAN/PHILLIPS,A2	ALL	2
5812-3M0PP08T	SCREW, M3x8,PAN/PHILLIPS,BLACK	ALL	4
5812-3M0VP08S	SCREW,M3 x8,OVAL C/S,PHILPS,A2	ALL	6
5812-5M0FP08S	SCREW, M5 x 8,FLAT/PHILLIPS,A2	ALL	4

VHF / UHF TRANSMITTER P25 DIGITAL FIRMWARE

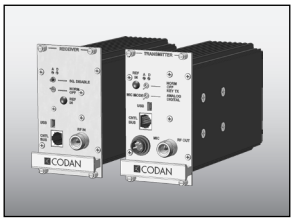
Part Number	Description	Product	Qty
APP-FIRM02-W-XX	FIRMWARE/MT4E TX/DIGITAL UPGRD	ALL	1



700 / 800 / 900MHZ RECEIVERS & TRANSMITTERS

The following section provides information that is primarily for 700 / 800 / 900MHz Receiver and Transmitter modules.

For specific information on VHF and UHF Receiver and Transmitter modules, see the VHF and UHF section.



700 / 800 / 900MHZ HARDWARE TUNING & TROUBLESHOOTING

REPAIR NOTE

MT-4E Receivers and Transmitters employ a high percentage of surface mount components which should not be removed or replaced using an ordinary soldering iron. Removal and replacement of surface mount components should be performed only with specifically designed surface mount rework and repair stations complete with Electrostatic Discharge (ESD) protection.

When removing surface mount solder jumpers, it is recommended to use solder wick braid in place of vacuum-type de-soldering tools. This will help prevent damage to the circuit boards.

RECOMMENDED TEST EQUIPMENT

Alignment of the 700 / 800 / 900MHz Receiver and Transmitter requires the following test equipment or its equivalent.

Power supply – Regulated +9.5VDC at 2A	Phillips PM 2811
Power Supply – Regulated +13.8VDC at 2A	Topward TPS-4000
Oscilloscope / Multimeter	Fluke 97 Scopemeter
Current Meter	Fluke 75 Multimeter
Communications Service Monitor (Analog)	Marconi Instruments 2965A or equivalent
Communications Service Monitor (P25 Digital and Analog)	IFR 2975
Alignment Tools	Codan A-TK-04

It is recommended that the radio communications test set be frequency-locked to an external reference (WWVH, GPS, Loran C) so that the high-stability local oscillator may be accurately set to within its ± 1ppm frequency tolerance.

Complete Receiver and Transmitter Alignment

A complete 700 / 800 / 900MHz Receiver and Transmitter alignment is performed at the factory and should not be required under normal circumstances. A large change in 700 / 800 / 900MHz receiver or transmitter operating frequency, as discussed in the next section, or a replacement of major receiver or transmitter subassembly modules, may require RSS reprogramming for the receiver or a complete realignment operation for the transmitter.

RECEIVER ASSEMBLY

The UDB, synthesizer and optional encryption module are mounted on the receiver mainboard. The 700 / 800 / 900MHz Preselector is attached with four screws through a front and a back bracket. An extruded aluminum shell that slides over the receiver assembly forms an enclosure. The enclosure is completed by the installation of side and front panel screws (see Figure 24: 700 / 800 / 900MHz Receiver Exploded View Diagram and Figure 25: 700 / 800 / 900MHz Receiver Top Component View).

RECEIVER FREQUENCY CHANGE

The receiver is initially aligned at the factory for the frequency stamped on the “Factory Set Operating Frequency” label (see Figure 23). This label should list the frequency at which the last complete receiver alignment was performed.

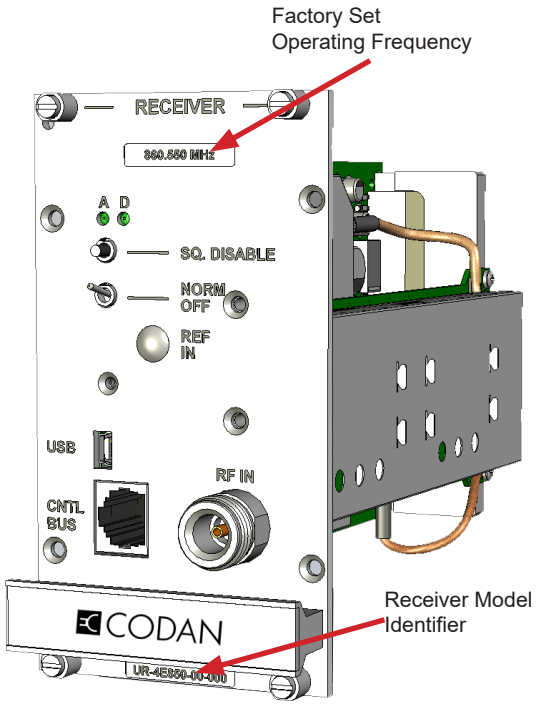


FIGURE 23: MT-4E Receiver Labeling

Change the frequency using the RSS. Verify that the receiver sensitivity is ≈ -116dBm for 12dB SINAD.

No additional tuning is required.

700 / 800 / 900MHZ RECEIVER EXPLODED VIEW DIAGRAM

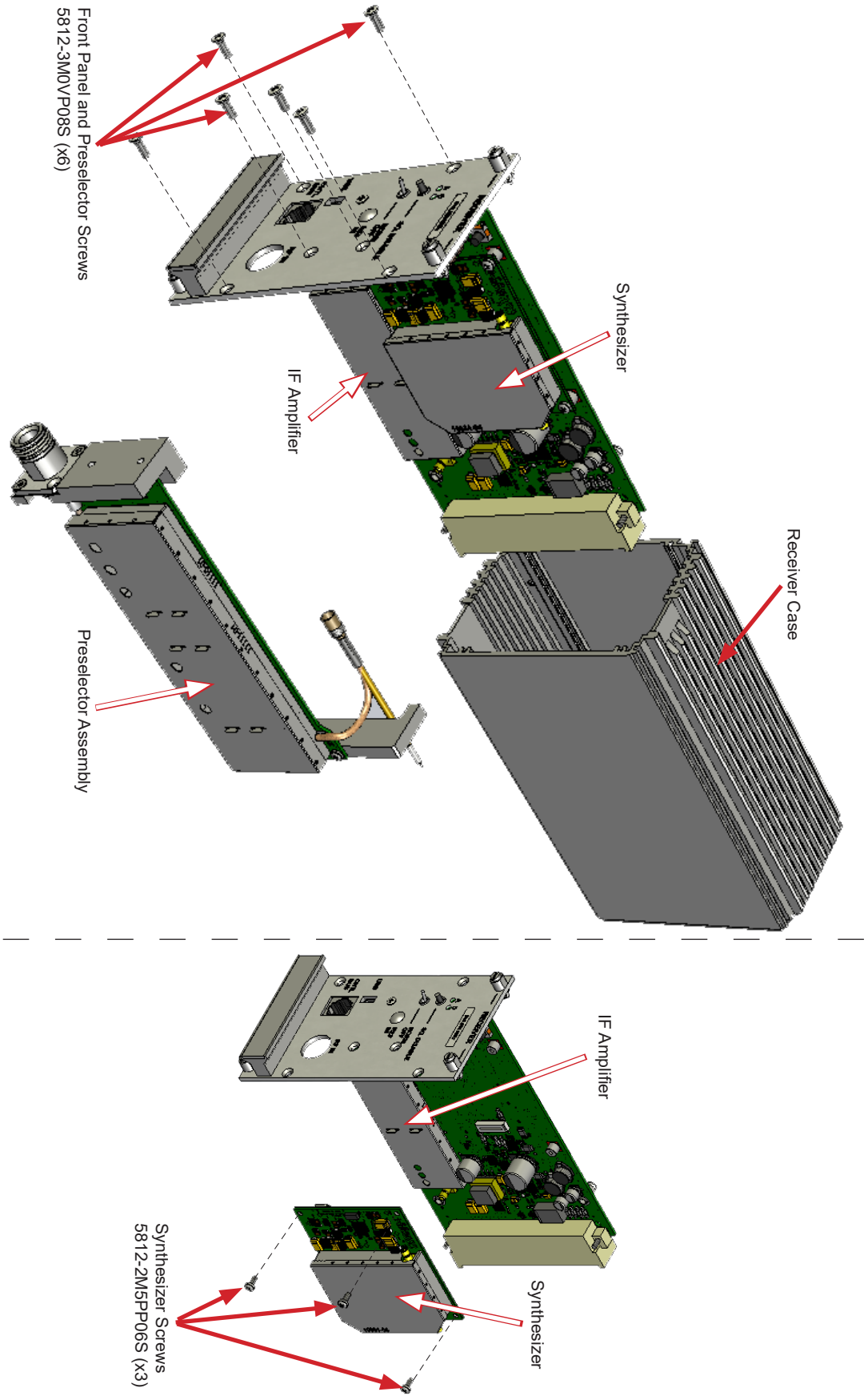


FIGURE 24: 700 / 800 / 900MHz Receiver Exploded View Diagram

700 / 800 / 900MHZ RECEIVER TOP COMPONENT VIEW

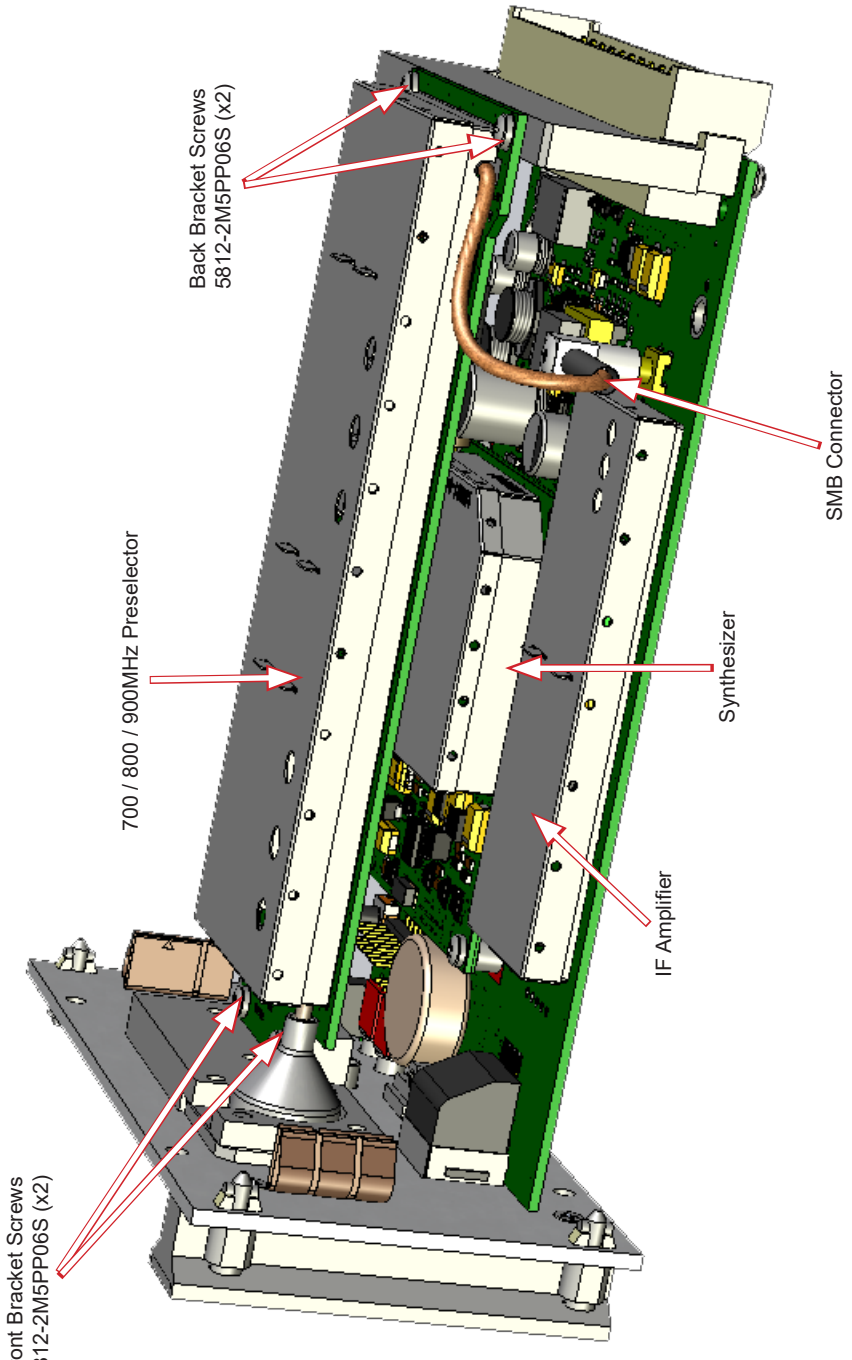


FIGURE 25: 700 / 800 / 900MHz Receiver Top Component View

TRANSMITTER ASSEMBLY

The UDB, synthesizer and optional encryption module are mounted to the transmitter mainboard. The RF Amplifier is mounted on an aluminum heatsink with the mainboard mounted on top (see Figure 27: 700 / 800 / 900MHz Transmitter Exploded View Diagram, Figure 28: 700 / 800 / 900MHz Transmitter Top Component View and Figure 29: 700 / 800 / 900MHz Transmitter Bottom Component View). An extruded aluminum shell slides over this assembly to complete the case.

This shell also serves as a heatsink to remove heat from the amplifier module. It is important that the four screws that bond the shell to the amplifier module be installed before prolonged operation of the transmitter. Moreover, the surface of the amplifier module that contacts the shell should be clean and free of foreign material.

Transmitter hardware alignment is performed solely on the amplifier. Alignment is simplified by using a SR-3 Subrack, SM-3 System Regulator, and extender card or cable to provide transmitter power and signal interconnection.

TRANSMITTER FREQUENCY CHANGE

The transmitter is initially aligned at the factory for the frequency stamped on the “Factory Set Operating Frequency” label (see Figure 26). This label should list the frequency at which the last complete transmitter alignment was performed.

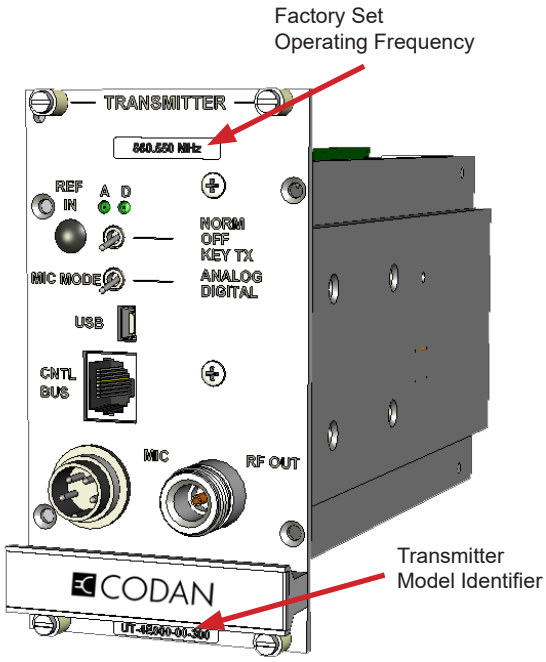


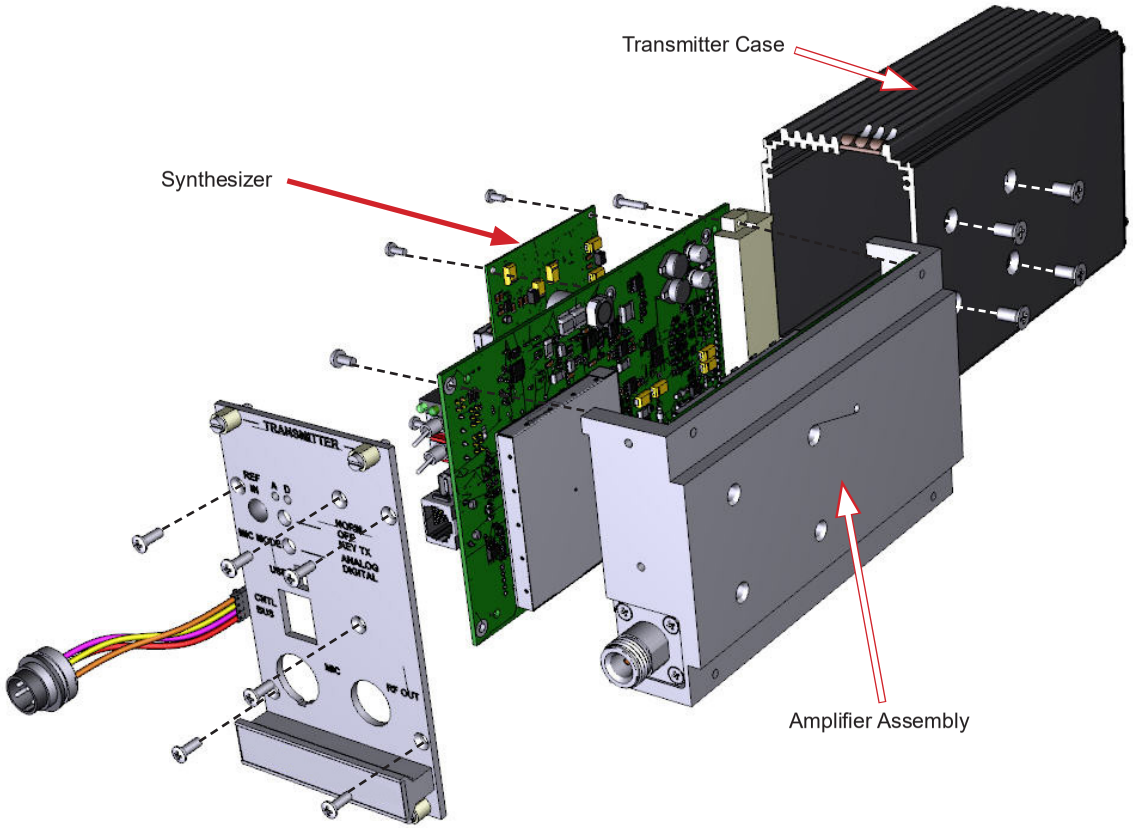
FIGURE 26: MT-4E Transmitter Labels

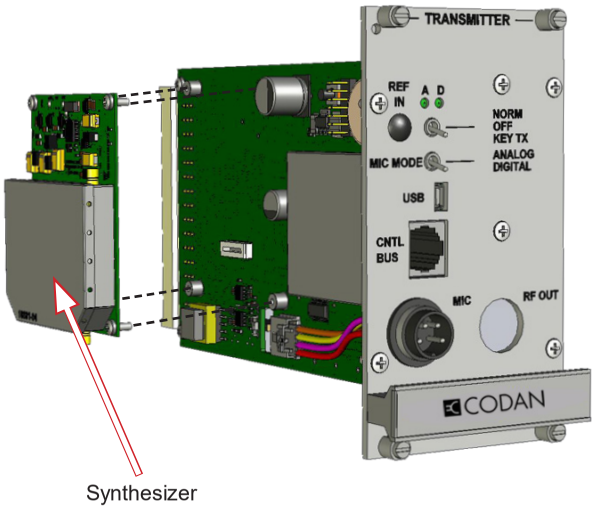
A small frequency change of less than ± 0.5MHz from the frequency stamped on the frequency label will generally not require any adjustment.

Changes greater than ± 0.5MHz from a previously tuned working transmit frequency will require VSWR alarm / overload alignment using the Transmitter Amplifier Tuning Procedure outlined below (after Figure 29). Failure to perform a realignment after a large frequency change could result in unreliable transmitter VSWR alarm / overload operation.

When the VSWR alarm / overload circuit is properly set, the amplifier is protected from excessive antenna VSWR by reducing the amplifier’s gain when an overload condition occurs. If the VSWR alarm is not set, the amplifier is still protected by a current limiter circuit. The circuit limits amplifier current draw to 2.0A and protects the amplifier from damage.

700 / 800 / 900MHZ TRANSMITTER EXPLODED VIEW





NOTE: The transmitter PCB colour and board number may be different from what is shown here.

FIGURE 27: 700 / 800 / 900MHz Transmitter Exploded View

700 / 800 / 900MHZ TRANSMITTER TOP COMPONENT VIEW

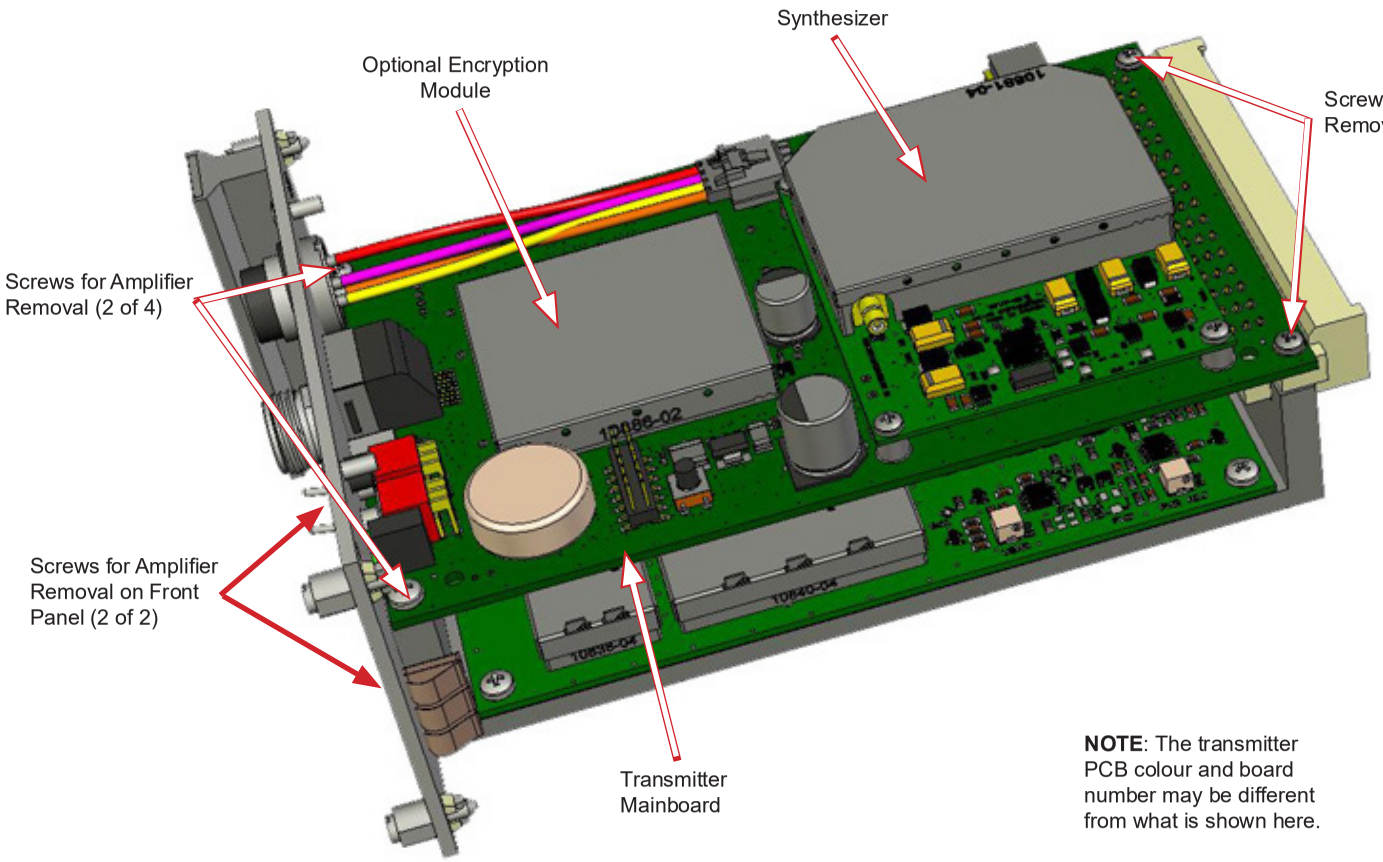
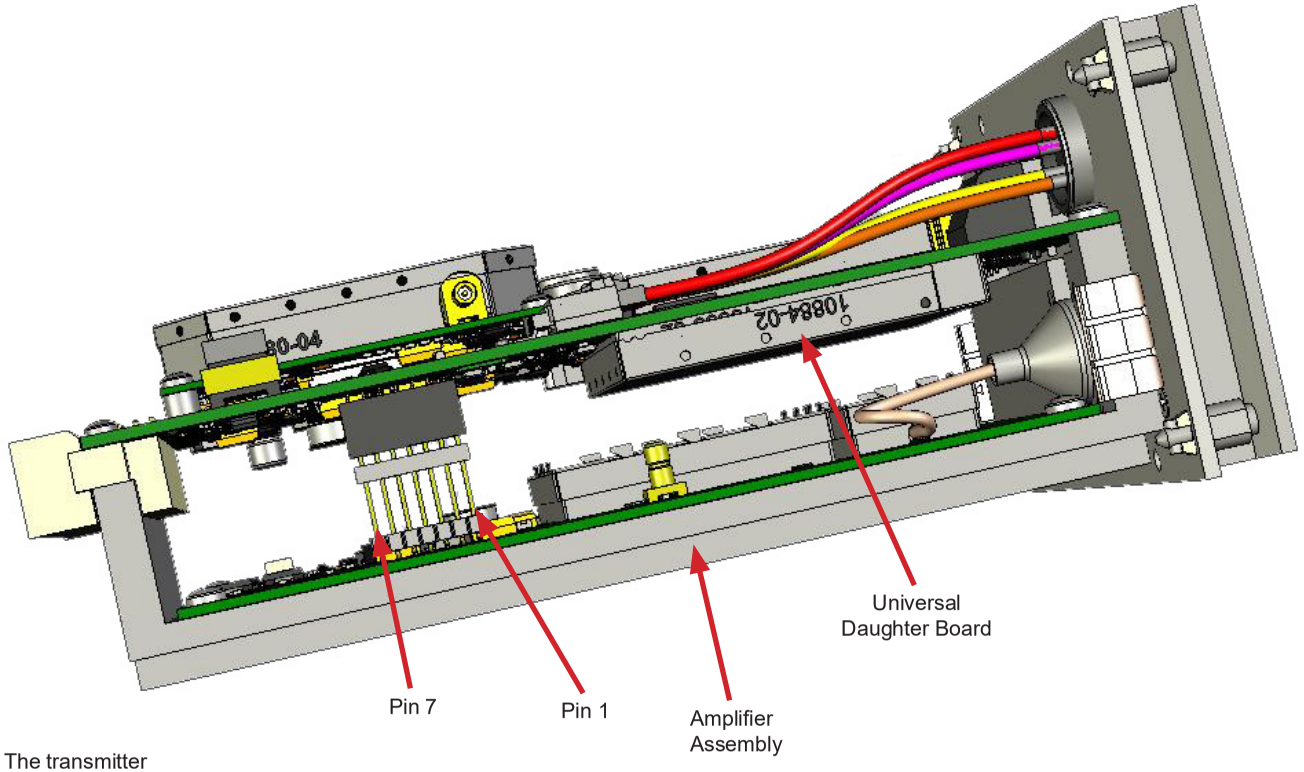


FIGURE 28: 700 / 800 / 900MHz Transmitter Top Component View

700 / 800 / 900MHZ TRANSMITTER BOTTOM COMPONENT VIEW



NOTE: The transmitter PCB colour and board number may be different from what is shown here.

FIGURE 29: 700 / 800 / 900MHz Transmitter Bottom Component View

TRANSMITTER AMPLIFIER TUNING PROCEDURE

The transmitter amplifier requires 0dBm of input power and is continuously adjustable over its power range (see the 700 / 800 / 900MHz Transmitter Performance Specifications data sheet). The amplifier provides Output Power and Antenna VSWR Alarm outputs (open drain; active low) through the transmitter to the motherboard. The amplifier's output power level and alarm level can be set without detaching the amplifier from the transmitter board.

Transmitter Amplifier Adjustment

The RF output and VSWR alarm threshold level adjustments are easily accessible so that fine adjustments can be made in the field. The forward power alarm does not require alignment. Depending on user requirements, the VSWR alarm threshold level should be checked whenever a significant change in operating frequency (± 0.5MHz) is made. As the antenna VSWR alarm is dependent on the output power, the output power should always be set first. The order of adjustment should be:

1. Set the output power.
2. Set the desired transmitter VSWR alarm level.
3. Test the transmitter overload.
4. Verify the transmitter current draw.

Initial Set-Up

Step	Reference	Action	Desired Results	Notes
1.	Transmitter RF output connector	Connect to radio communications test set.		Use Type-N cable.
2.	Potentiometers RV1 and RV3	Turn fully counter clockwise (CCW).		Initial starting point.
3.	MT-4E RSS	Click **Key Tx**. Adjust slider.	Desired Transmitter RF Output Power	Transmitter RSS Service > Power Levels

Transmitter VSWR Alarm (Reverse Power)

Step	Reference	Action	Desired Results	Notes
1.	Transmitter	Disconnect the test set and terminate the transmitter with a 3:1 mismatch load.		
2.	Connector J1	Connect a voltmeter to Pin 5 of connector J1.		
3.	RV3	Adjust clockwise (CW).	Until Pin 5 reads +2.5VDC, or a noticeable drop (10% of total curent or more) in current on the +13.8VDC line occurs.	
4.	Transmitter RF Output Connector	Connect to radio communications test set.	Pin 5 should read approximately 0VDC.	Use short section of low-loss 50Ω coaxial cable.

Transmitter Overload Test

Step	Reference	Action	Desired Results	Notes
1.	Amplifier	Disconnect radio communications test set.	Amplifier terminated with an open circuit.	
2.	+13.8VDC Supply	Monitor current.	Noticeable drop (10% of total current or more) in +13.8VDC current.	
3.	Amplifier	Reconnect radio communications test set.	+13.8VDC current should return to previous level.	

Transmitter Current Draw Verification

Step	Reference	Action	Desired Results	Notes
1.	+13.8VDC supply		Current < 2.0A	When transmitting.
2.	+9.5VDC supply		Current < 0.2A	When transmitting.

RECEIVER JUMPERS

700 / 800 / 900MHz Receiver Mainboard

PCB 50171

Jumper	Default Position	Function / Description
JU1	OUT	Enable Clear Keys 1*
JU2	OUT	Enable Clear Keys 2*
JU3	IN	Enable Power to Discriminator Circuit
JU4	OUT	Enable Pullup Resistor on RX_Mode
JU5	OUT	Bypass 330nF Audio Cap
JU6	IN	Enable Pullup Resistor on RX_Sec / Clr
JU7	OUT	Enable Power to Discriminator LPF Circuit
JU8	OUT	Enable Pullup Resistor on COR
JU9	X	Balanced Audio Transformer (X=Active, Y=Bypass)
JU10	X	Balanced Audio Transformer (X=Active, Y=Bypass)
JU11	IN	Enable Power to Balanced Audio Amplifier
JU12	OUT	Increase Balanced Audio Gain by 2.7dB
JU13	OUT	Disable 600Ω Audio Output; Enable Low Impedance Audio Output
JU14	X	Balanced Audio Transformer (X=Active, Y=Bypass)
JU15	X	X: Synchronous Y: Asynchronous
JU16	X	X: Synchronous Y: Asynchronous
JU17	X	X: Synchronous Y: Asynchronous
JU18	IN	Synthesizer Programming
JU19	IN	Synthesizer Programming
JU20	IN	Synthesizer Programming
JU21	IN	Enable 3.3V Power to IF VREF
JU22	IN	For factory testing purposes only (Isolate XTAL filter stage from circuit)
JU23	IN	Jumper for D&D Testing (Disconnect SSI Data Line)
JU24	IN	Jumper for D&D Testing (Disconnect SSI Data Line)
JU25	IN	Jumper for D&D Testing (Disconnect SSI Data Line)
JU26	IN	Jumper for D&D Testing (Disconnect SPI Data Line)
JU27	IN	Jumper for D&D Testing (Disconnect SPI Data Line)
JU28	IN	Jumper for D&D Testing (Disconnect SPI Data Line)
JU29	IN	Jumper for D&D Testing (Disconnect SPI Data Line)
JU30	Y	X: ENCRYPT-MT4E-XX Selects which UCM installed Y: ENCRYPT-MT4E-XX1

*Installed only if encryption is enabled.

Jumpers are shown on the 700 / 800 / 900MHz Receiver Jumper and Test Point Locator Illustrations (see Figure 30 and Figure 31).

700 / 800 / 900MHZ RX TOP JUMPER & TEST POINT LOCATOR

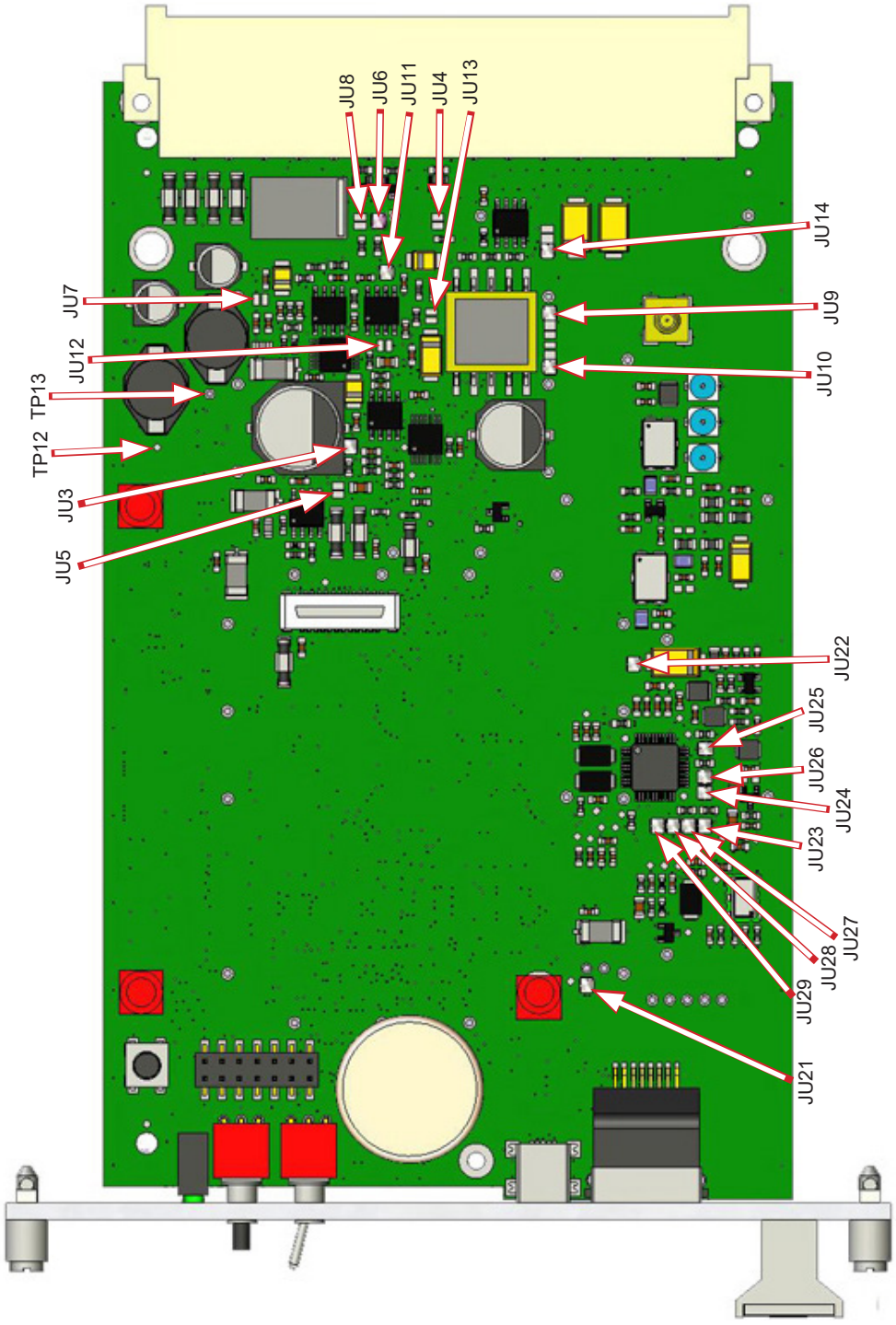


FIGURE 30: 700 / 800 / 900MHz Receiver Top Jumper & Test Point Locator

700 / 800 / 900MHZ RX BOTTOM JUMPER & TEST POINT LOCATOR

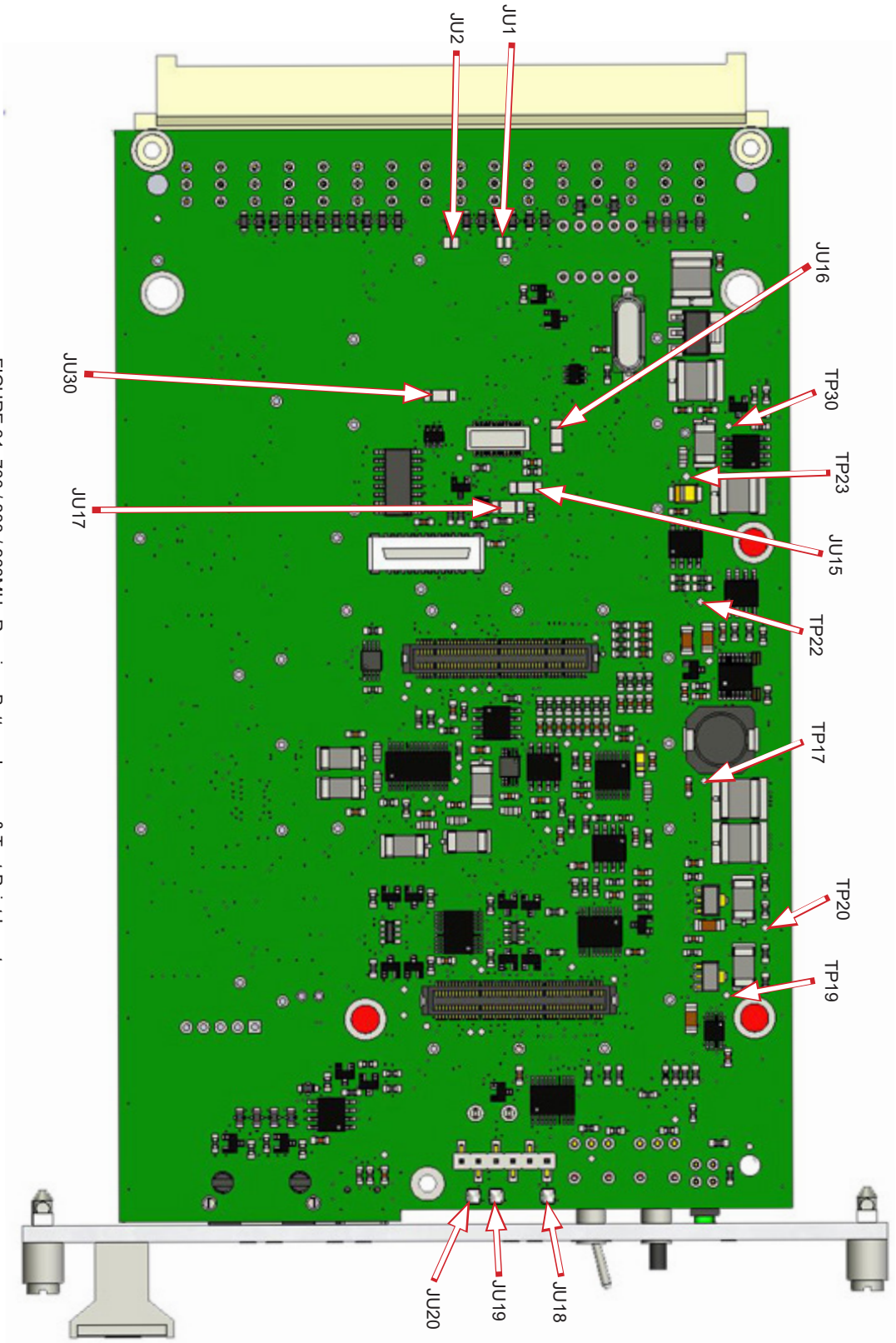


FIGURE 31: 700 / 800 / 900MHz Receiver Bottom Jumper & Test Point Locator

TRANSMITTER JUMPERS

VHF / UHF Transmitter Mainboard

PCB 50156

Jumper	Default Position	Function / Description
JU1	X	X: A/D Front Panel Switch selects transmitter A/D mode Y: A/D External Input selects transmitter A/D mode
JU2	Y	X: MIC OUT connects to Microphone Audio Input Y: MIC IN connects to Microphone Audio input
JU3	X	X: 600Ω Audio Transformer Y: Bypass Transformer
JU4	IN	When installed selects 600Ω resistance to ground
JU5	OUT	When installed enables Clear Keys 1 Input*
JU6	OUT	When installed enables Clear Keys 2 Input*
JU7	X	X: 600Ω Audio Transformer Y: Bypass Transformer
JU8	X	X: 600Ω Audio Transformer Y: Bypass Transformer
JU9	OUT	When installed bypasses AC coupling on the Subtone / Direct Modulation input.
JU10	X	X: ENCRYPT-MT4E-XX Selects which UCM installed Y: ENCRYPT-MT4E-XX1
JU11	OUT	Changes gain for TCXO / MOD input to synthesizer
JU12	X	X: Direct MOD input Select source for the Spare input. Y: Spare
JU13	X	X: Spare Select source for the Spare input. Y: DSP Bypass
JU14	OUT	When installed disables AC coupling of direct MOD input.

*Installed only if encryption is enabled.

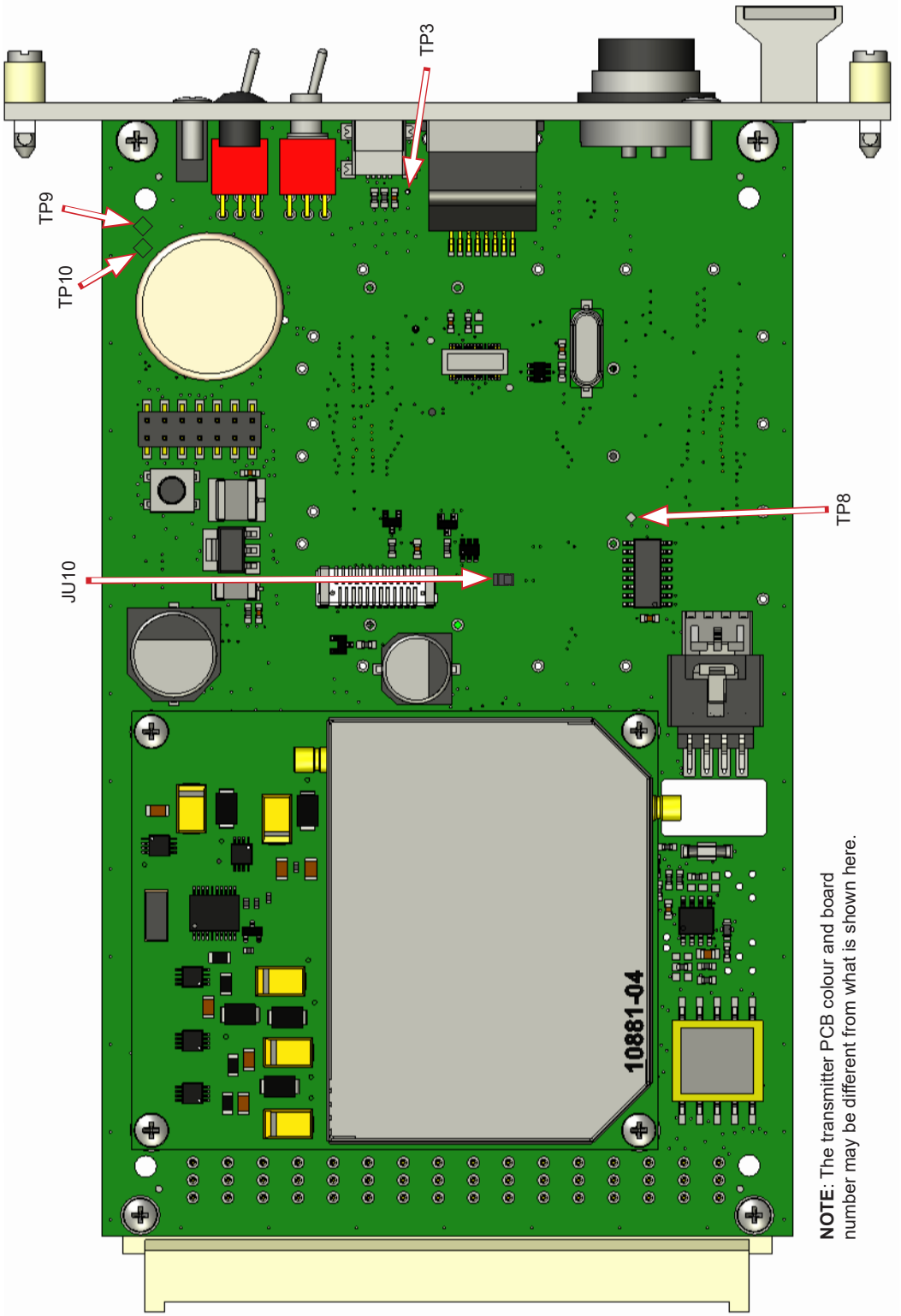
Jumpers are shown on the 700 / 800 / 900MHz Transmitter Jumper and Test Point Locator illustrations (see Figure 32 and Figure 33).

FRONT PANEL RJ45 JACK PINOUTS

A single, 8-position RJ45 jack is mounted on the front panel of the receiver and transmitter. The following are the connections on the RJ45 jack.

Pin	Signal
1	CRYPTO MODULE KEYLOAD
2	UNUSED SPARE
3	ANALOG COR / PTT
4	LVDS DATA A
5	LVDS DATA B
6	DIGITAL COR / PTT
7	NO CONNECTION
8	NO CONNECTION
Shield	GROUND

700 / 800 / 900MHZ TX TOP JUMPER & TEST POINT LOCATOR



NOTE: The transmitter PCB colour and board number may be different from what is shown here.

FIGURE 32: 700 / 800 / 900MHz Transmitter Top Jumper & Test Point Locator

700 / 800 / 900MHZ TX BOTTOM JUMPER & TEST POINT LOCATOR

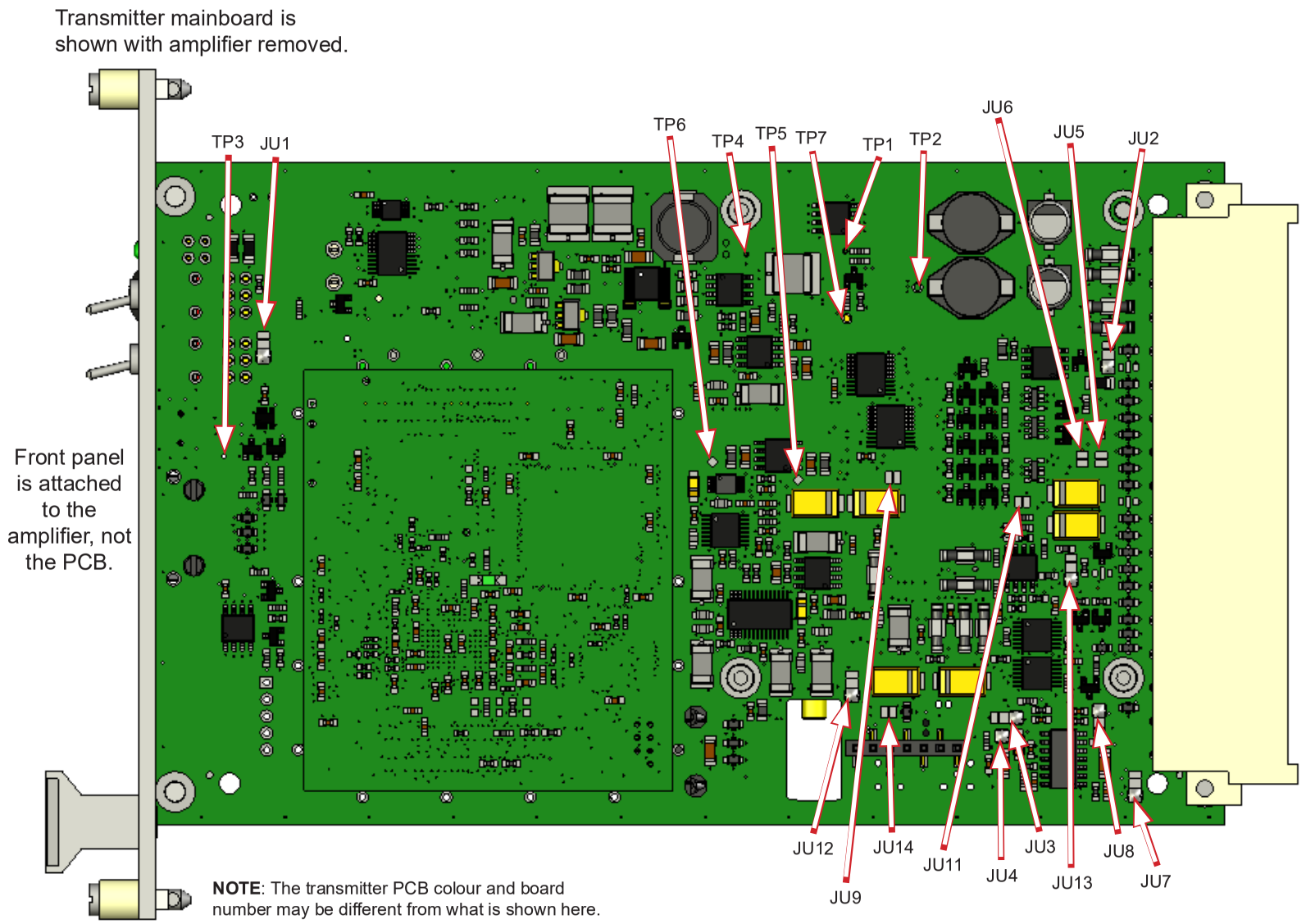


FIGURE 33: 700 / 800 / 900MHz Transmitter Bottom Jumper & Test Point Locator

RECEIVER TROUBLESHOOTING

The receiver is composed of four main components; the receiver mainboard, the synthesizer module, the RF Preselector and the universal daughter board (UDB). An optional decryption module can be installed in the receiver.

The following steps will help you determine the most likely subassembly that contains a fault and may help you determine why the fault is occurring.

NOTE: The subassembly or the entire receiver can be sent back to Codan Service department for repair. Contact the Codan Service department before returning any product.

Before testing, the receiver should be in the subrack or connected to the subrack by means of an Extender Card or kit. The main power to the system (+13.8VDC) and the +9.5VDC regulated supply should be checked via the front panel of the system regulator.

Testing the Receiver

1. Flip the receiver front panel ON/OFF switch from the OFF position to the NORM position.
2. Check the front panel LED lights.
   Typically the front panel LED lights will turn ON for less than 10 seconds and then turn OFF. On occasion these lights may stay on for up to one minute.

   If this is not the case, the most likely fault is on the UDB, on the synthesizer, or on the receiver mainboard. Make sure that the UDB and the synthesizer module are properly seated on the receiver mainboard.
3. Connect the receiver to the PC with the USB cable and run the Radio Service Software (RSS).
4. Select **[P25 or Analog]** Receiver.
5. Click **Configure**.
6. Click **Read** to read the receiver configuration.

   If the RSS returns "Device Not Responding," physically inspect the USB cable and connection, check the USB driver and check that the USB is recognized by the Device Manager. If the RSS still does not recognize the connection, the most likely fault is on the UDB.
7. Click **Rx ID**.
8. Click **Read** to read the Receiver ID.

   Confirm that the Frequency Band in the Synthesizer ID Model Number matches the Frequency Band in the Receiver Global Settings. If the settings do not match, the synthesizer will require replacement or the programming will need to be changed to match the synthesizer hardware.

If the receiver is still not functioning properly, the module level check should be conducted in the following order.

Receiver Mainboard

Check the following test points on the receiver mainboard:

Test Point	Signal Monitored	Typical Voltage
TP12	Voltage from motherboard +13.8VDC line	+13.8VDC**
TP13	Regulated +9.5VDC from motherboard	+9.5VDC ± 5%
TP17	+3.3VDC supply	+3.3VDC ± 5%
TP19	+1.2VDC supply	+1.2VDC ± 5%
TP20	+1.8VDC supply	+1.8VDC ± 5%
TP22	+2.5VDC supply	+2.5VDC ± 5%
TP23	+5.0VDC supply	+5.0VDC ± 5%
TP30	Switched +13.8VDC line from front panel switch	+13.8VDC**

Test Points are shown on the 700 / 800 / 900MHz Receiver Jumper and Test Point Locator illustrations (see Figure 30 and Figure 31).

If the test point voltages are within the specified range, the following tests further verify that the receiver mainboard may not be the source of problem.

Apply a -30dBm IF signal of frequency specified (see the 700 / 800 / 900MHz Receiver Performance Specifications data sheet) to the receiver mainboard IF Input SMB connector. If the signal is properly demodulated it proves that the receiver mainboard and UDB are in working condition. Proceed to troubleshooting the synthesizer and UDB module.

Synthesizer and UDB Module

The Synthesizer Module and UDB are bound together through their communication link. At this level of fault diagnosis it is not feasible to isolate the problem between the two units.

If the receiver front panel ON/OFF switch is set from the OFF position to the NORM position and the two front panel LEDs stay on, check that the synthesizer and the UDB are properly seated on the receiver mainboard and check the following test points on the UDB. To access the UDB, the shield lid covering the UDB will need to be removed.

Test Point	Signal Monitored	Typical Voltage
TP1	Ground	0VDC
TP2	Logic power	+3.3VDC ± 5%
TP3	DSP core supply	+1.2VDC ± 5%
TP4	CPLD core supply	+1.8VDC ± 5%

Test Points are shown on the 700 / 800 / 900MHz Receiver Jumper and Test Point Locator illustrations. See Figure 30 and Figure 31.

If the UDB voltages are not within the specified tolerances, it is most likely that the receiver mainboard (which supplies these voltages to the UDB) is faulty or there is a short within the UDB.

If the UDB voltages are within the specified tolerances, inject a -70dBm carrier on the receiver frequency into the RF input and check the RSSI meter reading in RSS Service mode. If the RSSI meter reading is low (< 30), perform the following test to verify that the synthesizer is locked on frequency:

Disconnect the LO output SMB connector from the synthesizer to the RF Preselector and connect the synthesizer LO output to a radio communication test set. Confirm that the synthesizer LO frequency matches the Target Synthesizer RF OUT found in the “Reference Oscillator” field of RSS Service mode. The LO output should be approximately 0 dbm.

The frequency of the synthesizer LO output will be above or below the receiver frequency by the IF frequency (see the 700 / 800 / 900MHz Receiver Performance Specifications data sheet). Check that this output is within tolerance for the local oscillator frequency stability (see the 700 / 800 / 900MHz Receiver Performance Specifications data sheet). Also ensure that the RF frequency will change with programmed channel changes in the RSS.

Receiver RF Preselector

Check the +9.5VDC power supply wire to the RF Preselector. The supply voltage should be +9.5VDC ± 5%. Inject a -30dBm carrier on the receiver frequency into the RF Input and check that the IF Output of the RF Preselector is equal to the specified IF frequency (see the 700 / 800 / 900MHz Receiver Performance Specifications data sheet) and it can be demodulated through the communication test set. If this is not the case, the synthesizer or the RF Preselector, or both, are faulty.

To further isolate the fault, the LO output can be disconnected from the synthesizer and an LO input can be applied from an external source to the RF Preselector. The frequency of the external LO input should be the Target Synthesizer RF OUT found in the “Reference Oscillator” field of RSS Service mode. Check the frequency output of the module using a communication test set.

TRANSMITTER TROUBLESHOOTING

The transmitter is composed of four main components: the transmitter mainboard, the synthesizer module, RF power amplifier and universal daughter board (UDB). An optional encryption module can also be installed in the transmitter.

The following steps will help you determine the most likely subassembly that contains a fault and may help you determine why the fault is occurring.

NOTE: The subassembly or the entire transmitter can be sent back to Codan Service department for repair. Contact the Codan Service department before returning any product.

Before testing, the transmitter should be in the subrack or connected to the subrack by means of an extender card or kit, and the main power to the system (+13.8VDC) and the +9.5VDC regulated supply should be checked via the front panel of the system regulator. The transmitter RF output should be terminated with a 50Ω load of appropriate wattage or communications test set.

Testing the Transmitter

1. Flip the transmitter front panel NORM/ OFF/ KEY TX switch from the OFF position to the NORM position .
2. Check the two front panel LED lights.
   Typically the front panel LED lights will turn ON for less than 10 seconds and then turn OFF. On occasion these lights may stay on for up to one minute.

If this is not the case or the transmitter is not functioning properly, the following steps can be taken:

3. Connect the transmitter to the PC with the USB cable.
4. Run the Radio Service Software (RSS).
5. Select [P25 or Analog] Transmitter.
6. Click **Configure**.
7. Click **Read** to read the transmitter configuration.

   If the RSS returns "Device Not Responding," physically inspect the USB cable and connection, check the USB driver and check the connection through the Device Manager. If the RSS still does not recognize the connection, the most likely fault is on the UDB.
8. Click **Tx ID**.
9. Click Read to read the transmitter ID.

   Confirm that the frequency band in the Synthesizer ID Model Number matches the frequency band in the Transmitter Global Settings. If the settings do not match, the synthesizer will require replacement or the programming will need to be changed to match the synthesizer hardware.

If the transmitter is still not functioning properly, the module level check should be conducted in the following order.

Transmitter Mainboard

Check the following test points on the transmitter mainboard:

Test Point	Signal Monitored	Typical Voltage
TP1	Voltage from Motherboard +13.8VDC Line	+13.8VDC**
TP2	Regulated +9.5VDC from Motherboard	+9.5VDC ± 5%
TP3	USB Connector Power	+3.0VDC or +5.0VDC
TP4	Switched +13.8VDC Line from Front Panel Switch	+13.8VDC**

Test Points are shown on the 700 / 800 / 900MHz Transmitter Jumper and Test Point Locator Illustrations (see Figure 32 and Figure 33).

If the test point voltages are within the specified range, it is most likely that the transmitter mainboard is not the source of problem. Proceed to Synthesizer and UDB Modules.

Synthesizer and UDB Modules

The synthesizer module and UDB are bound together through their communication link. At this level of fault diagnosis it is not feasible to isolate the problem between two units.

If the transmitter front panel NORM/ OFF/ KEY TX switch is set from the OFF position to the NORM position and the two front panel LEDs stay on for typically longer than ten seconds but occasionally longer than one minute, check that the synthesizer and the UDB are properly seated on the transmitter mainboard and check the following test points on the UDB. To access the UDB, the amplifier subassembly and the shield lid covering the UDB will need to be removed:

Test Point	Signal Monitored	Typical Voltage
TP1	Ground	0VDC
TP2	Logic power	+3.3VDC ± 5%
TP3	DSP core supply	+1.2VDC ± 5%
TP4	CPLD core supply	+1.8VDC ± 5%

Test Points are shown on the 700 / 800 / 900MHz Transmitter Jumper and Test Point Locator illustrations (see Figure 32 and Figure 33).

If the UDB voltages are not within the specified tolerances, the transmitter mainboard (that supplies these voltages to the UDB) is most likely faulty. If the UDB voltages are within the specified tolerances, it is likely that the synthesizer or the UDB, or both, are faulty.

If the transmitter front panel NORM/ OFF/ KEY TX switch is set from the OFF position to the NORM position; the two front panel LEDs both turn ON for typically less than ten seconds, but occasionally up to one minute and then turn OFF; the RF output power control is set at its maximum in the “Power Levels” field of RSS Service mode, but the RF output power is considerably less than 3.0 W:

- Ensure that Synthesizer Reference is set to Internal in the “Jumper Settings” field of RSS Service mode.
- Ensure that the output of the synthesizer module on the RF cable as shown on 700 / 800 / 900MHz Transmitter Bottom Component View (see Figure 29) is 0dBm (± 2dBm) and that the RF frequency will change with programmed channel changes in the RSS. If this is not the case, it is likely that the synthesizer or the UDB, or both, are faulty.

** Refer to the Specification Data Sheets (located in this manual) for the full Supply Voltage range.

Transmitter Amplifier

Set the front panel switch on the transmitter to KEY TX and check that the transmitter output power is approximately 0.5 to 3.0W.

Check the following pins on Connector J1 (while the transmitter is keyed):

Pin	Signal Monitored	Typical Voltage
1	RF Enable / Power Control (0.5 to 3.0W RF Power Output)	+0.5 to +3.0VDC
2	Regulated +9.5VDC from motherboard	+9.5VDC ± 5%
3	Voltage from motherboard +13.8VDC line	+13.8VDC**

Pins are shown on the 700 / 800 / 900MHz Transmitter Bottom Component View (see Figure 29).

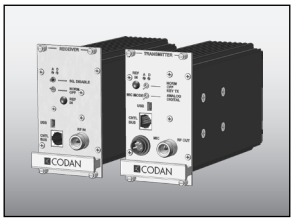
If Pin 1 is not within tolerance, the most likely fault is in the UDB or synthesizer module. If Pins 1, 2 and 3 are within the specified tolerances and there is no RF output power, the most likely fault is in the amplifier.

To test the RF power amplifier subassembly, set the output power to maximum using “Power Level Adjustments” in RSS Service mode, and drive the input of the amplifier with a 0dBm RF signal. Check that the RF power output is 3.0W when the front panel switch is set to KEY TX.

SYSTEM TROUBLESHOOTING

If the radio system is not receiving or transmitting on the right frequency, check the Channel and Bank Select lines of the receiver and transmitter modules. The motherboard has jumpers that allow one of 16 channels to be selected on a receiver or transmitter. The factory default is to have them all set to ground the channel select lines CSEL0 through CSEL3, forcing the radio to operate on Channel 1. On the receiver and transmitter there is also a Bank A/B line which usually floats high – selecting Bank A. The receiver and transmitter have separate control of their channel select lines, so changing receiver channels does not affect the transmitter channel.

If the Radio Service Software is failing partway through a read or write of the receiver, check that the receiver is not squelching and unsquelching repeatedly during the read or write operation. Disconnect the RF signal source from the receiver during the read or write operation.



700 / 800 / 900MHZ HARDWARE INTERFACING

RECEIVER AND TRANSMITTER INTERFACING

Interface connection to the MT-4E Receiver and Transmitter is available through the 48-pin motherboard interface connectors. These connectors allow interfacing to the audio, COR-PTT, Channel / Bank Select signals and many other signal lines in the receiver and transmitter.

Easy access to most of these signal lines can be achieved through the use of the optional A-PNL-AUX96-3 Auxiliary Connector, which uses plug-in screw connectors to secure 22AWG to 16AWG wires for external interface. Some of the interface lines are accessible through both jumpers and solder points on the motherboard. More information is available in the Subrack / Motherboard Instruction Manual.

Interfacing is not recommended to the RJ45 jack on the front of the receiver and transmitter modules.

Schematic diagrams are included at the end of this chapter (see Figure 34: 700 / 800 / 900MHz RX Interface Components Schematic Diagram and Figure 36: 700 / 800 / 900MHz TX Interface Components Schematic Diagram).

700 / 800 / 900MHz Receiver (see Figure 35) and 700 / 800 / 900MHz Transmitter (see Figure 37) block diagrams are also included in this manual.

700 / 800 / 900MHZ RX INTERFACE COMPONENTS SCHEMATIC

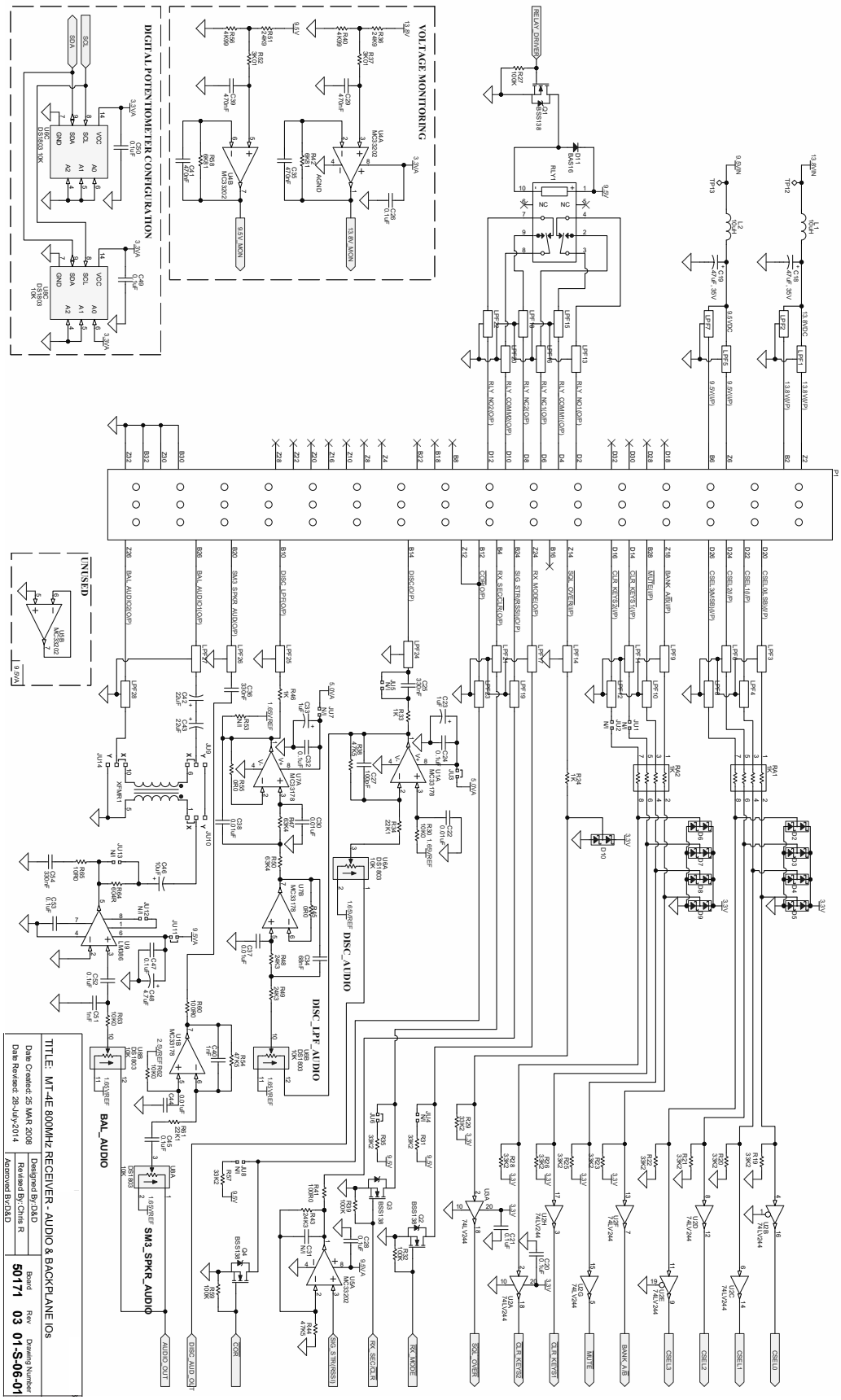


FIGURE 34: 700 / 800 / 900MHz Receiver Interface Components Schematic

700 / 800 / 900MHZ RECEIVER BLOCK DIAGRAM

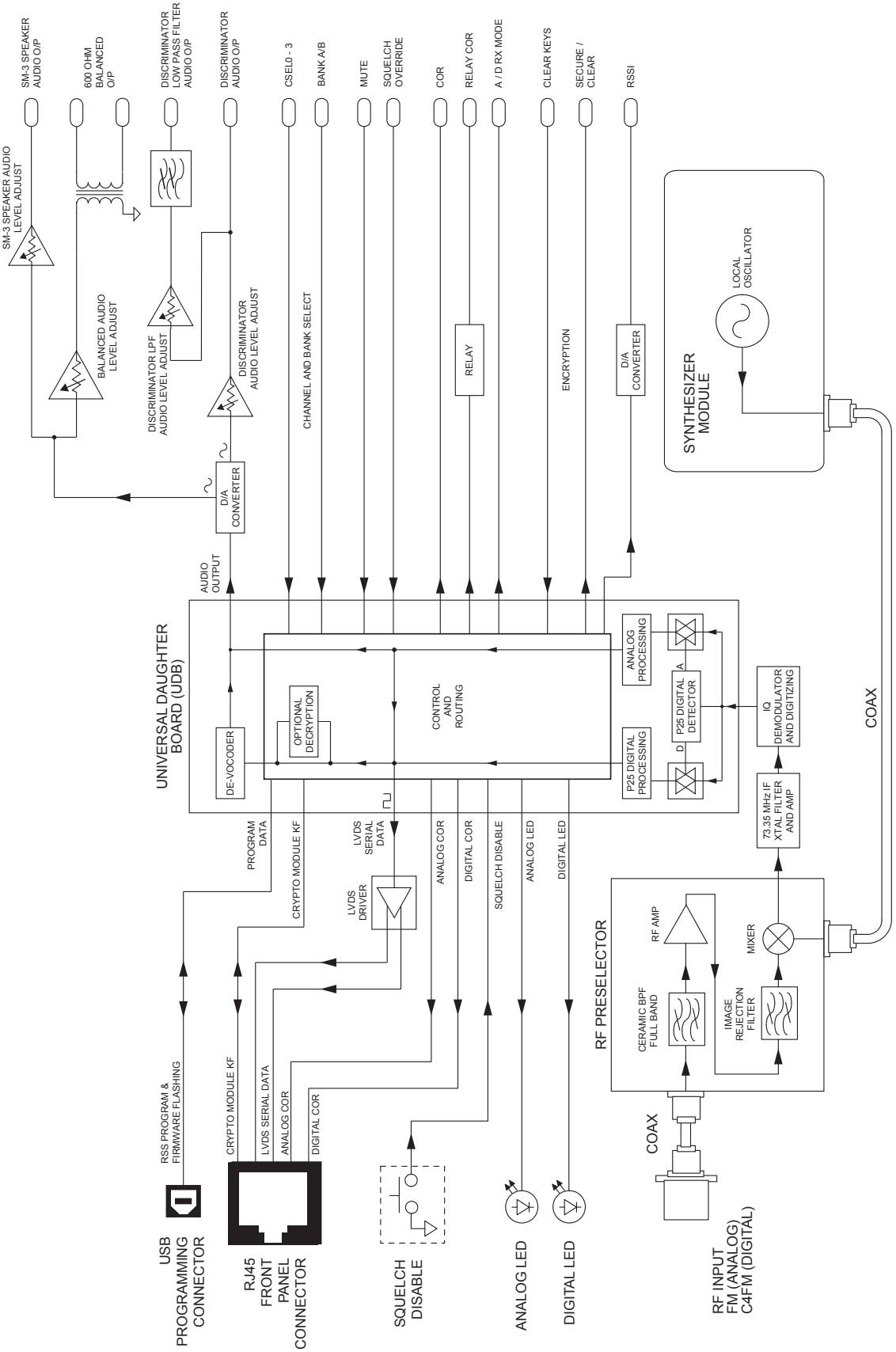


FIGURE 35: 700 / 800 / 900MHz Receiver Block Diagram

700 / 800 / 900MHZ TX INTERFACE COMPONENTS SCHEMATIC

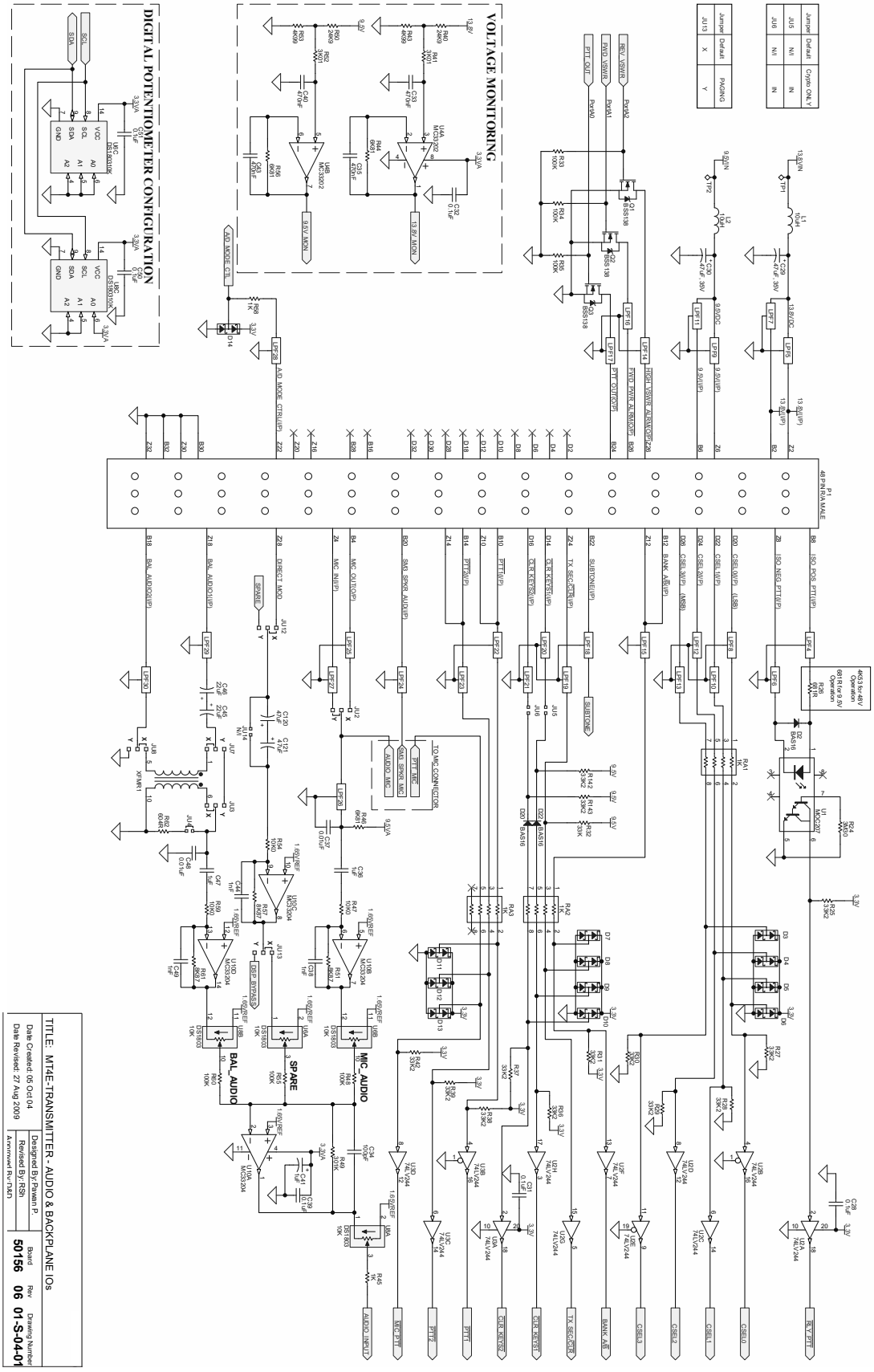


FIGURE 36: 700 / 800 / 900MHz Transmitter Interface Components Schematic

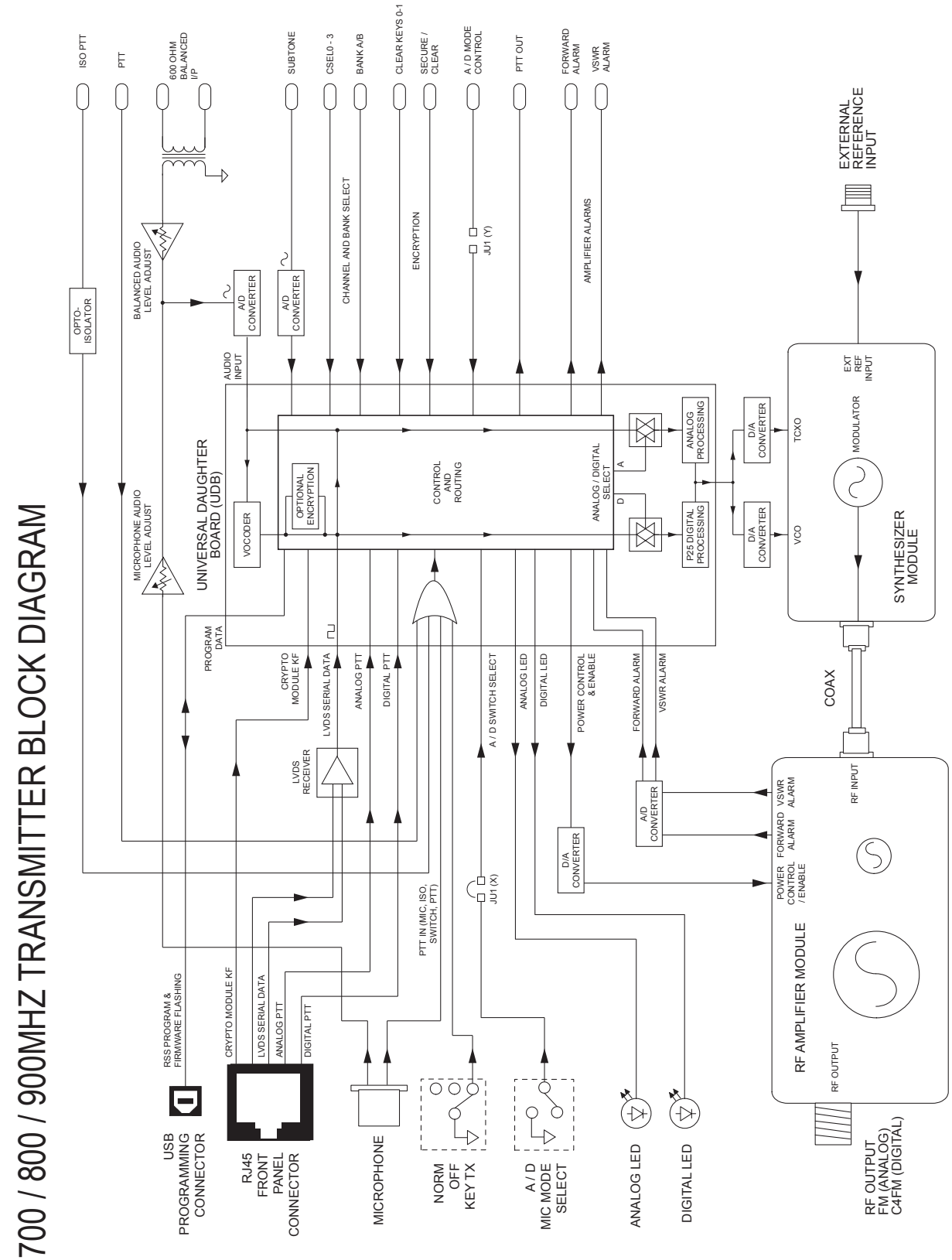


FIGURE 37: 700 / 800 / 900MHz Transmitter Block Diagram

48-PIN MOTHERBOARD INTERFACE CONNECTOR

A 48-pin connector is used for interfacing the MT-4E Motherboard to the receiver and transmitter.

Receiver

Pin	Name	Pin	Name	Pin	Name
D2	Relay Normally Open 1 [O/P]	B2	+13.8VDC [I/P]	Z2	+13.8VDC [I/P]
D4	Relay Common 1 [O/P]	B4	Receive Secure / Clear* [O/P]	Z4	No Connect
D6	Relay Normally Closed 1 [O/P]	B6	+9.5VDC [I/P]	Z6	+9.5VDC [I/P]
D8	Relay Normally Closed 2 [O/P]	B8	No Connect	Z8	No Connect
D10	Relay Common 2 [O/P]	B10	Discriminator LPF [O/P]	Z10	No Connect
D12	Relay Normally Open 2 [O/P]	B12	COR* [O/P]	Z12	COR* [O/P]
D14	Clear Keys 1* [I/P]	B14	Discriminator [O/P]	Z14	Squelch Override* [I/P]
D16	Clear Keys 2* [I/P]	B16	No Connect	Z16	No Connect
D18	No Connect	B18	No Connect	Z18	Bank A/B* [I/P]
D20	Channel Select 0 (LSB) [I/P]	B20	SM-3 Speaker Audio [O/P]	Z20	No Connect
D22	Channel Select 1 [I/P]	B22	No Connect	Z22	No Connect
D24	Channel Select 2 [I/P]	B24	Signal Strength (RSSI) [O/P]	Z24	RX Mode [O/P]
D26	Channel Select 3 (MSB) [I/P]	B26	Balanced Audio 1 [O/P]	Z26	Balanced Audio 2 [O/P]
D28	No Connect	B28	Mute* [I/P]	Z28	No Connect
D30	No Connect	B30	Ground	Z30	Ground
D32	No Connect	B32	Ground	Z32	Ground

Transmitter

Pin	Name	Pin	Name	Pin	Name
D2	No Connect	B2	+13.8VDC [I/P]	Z2	+13.8VDC [I/P]
D4	No Connect	B4	Microphone Audio Output [O/P]	Z4	Microphone Audio Input [I/P]
D6	No Connect	B6	+9.5VDC [I/P]	Z6	+9.5VDC [I/P]
D8	No Connect	B8	Isolated Positive PTT [I/P]	Z8	Isolated Negative PTT [I/P]
D10	No Connect	B10	PTT* [I/P]	Z10	PTT* [I/P]
D12	No Connect	B12	Bank A/B* [I/P]	Z12	Bank A/B* [I/P]
D14	Clear Keys 1* [I/P]	B14	PTT* [I/P]	Z14	PTT* [I/P]
D16	Clear Keys 2* [I/P]	B16	No Connect	Z16	No Connect
D18	No Connect	B18	Balanced Audio 2 [I/P]	Z18	Balanced Audio 1 [I/P]
D20	Channel Select 0 (LSB) [I/P]	B20	SM-3 Speaker Audio [I/P]	Z20	No Connect
D22	Channel Select 1 [I/P]	B22	Subtone [I/P]	Z22	A*/D Mode Control [I/P]
D24	Channel Select 2 [I/P]	B24	PTT Output* [O/P]	Z24	Transmit Secure / Clear* [I/P]
D26	Channel Select 3 (MSB) [I/P]	B26	Forward Power Alarm* [O/P]	Z26	High VSWR Alarm* [O/P]
D28	No Connect	B28	No Connect	Z28	Direct Modulation
D30	No Connect	B30	Ground	Z30	Ground
D32	No Connect	B32	Ground	Z32	Ground

* Indicates an active low signal.

RECEIVER PIN INTERFACE INFORMATION

COR Relay Outputs

D2	Relay Normally Open 1 [O/P]
D4	Relay Common 1 [O/P]
D6	Relay Normally Closed 1 [O/P]
D8	Relay Normally Closed 2 [O/P]
D10	Relay Common 2 [O/P]
D12	Relay Normally Open 2 [O/P]

The COR dual output is a general purpose output that can be used to drive various connection requirements, such as E&M. This output can be enabled and disabled in RSS.

The COR outputs are accessible through solder points on the motherboard. Solder connections for motherboard Version 50016-02 are labelled J41 through J46 for Receiver A and J35 through J40 for Receiver B. Solder connections for motherboard Version 50016-03 are labelled J17, J20, J24, J29, J36 and J40 for Receiver A and J16, J22, J26, J28, J37 and J42 for Receiver B.

Motherboard Version 50016-07 and newer contain three jumpers (JU80, JU81 and JU82) that can be enabled in the A position to allow Receiver A Relay 1 outputs to be routed to the A-PNL-AUX96-3 Auxiliary Connector for easier access on Pins J6-9 (N/O), J6-2 (COM) and J6-6 (N/C).

COR Output

B12	COR* [O/P]

Z12	COR* [O/P]

The COR output is an active low indication of an incoming decoded RF signal (analog or digital). This output pulls to ground through an open collector output Q4. The COR output also has jumper JU8 that can be installed to provide a pull up through R39 to 9.5VDC.

Mute Input

B28	Mute* [I/P]

The Mute input is an active low input that will keep the receiver from unsquelching when grounded. This input is internally pulled high through R7 to 3.3VDC to allow the receiver to unsquelch when this line is not connected to an external input.

When the receiver RSS is configured for internal CTCSS or DCS decode, this line should not be used or a conflict between internal and external mute functions will occur.

Squelch Override Input

Z14	Squelch Override* [I/P]

The Squelch Override input is an active low input that will unsquelch the receiver to monitor any RF signal present on the receiver frequency. This input is internally pulled high through R11 to 3.3VDC when this line is not connected to an external input.

* Indicates an active low signal.

Channel Select Inputs

D20	Channel Select 0 (LSB) [I/P]
D22	Channel Select 1 [I/P]
D24	Channel Select 2 [I/P]
D26	Channel Select 3 (MSB) [I/P]

The Channel Select inputs are used as binary logic inputs that allow the receiver to change up to 16 channels. A pull to ground represents a Logic 0 and a pull to high (5.0VDC–9.5VDC) is a Logic 1. When there are no connections to these inputs, the channel select lines are internally pulled to ground.

Important Note: Channel Select lines must be set before the receiver unsquelches or the receiver will not change channel.

Bank A/B Input

Z18	Bank A/B* [I/P]

The Bank A/B input is used as a binary logic input that allows the receiver to change between two banks of 16 channels. A pull to ground represents a Logic 0 (Bank B) and a pull to high (5.0VDC--9.5VDC) is a Logic 1 (Bank A). When there are no connections to these inputs, the channel select lines are internally pulled to high (Bank A).

Important Note: The Bank A/B line must be set before the receiver unsquelches or the receiver will not change bank.

Clear Keys Inputs

D14	Clear Keys 1* [I/P]
D16	Clear Keys 2* [I/P]

The Clear Keys inputs are used to zeroize all encryption keys in a receiver with an optional decryption module installed. Jumpers JU1 and JU2 need to be installed in order to enable the Clear Keys inputs. The activation sequence to zeroize encryption keys is generated by the CI-BC-4E Base Controller when the zeroize keys push button is pushed.

NOTE: The duration of the key press must be greater than one second to ensure that the keys have been zeroized.

Receive Secure / Clear Output

B4	Receive Secure / Clear* [O/P]

The receive Secure / Clear output is a logic output that indicates whether the received signal is encrypted or clear. A logic low level (Ground) is an indication of a non-encrypted signal received.

* Indicates an active low signal.

When an encrypted signal has been received, this output will be floating or pulled to +9.5VDC through R17. The jumper JU6 will determine if the output is pulled up to +9.5VDC (JU6 IN) or floating (JU6 OUT).

When the receiver is not receiving, the signal line sits low (Ground). This signal line should be used in conjunction with the COR output.

Analog / Digital Receive Mode Output

Z24 RX Mode [O/P]

The receiver mode output is a logic output that indicates whether the received signal is analog or P25 Digital. A logic low level (Ground) is an indication of an analog signal received. When a P25 Digital signal has been received this output will be floating or pulled to +9.5VDC through R13. The jumper JU4 will determine if the output is pulled up to +9.5VDC (JU4 IN) or floating (JU4 OUT).

When the receiver is not receiving, the signal line sits low (Ground). This signal line should be used in conjunction with the COR output.

Receive Signal Strength Indicator (RSSI) Output

B24 Signal Strength (RSSI) [O/P]

The Receive Signal Strength Indicator (RSSI) output is a 0VDC to +5VDC voltage indication of the received RF signal strength. The RSS also has an RSSI meter built into the Service mode that gives a numerical representation from (0 to 5000) of the received RF signal strength. These values are not calibrated to any standard and may vary from receiver to receiver.

The graph below (see Figure 38) shows a typical representation of RF Signal Strength versus RSSI voltage. (Data results are from receiver PCB mainboard 50164-05 and Firmware Version 2.1.1.)

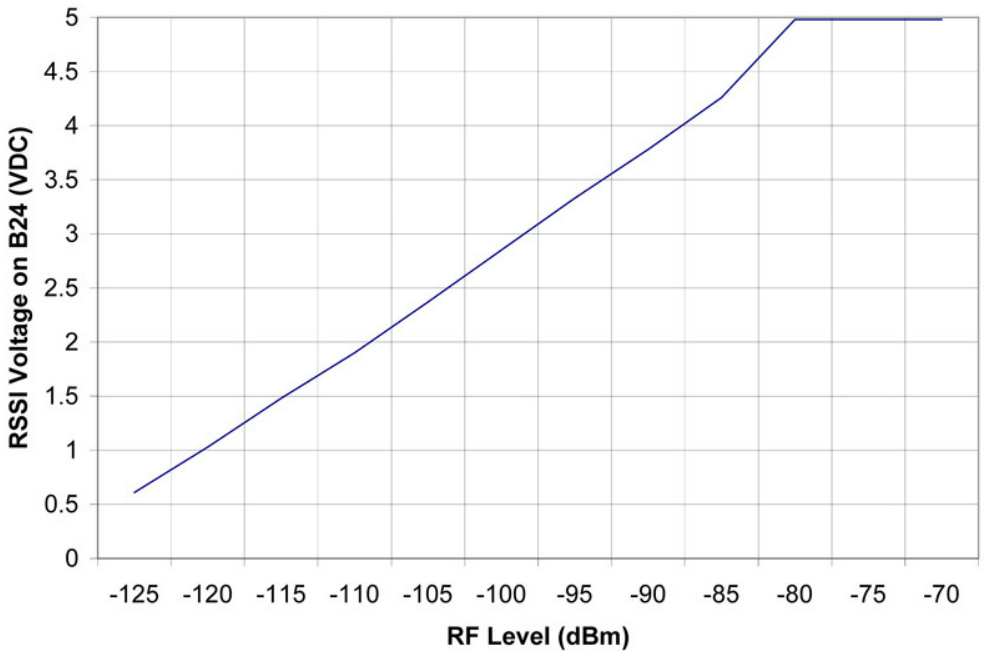


FIGURE 38: RF Signal Strength Versus RSSI Voltage

Discriminator Audio Output

B14	Discriminator [O/P]

The Discriminator Audio output is an audio output that is not gated by the squelch circuitry. Audio, CTCSS and DCS present on the received signal will be output. The Discriminator Audio output is typically used to connect to external CTCSS, DCS or DTMF decoder modules. The Discriminator Audio output level is adjustable via a softpot in the RSS.

Discriminator LPF Output

B10	Discriminator LPF [O/P]

The Discriminator Low Pass Filter (LPF) output is a 250Hz low pass filtered audio output that is not gated by the squelch circuitry. CTCSS and DCS present on the received signal will be output. The Discriminator LPF output is typically used to pass CTCSS tones directly through to the transmitter or to connect to external CTCSS or DCS decoder modules. The Discriminator LPF output level is adjustable via a softpot in the RSS.

SM-3 Speaker Audio Output

B20	SM-3 Speaker Audio [O/P]

The SM-3 Speaker Audio output is an unbalanced audio output that is used to drive the speaker audio on the SM-3 System Regulator. The SM-3 Speaker Audio output level is adjustable via a softpot in the RSS.

Balanced Audio Output

B26 Balanced Audio 1 [O/P]	Z26 Balanced Audio 2 [O/P]

The Balanced Audio output is a 600-ohm balanced (via transformer) audio output that is capable of various configurations. The Balanced Audio output can be configured for de-emphasis or flat audio response via RSS programming options. De-emphasis (6dB / octave) audio is the standard setting and is typically used for all voice communications. Flat audio (de-emphasis disabled) is typically used for analog data formats such as frequency shift keying (FSK).

The Balanced Audio output can be configured by setting jumpers JU9, JU10, JU13 and JU14 (JU9, JU10 and JU14 can be installed in position X or Y; JU13 is installed or not installed):

	JU9	JU10	JU13	JU14
Balanced 600 ohm with transformer (standard configuration)	X	X	OUT	X
Balanced low impedance with transformer	X	X	IN	X
Unbalanced 600 ohm without transformer	Y	Y	OUT	Y
Unbalanced low impedance without transformer	Y	Y	IN	Y

NOTE: Z26 Balanced Audio 2 [O/P] is grounded when the receiver is set for unbalanced.

The Balanced Audio output level is adjustable via a softpot in the RSS. Maximum audio output level for the 600-ohm balanced output configuration is +4.0dBm with a 1.0kHz tone @ 1.5kHz deviation at the receiver RF input.

Status Tone Encoder

The Balanced Audio output can be configured with a status tone encoder option for “Voting.” The Status Tone Encoder is not a standard feature; the receiver is ordered or upgraded to enable this custom feature.

In the “Extra Features” field of RSS Service mode, the frequency and amplitude of the status tone can be adjusted. The typical frequency of a status tone is 1950Hz or 2175Hz, but the frequency can be selected for any non-standard tone (300Hz–3000Hz). The status tone is active when the receiver is squelched and will revert to audio only when the receiver is unsquelched.

TRANSMITTER PIN INTERFACE INFORMATION

Isolated PTT Inputs

B8	Isolated Positive PTT [I/P]	Z8	Isolated Negative PTT [I/P]

The Isolated PTT (Push-To-Talk) input is a general purpose input for keying the transmitter that is capable of handling many different voltage inputs. To activate the Isolated PTT input, connect the positive voltage to pin B8 and the open collector output of the external device being used to activate the Isolated PTT will connect to pin Z8. When the open collector input is pulled to ground the transmitter PTT will be activated through U1.

When using lower voltages such as +9.5VDC on the positive pin, R26 will remain as a standard 681-ohm resistor. When the positive input is connected to higher voltage such as an E&M +48VDC or -48VDC, R26 will need to be changed to a 4K53 ohm resistor to prevent damage by limiting the current draw through U1.

PTT Inputs

B10	PTT* [I/P]	Z10	PTT* [I/P]
B14	PTT* [I/P]	Z14	PTT* [I/P]

The PTT inputs are the standard inputs for keying the transmitter. These inputs are typically controlled using an open collector input from a receiver COR output. A pull to ground will activate the transmitter PTT.

PTT Output

B24	PTT Output* [O/P]

The PTT output is a logic output that indicates when the transmitter is active (keyed). This is an open collector output that will pull to ground through Q3 when the transmitter is active.

Alarm Outputs

B26	Forward Power Alarm* [O/P]	Z26	High VSWR Alarm* [O/P]

The Alarm outputs are open collector outputs that will pull to ground with a failure condition. The Forward Power Alarm will pull to ground through Q2 with a condition of half power or less out of the transmitter amplifier. The high VSWR Alarm will pull to ground through Q1 with a high VSWR condition of 3:1 or more reflected power.

* Indicates an active low signal.

Channel Select Inputs

D20	Channel Select 0 (LSB) [I/P]
D22	Channel Select 1 [I/P]
D24	Channel Select 2 [I/P]
D26	Channel Select 3 (MSB) [I/P]

The Channel Select inputs are used as binary logic inputs that will allow the transmitter to change up to 16 channels. A pull to ground represents a Logic 0 and a pull to high (5.0VDC–9.5VDC) is a Logic 1. When there are no connections to these inputs, the channel select lines are internally pulled to ground.

Important Note: Set the Channel Select lines before the transmitter is keyed (PTT) or the transmitter will not change channel.

Bank A/B Inputs

B12	Bank A/B* [I/P]

Z12	Bank A/B* [I/P]

The Bank A/B inputs are used as a binary logic input that allows the transmitter to change between two banks of 16 channels. A pull to ground represents a Logic 0 (Bank B) and a pull to high (5.0VDC--9.5VDC) is a Logic 1 (Bank A). When there are no connections to these inputs, the channel select lines are internally pulled to high (Bank A).

Important Note: The Bank A/B line must be set before the transmitter is keyed (PTT) or the transmitter will not change bank.

Clear Keys Inputs

D14	Clear Keys 1* [I/P]
D16	Clear Keys 2* [I/P]

The Clear Keys inputs are used to zeroize all encryption keys in a transmitter with an optional encryption module installed. Jumpers JU5 and JU6 need to be installed in order to enable the Clear Keys inputs. The activation sequence to zeroize encryption keys is generated by the CI-BC-4E Base Controller when the zeroize keys push button is pushed.

NOTE: The duration of the key press must be greater than one second to ensure that the keys have been zeroized.

* Indicates an active low signal.

Transmit Secure / Clear Input

Z24 Transmit Secure / Clear* [I/P]

The Transmit Secure / Clear input is a logic control input for a transmitter in base station configuration with an encryption module installed. To transmit in secure mode, the Transmit Secure / Clear input needs to be pulled high using a 5.0VDC – 9.5VDC input. To transmit in clear mode, the Transmit Secure / Clear input needs to pulled low (Ground). If the input is not connected, the transmitter will transmit in clear mode.

Analog / Digital Mode Control Input

Z22 A*/D Mode Control [I/P]

The Analog / Digital Mode Control input provides added control over the transmitter when used in a base station configuration. To use this input, the transmitter must be programmed by the RSS for mixed mode.

If the transmitter is keyed up through the connections on the front panel RJ45 jack, the incoming PTT signal is separated into Analog PTT and Digital PTT. This is typically being driven by the mode the receiver received the RF signal in, and is used for most repeat functions. When the transmitter is keyed up by the 48-pin motherboard interface PTT or Isolated PTT, or by the Microphone PTT or test switch “Key TX” on the front panel, the transmitter mode needs to be selected. When JU1 is in the X position, the front panel “MIC MODE” switch position will determine the transmitter mode. When JU1 is in the Y position, a ground on the Analog / Digital Mode Control input will key the transmitter in analog mode. To key the transmitter in P25 Digital mode, leave the input open and the internal pull up Resistor R72 will provide the logic high required.

When the transmitter is configured for paging applications, JU1 will be in the Y position and “Paging**”** will be selected in RSS for the Transmitter Channel Type. With this paging configuration, the following changes occur:

- The front panel A/D LEDs no longer indicate the transmitter’s true A/D status.
- The front panel Analog / Digital switch SW2 is disabled.
- The A/D_MODE_CTRL (I/P) at Z22 is now externally controlled by the paging modulator. When this line goes ‘LOW’, the transmitter balanced input is turned off.

Important Note: The Analog / Digital Mode Control line must be set before the transmitter is keyed (PTT) or the transmitter will not change mode.

Balanced Audio Input

B18 Balanced Audio 2 [I/P]	Z18 Balanced Audio 1 [I/P]

The Balanced Audio input is a 600-ohm balanced (via transformer) audio input that is capable of various configurations. The Balanced Audio input can be configured for pre-emphasis or flat audio response via RSS programming options. Pre-emphasis (6dB / octave) audio is the standard setting and is typically used for all voice communications. Flat audio (pre-emphasis disabled) is typically used for analog data formats such as frequency shift keying (FSK).

The Balanced Audio input can be configured the following ways by setting jumpers JU3, JU4, JU7 and JU8 (JU3, JU7 and JU8 can be installed in position X or Y; JU4 is installed or not installed).

	JU3	JU4	JU7	JU8
Balanced 600 ohm with transformer (standard configuration)	X	IN	X	X
Balanced low impedance with transformer	X	OUT	X	X
Unbalanced 600 ohm without transformer	Y	IN	Y	Y
Unbalanced low impedance without transformer	Y	OUT	Y	Y

NOTE: B18 Balanced Audio 2 [I/P] is ground when the receiver is set for unbalanced.

The Balanced Audio input level is adjustable via a softpot in the RSS. Audio input level for the 600-ohm balanced input configuration is adjustable from -18dBm to +4.0dBm with a 1.0kHz tone @ 1.5kHz deviation at the transmitter RF output.

Microphone Audio Input / Output and SM-3 Speaker Audio Input

B4 Microphone Audio Output [O/P]	Z4 Microphone Audio Input [I/P]
B20 SM-3 Speaker Audio [I/P]	

The Microphone Audio input / output is an unbalanced high impedance audio input. Using jumper JU2 in the X position, the microphone audio output can be directly routed out to the subrack. Using jumper JU2 in the Y position, the microphone audio input can be used as a spare high-impedance unbalanced audio input. The SM-3 Speaker Audio input is used as an input for custom microphones.

The Microphone Audio input level is adjustable via a softpot in the RSS. Audio input level for the unbalanced high-impedance input is adjustable from -20dBm to +4.0dBm with a 1.0kHz tone @ 1.5kHz deviation at the transmitter RF output.

Subtone Input

B22 Subtone [I/P]

The Subtone input supports external modulation sources for CTCSS, DCS and Logic Trunked Radio (LTR®).

The RSS enables the Subtone input in the “Analog Settings” field of the MT-4E Transmitter configuration screen. Set the “Signaling” field to “External Input” on a per-channel basis.

The Subtone input level is adjustable via a softpot in the RSS. Level adjustments can also be made on the external encoder device. Subtone input level for the CTCSS configuration is -18dBm input level for a 100Hz tone @ 0.35kHz deviation (narrow band) or 0.50kHz deviation (wide band) at the transmitter RF Output.

CTCSS

Continuous Tone Coded Squelch System (CTCSS) can be injected from an external device rather than using the internal CTCSS generator in the RSS.

DCS / LTR®

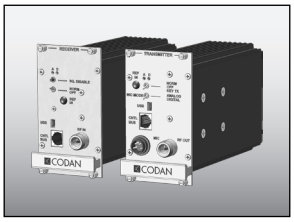
Digitally Coded Squelch (DCS) and LTR® can be connected to the transmitter Subtone input.

Direct Modulation Audio

Z28 Direct Modulation

The Direct Modulation Audio is configured for a flat audio response. This input can be used for audio or analog in-band data formats (such as 2400 baud FSK) or Flex™ paging formats. The Direct Modulation input should be AC coupled with JU14 (OUT – default) and JU13 in the Y position for paging applications. For formats such as FSK, JU13 can be set in the X position.

LTR® is a registered trademark of E.F. Johnson Co.
FLEX™ is a trademark of Motorola, Inc.



700 / 800 / 900MHZ PARTS LIST

700 / 800 / 900MHZ RECEIVER ELECTRICAL PARTS

Part Number	Description	Product	Qty
A14-RX4E-MAIN-01	RX|MB|MT4E|768-869MHz|INCL UDB	ALL	1
A11-UDB	UNIVERSAL DAUGHTER BOARD	ALL	1
A14-FE4-770	PRESELECTOR/4E|CLASS B|768-776		1
A15-FE4-800	PRESELECTOR/UR-4E| 799-824MHz		1
A15-FE4-850	PRESELECTOR/UR-4E| 851-869MHz		1
A16-FE4-950	PRESELECTOR/UR-4E| 930-960MHz		1
A64-OR4-800	SYNTHESIZER| RX/4E|UHF|764-960		1
5047-1RJ45JPS	JACK,SINGLE,RJ45,8POS,PCB,SHLD		1

700 / 800 / 900MHZ RECEIVER MECHANICAL PARTS

Part Number	Description	Product	Qty
3702-62501010	CASE,14HP RF PLUG-IN,MT-4 RX	ALL	1
3702-10000120	FASTENER, QUICK RELEASE, GRAY	ALL	4
5630-12023250	GASKET, BeCu,4FINGER,.71",CLIP	ALL	2
3702-10000614	HANDLE, FRONT PANEL,14HP, GRAY	ALL	1
3702-33160011	HOLDER, PCB/EXTRACTOR,DIE CAST	ALL	1
5671-250N062B	HOLE PLUG,.250" HOLE,NYLON,BLK	ALL	1
5783-15110977	INSULATOR,2.56” x 2.27”,MYLAR	ALL	1
5814-3M0LK00S	LOCKWASHER,SPLIT, M3, A2 STEEL	ALL	4
5813-2M5SQ50S	NUT, M2.5, SQUARE-5mm, A2 S/S	ALL	2
3702-63001101	PANEL,REAR,POS1,14HP EXTRUSION	ALL	1
3802-61001082	PANEL,FRONT,W/IDENT: MT-4E RX	ALL	1

700 / 800 / 900MHz Receiver Mechanical Parts (continued)

Part Number	Description	Product	Qty
3802-12141030	PLATE,BLACK IDENT,14HP MODULES	ALL	1
5812-2M5PP12S	SCREW,M2.5 x12 PAN/PHILLIPS,A2	800MHz, 700MHz	2
5812-2M5PP12S	SCREW,M2.5 x12 PAN/PHILLIPS,A2	850MHz	1
5812-2M5FP14S	SCREW,M2.5x14 FLAT/PHILLIPS,A2	ALL	2
5812-2M5PP06S	SCREW,M2.5 X 6,PAN/PHILLIPS,A2	ALL	4
5812-2M5VP08S	SCREW,M2.5x8,OVAL CS/PHILPS,A2	ALL	1
5812-3M0PP08T	SCREW, M3x8,PAN/PHILLIPS,BLACK	ALL	4
5812-3M0VP08S	SCREW,M3 x8,OVAL C/S,PHILPS,A2	800MHz, 700MHz	8
5812-3M0VP08S	SCREW,M3 x8,OVAL C/S,PHILPS,A2	850MHz	6
5917-7B4BM30T	STANDOFF, 7/32OD,1/4L,M3,SWAGE	ALL	2
5927-5SFBM25T	STANDOFF,5.5mm OD,4.5mm L,M2.5	ALL	3

700 / 800 / 900MHZ RECEIVER P25 DIGITAL FIRMWARE

Part Number	Description	Product	Qty
APP-FIRM01-W-XX	FIRMWARE/MT4E RX/DIGITAL UPGRD	ALL	1

700 / 800 / 900MHZ RECEIVER & TRANSMITTER ENCRYPTION

Part Number	Description	Product	Qty
ENCRYPT-MT4E-XX1	ENCRYPTION/MODULE, P25	ALL	1

700 / 800 / 900MHZ RECEIVER AND TRANSMITTER RSS

Part Number	Description	Product	Qty
APP-RSS03-WC-XX	RADIO SERVICE SOFTWARE|USB|4E	ALL	1

700 / 800 / 900MHZ TRANSMITTER ELECTRICAL PARTS

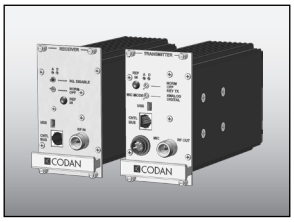
Part Number	Description	Qty
A21-TX4E-MAIN-01	TX MAIN| MT4E|136-869MHz|W/UDB	1
A11-UDB	UNIVERSAL DAUGHTER BOARD	1
A25-UPA835-03	POWER AMP| TX| 3W| 768-869MHz	1
A26-UPA930-03	POWER AMP,TX,3W,896-960MHz	1
A64-OT4-800	SYNTHESIZER| TX/4E|UHF|760-960	1

700 / 800 / 900MHZ TRANSMITTER MECHANICAL PARTS

Part Number	Description	Qty
A89-MIC4-08	CABLE/CONN ASSY|MICROPHONE CON	1
7910-WP0WP011	CABLE,SMB PLUG-PLUG,RG316,11cm	1
3702-62502010	CASE,14HP RF PLUG-IN,MT-4 TX	1
3702-10000120	FASTENER, QUICK RELEASE, GRAY	4
5630-12023250	GASKET, BeCu,4FINGER,.71",CLIP	2
3702-10000614	HANDLE, FRONT PANEL,14HP, GRAY	1
5671-250N062B	HOLE PLUG,.250" HOLE,NYLON,BLK	1
5783-15110977	INSULATOR,2.56" x 2.27",MYLAR	1
5814-3M0LK00S	LOCKWASHER,SPLIT, M3, A2 STEEL	4
5813-2M5SQ50S	NUT, M2.5, SQUARE-5mm, A2 S/S	2
3802-61002121	PANEL,FRONT, W/IDENT: MT-4E TX	1
3702-63002101	PANEL,REAR,POS4,14HP EXTRUSION	1
3802-12141030	PLATE,BLACK IDENT,14HP MODULES	1
5812-2M5FP14S	SCREW,M2.5x14 FLAT/PHILLIPS,A2	2
5812-2M5PP06S	SCREW,M2.5 X 6,PAN/PHILLIPS,A2	4
5812-2M5PP14S	SCREW,M2.5 x14 PAN/PHILLIPS,A2	2
5812-3M0PP06S	SCREW, M3 X 6, PAN/PHILLIPS,A2	2
5812-3M0PP08T	SCREW, M3x8,PAN/PHILLIPS,BLACK	4
5812-3M0VP08S	SCREW,M3 x8,OVAL C/S,PHILPS,A2	6
5812-5M0FP08S	SCREW, M5 x 8,FLAT/PHILLIPS,A2	4
5927-5SFBM25T	STANDOFF,5.5mm OD,4.5mm L,M2.5	4

700 / 800 / 900MHZ TRANSMITTER P25 DIGITAL FIRMWARE

Part Number	Description	Qty
APP-FIRM02-W-XX	FIRMWARE/MT4E TX/DIGITAL UPGRD	1



SPECIFICATION DATA SHEETS

This section contains detailed specifications for RF modules within the scope of this manual.

To verify performance and set-up test procedures, please refer to these data sheets when performing technical operations on the equipment.

VHF AND UHF RECEIVER PERFORMANCE SPECIFICATIONS

NOTE: 470–520MHz T-Band receiver specifications are **not included** in these VHF / UHF receiver specifications; see the separate 470–520MHz T-Band receiver specifications following.

General

Frequency Range:	136–174MHz (VHF) / 406–430MHz, 450–470MHz-(UHF)
Channel Spacing:	12.5, 15, 25 and 30kHz
Channel Selection:	VHF: 2.5kHz, 5.0kHz and 6.25kHz UHF: 6.25kHz
Number of Channels:	Preset capability for 2 banks of 16 channels
Channel Switching Range:	± 2.0MHz
Compatibility:	MT-4 Series Radio Systems; P25 interoperable*
System Impedance:	50Ω (Type-N connector)
Frequency Generation:	Internal Synthesizer
Reference Sensitivity:	≤ -118dBm (0.28uV) for 12dB SINAD / ≤ -118dBm (0.28uV) for 5% BER*
Local Oscillator Frequency Stability:	± 1.0ppm (VHF) / ± 0.5ppm (UHF)
Adjacent Channel Rejection (Selectivity):	Class A: ≥ 45dB; Narrow Band Analog ≥ 75dB; Wide Band Analog ≥ 60dB; Digital* Class B: ≥ 40dB; Narrow Band Analog ≥ 70dB; Wide Band Analog ≥ 60dB; Digital*
Intermodulation Rejection:	Class A: ≥ 75dB; Narrow Band and Wide Band Analog ≥ 80dB; Digital* Class B: ≥ 70dB; Narrow Band and Wide Band Analog ≥ 70dB; Digital*
Spurious Response Rejection:	Class A: ≥ 75dB; Narrow Band Analog ≥ 85dB; Wide Band Analog ≥ 90dB; Digital* Class B: ≥ 70dB; Narrow Band and Wide Band Analog ≥ 70dB; Digital*
Conducted Spurious Output Power:	Class A: ≤ -95dBm Class B: ≤ -57dBm
Hum and Noise Ratio:	≥ 34dB; Narrow Band Analog ≥ 40dB; Wide Band Analog

** P25 Digital specifications are applicable only for modules with the P25 Digital firmware upgrade.*

VHF and UHF Receiver Performance Specifications (Continued)

Audio Output:	600Ω balanced line output (configurable for unbalanced line); De-emphasis or Flat output, +3dBm maximum level
Audio Distortion:	Analog: ≤ 2.0% (25°C); ≤ 3.0% (-30˚C to +60˚C) Digital as per TIA / EIA 102.CAAB*
Front Panel Controls:	Receiver Power On (Norm) / Off Squelch Disable (Push button) Analog and Digital Receive LED indicators
COR Interface:	150mA, 50V open drain power MOSFET
Supply Voltage:	+13.8VDC Nominal (range +10 to +17VDC) +9.5VDC Regulated
Supply Current:	Class A: < 250mA; with no encryption module installed < 280mA; with encryption module installed Class B: < 115mA; with no encryption module installed < 145mA; with encryption module installed
Operating Temperature Range:	-30°C to +60°C
Operating Humidity:	95% RH (non-condensing) at +25°C
CTCSS Decode:	Programmable to any of 42 CTCSS tones
DCS Decode:	Programmable to any of 83 DCS sequences. Normal or inverted DCS can be selected.
IC Certification No.:	142A-VR4E150A (VHF Class A) 142A-UR4E420A (UHF Low Band Class A) 142A-UR4E460A (UHF High Band Class A) 142A-VR4E150 (VHF Class B) 142A-UR4E420 (UHF Low Band Class B) 142A-UR4E460 (UHF High Band Class B)
FCC ID:	n/a – Declaration of Conformity (DOC)

** P25 Digital specifications are applicable only for modules with the P25 Digital firmware upgrade.*

470–520MHZ T-BAND RECEIVER PERFORMANCE SPECIFICATIONS

General

Frequency Range:	470–520MHz
Channel Spacing:	12.5, 15, 25 and 30kHz
Channel Selection:	6.25kHz
Number of Channels:	Preset capability for 2 banks of 16 channels
Channel Switching Range:	± 2.0MHz
Compatibility:	MT-4 Series Radio Systems; P25 interoperable*
System Impedance:	50Ω (Type-N connector)
Frequency Generation:	Internal Synthesizer
Reference Sensitivity:	≤ -116dBm (0.35uV) for 12dB SINAD / ≤ -116dBm (0.35uV) for 5% BER*
Local Oscillator Frequency Stability:	± 0.5ppm
Adjacent Channel Rejection (Selectivity):	Class B: ≥ 40dB; Narrow band Analog ≥ 70dB; Wide Band Analog ≥ 60dB; Digital*
Intermodulation Rejection:	Class B: ≥ 70dB; Narrow band and Wide Band Analog ≥ 70dB; Digital*
Spurious Response Rejection:	Class B: ≥ 70dB; Narrow band and Wide Band Analog ≥ 70dB; Digital*
Conducted Spurious Output Power:	≤ -80dBm
Hum and Noise Ratio:	≥ 34dB; Narrow band Analog ≥ 40dB; Wide Band Analog
Audio Output:	600Ω balanced line output (configurable for unbalanced line); De-emphasis or Flat output, +3dBm maximum level
Audio Distortion:	Analog: ≤ 2.0% (25°C); ≤ 3.0% (-30˚C to +60˚C) Digital as per TIA / EIA 102.CAAB*
Front Panel Controls:	Receiver Power On (Norm) / Off Squelch Disable (Push button) Analog and Digital Receive LED indicators
COR Interface:	150mA, 50V open drain power MOSFET
Supply Voltage:	+13.8VDC Nominal (range +10 to +17VDC) +9.5VDC Regulated
Supply Current:	< 115mA; with no encryption module installed < 145mA; with encryption module installed
Operating Temperature Range:	-30°C to +60°C
Operating Humidity:	95% RH (non-condensing) at +25°C
CTCSS Decode:	Programmable to any of 42 CTCSS tones
DCS Decode:	Programmable to any of 83 DCS sequences. Normal or inverted DCS can be selected.
IC Certification No.:	n/a
FCC ID:	n/a - Declaration of Conformity (DOC)

** P25 Digital specifications are applicable only for modules with the P25 Digital firmware upgrade.*

VHF AND UHF TRANSMITTER PERFORMANCE SPECIFICATIONS

NOTE: 470–520MHz transmitter specifications are **not included** in these VHF / UHF transmitter specifications; see separate 470–520MHz T-Band transmitter specifications following.

General

Frequency Range:	136–174MHz, 406–470MHz
Carrier Frequency Stability:	± 1.0ppm (VHF) / ± 0.5ppm (UHF)
Channel Spacing:	12.5, 15, 25 and 30kHz
Channel Selection:	VHF: 2.5kHz, 5.0kHz and 6.25kHz UHF: 6.25kHz
Number of Channels:	Preset capability for 2 banks of 16 channels
Compatibility:	MT-4 Series Radio Systems; P25 interoperable*
RF Output Power:	0.5 to 8.0W Continuous
Emission Designators:	Analog: 11K0F3E (Narrow Band); 16K0F3E (Wide Band) Digital Paging: 9K2F1D P25 Digital: 8K10F1E (Digital Voice); 8K10F1D (Digital Data)*
System Impedance:	50Ω; Type-N connector
Duty Cycle:	100%; Continuous operation
Undesired Emissions: (Adjacent Channel Power Ratio)	< -70dBc, Analog < -70dBc, Digital*
Undesired Emissions: (Conducted Spurious)	< -70dBc @ 8.0W RF Output
Intermodulation Attenuation:	≥ 45dB Analog ≥ 45dB Digital*
VSWR Protection:	< 20:1 VSWR at all phase angles
Operating Temperature Range:	-30˚C to +60˚C
Operating Humidity:	95% RH (non-condensing) at +25°C
Operating Voltage:	+13.8VDC Nominal (range +10 to +17VDC) +9.5VDC Regulated
Transmit Current:	< 1.7A @ 2.0W RF Output < 2.8A @ 8.0W RF Output
Standby Current:	<50mA (no encryption installed) <80mA (with encryption modules installed)
Front Panel Controls:	Transmitter Power On (Norm) / Off / Key Tx Mic Mode: Analog / Digital Analog and Digital Transmit LED indicators
PTT Time-Out-Timer:	Programmable from 15 to 465 sec. (in increments of 15 sec., ± 5 sec.) or infinity (default 300 sec.) using Radio Service Software.
Audio Input:	600Ω balanced or unbalanced input (600Ω or 15kΩ selectable input impedance)
Audio Frequency Response:	Pre-emphasis; +1, -3dB (300 to 3000Hz)

** P25 Digital specifications are applicable only for modules with the P25 Digital firmware upgrade.*

VHF and UHF Transmitter Performance Specifications (Continued)

Audio Deviation Limiting:	± 2.5kHz Narrow Band, ± 5.0kHz Wide Band
Audio Distortion:	< 3% THD; 1kHz tone at 1.5kHz or 3kHz deviation
FM Hum and Noise Ratio: 0.3 to 3.4kHz De-emphasis off	≥ 42dB; VHF Narrow Band ≥ 48dB; VHF Wide Band ≥ 38dB; UHF Narrow Band ≥ 44dB; UHF Wide Band
CTCSS Decode / Encode:	Programmable to any of 42 CTCSS tones.
DCS Decode / Encode:	Programmable to any of 83 DCS sequences. Normal or inverted DCS can be selected. Turn-Off code optional.
IC Certification No.:	142A-VT4E150 (VHF) 142A-UT4E450 (UHF 406–470MHz only)
FCC ID:	H4JVT-4E150 (VHF) H4JUT-4E450 (UHF)

** P25 Digital specifications are applicable only for modules with the P25 Digital firmware upgrade.*

VHF AND UHF PHYSICAL SPECIFICATIONS

Physical Dimensions:	Width: 7.1cm (2.8in)	Height: 12.8cm (5.05in)	Depth: 19cm (7.5in)
Module Weight:	Receiver: 1.2kg (2.5lbs) Transmitter: 1.4kg (3.0lbs)		
Corrosion Prevention:	• Anodized aluminum construction • Stainless steel hardware • Gold-plated module connectors		
Module Design:	• Compact Eurostandard modular design • Plug-in modules mate with the standard MT-4E repeater subrack • Subracks / modules comply with IEEE 1101, DIN 41494 and IEC 297-3 (mechanical size / modular arrangement)		
External Connections:	RF Connection: Type-N connector located on the module front panel. Digital I/O interface is made via RJ45 modular jack located on the front panel. Programming interface is made via Mini-B USB 1.1 jack located on the front panel. Motherboard connections (Audio, Power, and Control) are made through a 48-pin, gold-plated, Type-F connector on the rear of the module. User connection made through mated motherboard assembly of the radio subrack. Type-F connector complies with DIN 41612 Level 2 (200 mating cycles, 4-day 10ppm SO_2 gas test with no functional impairment and no change in contact resistance).		

470–520MHZ T-BAND TRANSMITTER PERFORMANCE SPECIFICATIONS

General

Frequency Range:	470–520MHz
Carrier Frequency Stability:	± 0.5ppm
Channel Spacing:	12.5, 25kHz
Channel Selection:	6.25kHz
Number of Channels:	Preset capability for 2 banks of 16 channels
Compatibility:	MT-4 Series Radio Systems; P25 interoperable*
RF Output Power:	0.5 to 6.0W Continuous
Emission Designators:	Analog: 11K0F3E (Narrow band); 16K0F3E (Wide Band) Digital Paging: 9K2F1D P25 Digital: 8K10F1E (Digital Voice); 8K10F1D (Digital Data)*
System Impedance:	50Ω; Type-N connector
Duty Cycle:	100%; Continuous operation
Undesired Emissions: (Adjacent Channel Power Ratio)	< -70dBc, Analog < -70dBc, Digital*
Undesired Emissions: (Conducted Spurious)	< -70dBc @ 6.0W RF Output
Intermodulation Attenuation:	≥ 45dB Analog ≥ 45dB Digital*
VSWR Protection:	< 20:1 VSWR at all phase angles
Operating Temperature Range:	-30˚C to +60˚C
Operating Humidity:	95% RH (non-condensing) at +25°C
Operating Voltage:	+13.8VDC Nominal (range +10 to +17VDC) +9.5VDC Regulated
Transmit Current:	< 1.7A @ 2.0W RF Output < 2.8A @ 6.0W RF Output
Standby Current:	<50mA (no encryption installed) <80mA (with encryption modules installed)
Front Panel Controls:	Transmitter Power On (Norm) / Off / Key Tx Mic Mode: Analog / Digital Analog and Digital Transmit LED indicators
PTT Time-Out-Timer:	Programmable from 15 to 465 sec. (in increments of 15 sec., ± 5 sec.) or infinity (default 300 sec.) using Radio Service Software.
Audio Input:	600Ω balanced or unbalanced input (600Ω or 15kΩ selectable input impedance)
Audio Frequency Response:	Pre-emphasis; +1, -3dB (300 to 3000Hz)
Audio Deviation Limiting:	± 2.5kHz Narrow band, ± 5.0kHz Wide Band
Audio Distortion:	< 3% THD; 1kHz tone at 1.5kHz or 3kHz deviation
FM Hum and Noise Ratio:	≥ 40dB (0.3 to 3.4kHz De-emphasis off)
CTCSS Decode / Encode:	Programmable to any of 42 CTCSS tones.

** P25 Digital specifications are applicable only for modules with the P25 Digital firmware upgrade.*

470–520MHz T-Band Transmitter Performance Specifications (Continued)

DCS Decode / Encode:	Programmable to any of 83 DCS sequences. Normal or inverted DCS can be selected. Turn-Off code optional.
IC Certification No.:	n/a
FCC ID:	H4JUT-4E500

** P25 Digital specifications are applicable only for modules with the P25 Digital firmware upgrade.*

470–520MHZ T-BAND PHYSICAL SPECIFICATIONS

Physical Dimensions:	Width: 7.1cm (2.8in)	Height: 12.8cm (5.05in)	Depth: 19cm (7.5in)
Module Weight:	Receiver: 1.2kg (2.5lbs) Transmitter: 1.4kg (3.0lbs)		
Corrosion Prevention:	• Anodized aluminum construction • Stainless steel hardware • Gold-plated module connectors		
Module Design:	• Compact Eurostandard modular design • Plug-in modules mate with the standard MT-4E repeater subrack • Subracks / modules comply with IEEE 1101, DIN 41494 and IEC 297-3 (mechanical size / modular arrangement)		
External Connections:	RF Connection: Type-N connector located on the module front panel. Digital I/O interface is made via RJ45 modular jack located on the front panel. Programming interface is made via Mini-B USB 1.1 jack located on the front panel. Motherboard connections (Audio, Power, and Control) are made through a 48-pin, gold-plated, Type-F connector on the rear of the module. User connection made through mated motherboard assembly of the radio subrack. Type-F connector complies with DIN 41612 Level 2 (200 mating cycles, 4-day 10ppm SO_2 gas test with no functional impairment and no change in contact resistance).		

700 / 800 / 900MHZ RECEIVER PERFORMANCE SPECIFICATIONS

General

Frequency Range:	768–776MHz, 798–824MHz, 851–869MHz, 896–960MHz
Channel Spacing:	12.5 and 25kHz
Channel Selection:	6.25kHz
Number of Channels:	Preset capability for 2 banks of 16 channels
Channel Switching Range:	unlimited
Compatibility:	MT-4 Series Radio Systems; P25 interoperable*
System Impedance:	50Ω (Type-N connector)
Frequency Generation:	Internal Synthesizer
IF Frequency	73.35MHz
Reference Sensitivity:	≤ -116dBm (0.35uV) for 12dB SINAD ≤ -116dBm (0.35uV) for 5% BER*
Local Oscillator Frequency Stability:	± 0.1ppm
Adjacent Channel Rejection (Selectivity):	Class B: ≥ 40dB; Narrow Band Analog ≥ 45dB; Wide Band Analog ≥ 60dB; Digital*
Intermodulation Rejection:	Class B: ≥ 70dB; Narrow Band and Wide Band Analog ≥ 70dB; Digital*
Spurious Response Rejection:	Class B: ≥ 70dB; Narrow Band and Wide Band Analog ≥ 70dB; Digital*
Conducted Spurious Output Power:	≥ 80dBm
Hum and Noise Ratio:	≥ 31dB; Narrow Band Analog ≥ 37dB; Wide Band Analog
Audio Output:	600Ω balanced line output (configurable for unbalanced line); De-emphasis or Flat output, +3dBm maximum level
Audio Distortion:	≤ 3.0% (25°C); ≤ 5.0% (-30˚C to +60˚C)
Front Panel Controls:	Receiver Power On (Norm) / Off Squelch Disable (Push button) Analog and Digital Receive LED indicators
COR Interface:	150mA, 50V open drain power MOSFET
Supply Voltage:	+13.8VDC Nominal (range +10 to +17VDC) +9.5VDC Regulated
Supply Current:	Class B: <200mA (no encryption modules installed) <230mA (with encryption modules installed)
Operating Temperature Range:	-30°C to +60°C
Operating Humidity:	95% RH (non-condensing) at +25°C
CTCSS Decode:	Programmable to any of 42 CTCSS tones

** P25 Digital specifications are applicable only for modules with the P25 Digital firmware upgrade.*

700 / 800 / 900MHz Receiver Performance Specifications (Continued)

DCS Decode:	Programmable to any of 83 DCS sequences. Normal or inverted DCS can be selected.
IC Certification No.:	142A-UR4E768 (768-776MHz) 142A-UR4E800 (798-824MHz) 142A-UR4E850 (851-869MHz) 142A-UR4E950 (930-960MHz)
FCC ID:	n/a – Declaration of Conformity (DOC)

** P25 Digital specifications are applicable only for modules with the P25 Digital firmware upgrade.*

700 / 800 / 900MHZ TRANSMITTER PERFORMANCE SPECIFICATIONS

General

Frequency Range:	768–869MHz , 896–960MHz
Carrier Frequency Stability:	± 0.1ppm
Channel Spacing:	12.5 and 25kHz
Channel Selection:	6.25kHz
Number of Channels:	Preset capability for 2 banks of 16 channels
Compatibility:	MT-4 Series Radio Systems; P25 interoperable*
RF Output Power:	0.5 to 3.0W Continuous
Emission Designators:	Analog: 11K0F3E (Narrow Band); 16K0F3E (Wide Band) Digital Paging: 9K2F1D (Narrow Band); 16K0F1D (Wide Band) P25 Digital: 8K10F1E (Digital Voice); 8K10F1D (Digital Data)*
System Impedance:	50Ω; Type-N connector
Duty Cycle:	100%; Continuous operation
Undesired Emissions: (Adjacent Channel Power Ratio)	< -50dBc, Narrow Band < -60dBc, Wide Band < -67dBc, Digital*
Undesired Emissions: (Conducted Spurious)	< -80dBc @ 3.0W RF Output < -80dBc @ 3.0W (700 / 800MHz) < -65dBc @ 3.0W (900MHz)
VSWR Protection:	< 20:1 VSWR at all phase angles
Operating Temperature Range:	-30˚C to +60˚C
Operating Humidity:	95% RH (non-condensing) at +25°C
Operating Voltage:	+13.8VDC Nominal (range +10 to +17VDC) +9.5VDC Regulated
Transmit Current:	< 1.8A @ 3.0W RF Output
Stand By Current:	< 115mA (no encryption modules installed) < 145mA (with encryption modules installed)
Front Panel Controls:	Transmitter Power On (Norm) / Off / Key Tx Mic Mode: Analog / Digital Analog and Digital Transmit LED indicators
PTT Time-Out-Timer:	Programmable from 15 to 465 sec. (in increments of 15 sec., ± 5 sec.) or infinity (default 300 sec.) using Radio Service Software.
Audio Input:	600Ω balanced or unbalanced input (600Ω or 15kΩ selectable input impedance)
Audio Frequency Response:	Pre-emphasis; +1, -3dB (300 to 3000Hz)
Audio Deviation Limiting:	± 2.5kHz Narrow Band, ± 5.0kHz Wide Band
Audio Distortion:	< 3% THD; 1kHz tone at 1.5kHz or 3kHz deviation
FM Hum and Noise Ratio: 0.3 to 3.4kHz De-emphasis off	≥ 34dB; 700-800MHz Narrow Band Analog ≥ 40dB; 700-800MHz Wide Band Analog ≥ 31dB; 900MHz Narrow Band Analog ≥ 37dB; 900MHz Wide Band Analog

** P25 Digital specifications are applicable only for modules with the P25 Digital firmware upgrade.*

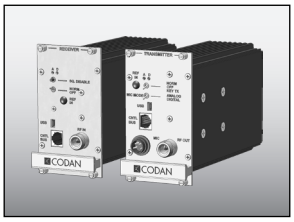
700 / 800 / 900MHz Transmitter Performance Specifications (Continued)

CTCSS Decode / Encode:	Programmable to any of 42 CTCSS tones.
DCS Decode / Encode:	Programmable to any of 83 DCS sequences. Normal or inverted DCS can be selected. Turn-Off code optional.
IC Certification No.:	142A-UT4E850 (768–869MHz) 142A-UT4E900 (896-960MHz)
FCC ID:	H4JUT-4E850 (768-869MHz) H4JUT-4E900 (896-960MHz)

** P25 Digital specifications are applicable only for modules with the P25 Digital firmware upgrade.*

700 / 800 / 900MHZ PHYSICAL SPECIFICATIONS

Physical Dimensions:	Width: 7.1cm (2.8in)	Height: 12.8cm (5.05in)	Depth: 19cm (7.5in)
Module Weight:	Receiver: 1.2kg (2.5lbs) Transmitter: 1.4kg (3.0lbs)		
Corrosion Prevention:	• Anodized aluminum construction • Stainless steel hardware • Gold-plated module connectors		
Module Design:	• Compact Eurostandard modular design • Plug-in modules mate with the standard MT-4E repeater subrack • Subracks / modules comply with IEEE 1101, DIN 41494 and IEC 297-3 (mechanical size / modular arrangement)		
External Connections:	RF Connection: Type-N connector located on the module front panel. Digital I/O interface is made via RJ45 modular jack located on the front panel. Programming interface is made via Mini-B USB 1.1 jack located on the front panel. Motherboard connections (Audio, Power and Control) are made through a 48-pin, gold-plated, Type-F connector on the rear of the module. User connection is made through mated motherboard assembly of the radio subrack. Type-F connector complies with DIN 41612 Level 2 (200 mating cycles, 4-day 10ppm SO_2 gas test with no functional impairment and no change in contact resistance).		



GLOSSARY OF TERMS

Bandwidth
: The difference between the limiting frequencies of a continuous frequency band. Typically measured in kilohertz. May be considered the amount in kilohertz required for a single communications channel.

BER
: BIT Error Rate

BPS
: BITs Per Second – a data rate measure

C4FM
: Constant Envelope 4-Level Frequency Modulation – an FM transmitter which uses Quadrature Phase Shift Keying (QPSK) modulation to work with a compatible frequency discriminator detection (CFDD) receiver.

Channel
: A single unidirectional or bidirectional path for transmitting or receiving, or both, of electrical or electromagnetic signals.

Channel Spacing
: Typically measured in kilohertz from the center of one channel to the center of the next-adjacent channel. May, or may not, be identical to bandwidth.

Common Air Interface (CAI)
: A radio-to-radio signal path defined in terms of Access Method, Modulation Scheme, Vocoding Method, Channel Data Rate and Channel Data Format.

CTCSS
: Abbreviation for “Continuous Tone-Controlled Squelch System”

DCS
: Digital Coded Squelch

DSP
: Digital Signal Processor – a specialized microprocessor

DTMF
: Dual-Tone Multi-Frequency – a signaling scheme used by the telephone system in which two voice band tones are generated for each keypad key press.

Encryption
: A coding of plain text (or clear voice) into unintelligible forms for secure transmission.

Error Correction
: Digital coding technique for detecting and correcting information transmission errors.

FCC
: Federal Communications Commission

Firmware
: Software that is permanently stored in a hardware device and allows reading and executing, but not writing or modifying the software.

I/O
: Input / Output

ISO
: International Standards Organization

LSB
: Least Significant BIT

Modulation
: A controlled variation of any property of a carrier wave for the purpose of transferring information.

MSB
: Most Significant BIT

NAC
: Network Access Code. A 12-bit field identifying the network of the radio message. Typically used to steer repeater functions.

PTT
: Push-to-Talk – the switch on a subscriber unit which, when pressed, causes the subscriber unit to transmit.

QPSK
: Quadrature Phase Shift Keying – modulation technique. PSK using four-phase states.

RF
: Radio Frequency

Signal
: The detectable transmitted energy which carries information from a transmitter to a receiver.

SINAD
: SIgnal-plus-Noise-and-Distortion to noise-and-distortion ratio

Squelch
: A radio circuit that eliminates noise from the speaker when no transmitted signal is present.

Subscriber Unit
: A mobile or portable radio unit used in a radio system.

TGID
: Talk Group Identifier. A 16-bit field identifying the talk group of the radio message.

TIA
: Telecommunications Industry Association

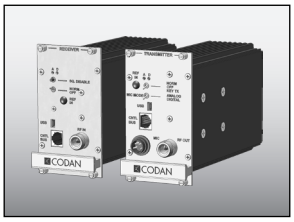
Time-Out Timer
: A function that limits the transmission period to a pre-defined time. The user will automatically stop transmitting when the timer goes off after the pre-defined time.

Trunked (System)
: Systems with full-feature sets in which all aspects of radio operation—including RF channel selection and access—are centrally managed.

UCM
: Universal Cryptographic Module

UDB
: Universal Daughter Board

VOCODER
: Short for Voice Encoder. Usually consisting of a speech analyzer and a speech synthesizer which convert analog speech into digital signals for transmission and digital signals back into artificial speech sounds for reception.



REVISION HISTORY

Revision	Date	Action No.	Description
	Revision history older than seven years has been removed. See previous versions of this manual for prior revision history		
6-0-0	Jan 2013	6773	Lexan Label replacement to ID Handle Plates • Updated VHF, UHF and 700–900 MHz BOMs • Removed handle text colour specifications
		6827	Remove encryption appendix to new stand-alone manual
		---	Update to Codan template
6-1-0	May 2013	6841	Update for speaker output – R36 • Updated A11-RX4E-MAIN schematic – Page 3 was 60021-01-02-S now 60021-01-03-S
		6854	Replacement for LC Filter • Updated A14-RX4E-MAIN schematic – Page 3 was 50171-03-01-S-04-01 now 50171-03-01-S-05-01
		6856	Mainboard Distortion – CV3 • Updated A11-RX4E-MAIN schematic – Page 3 was 60021-01-03-S now 60021-01-04-S
		6861	EOL Replacement of LC Filter • Updated A21-TX4E-MAIN schematic – Page 3 was 60012-02-01-S now 60012-02-02-S
6-2-0	Apr 2014	6938	Update FM Hum and Noise Specs • Updated specifications for VHF and UHF • Updated specifications for UHF 380 MHz • Updated specifications for UHF 700-800 MHz

Table of Contents continued on the next page.

Revision History (Continued)

Revision	Date	Action No.	Description
6-3-0	Sep 2014	6963	UHF Receivers • Updated jumper table – JU20 for the 21.4 MHz mainboard WAS not installed; NOW installed • Updated jumper table – JU26 for the 73.35 MHz mainboard WAS not installed; NOW installed • Upreved schematic PCB 60021 on Page 46 WAS 60021-01-04-S; NOW 60021-01-05-S • Upreved schematic PCB 50171 on Page 92 WAS 50171-03-01-S-05-01; NOW 50171-03-01-S-06-01
		---	Removed MT-3 references in the manual Introduction and in the Performance Specifications
7-0-0	Apr 2017	7096	RSS Release • Updated RSS sections with new screen shots • Corrected numerous text errors • Added formal Figure captions
		7101	• Updated Receiver RF Preselector Tuning Procedure values
08	Nov 2019	AR 7120	• Updated to SAP versioning WAS x-x-x NOW -xx • Updated branding to Codan Communications • New template and style guide
09	Dec 2019	AR 7206	• Corrected Test Point table - removed TP57
10	Jan 2021	CO 8148 AR 8159	• Removed UR-4E380-00-000, UR-4E440-00-000, UR-4E900-00-000 and UT-4E380-00-000 models from the front cover • Removed references of UR-4E380-00-000, UR-4E440-00-000, UR-4E900-00-000 and UT-4E380-00-000 from tables and text • Removed UHF 380, 440 and UT 900MHz references • Removed A13-FE4-440, A16-FE4-900, A64-OR4-380, A64-OT4-380 from Parts Lists • Removed Specification tables for UR-4E380, UR-4E440, UR-4E900 and UT-4E380